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Maintenance therapy for chronic lymphocytic leukaemia (Review)

Lee CH, Wu YY, Huang TC, Lin C, Zou YF, Cheng JC, Chen PH, Jhou HJ, Ho CL

Lee C-H, Wu Y-Y, Huang T-C, Lin C, Zou Y-F, Cheng J-C, Chen P-H, Jhou H-J, Ho C-L.
Maintenance therapy for chronic lymphocytic leukaemia.
Cochrane Database of Systematic Reviews 2024, Issue 1. Art. No.: CD013474.
DOI: 10.1002/14651858.CD013474.pub2.

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[Intervention Review]

Maintenance therapy for chronic lymphocytic leukaemia

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Editorial group: Cochrane Haematology Group.
Publication status and date: New, published in Issue 1, 2024.

Citation: Lee C-H, Wu Y-Y, Huang T-C, Lin C, Zou Y-F, Cheng J-C, Chen P-H, Jhou H-J, Ho C-L. Maintenance therapy for chronic lymphocytic leukaemia. *Cochrane Database of Systematic Reviews* 2024, Issue 1. Art. No.: CD013474. DOI: 10.1002/14651858.CD013474.pub2.

ABSTRACT

Background

Chronic lymphocytic leukaemia (CLL) is the most common lymphoproliferative disease in adults and currently remains incurable. As the progression-free period shortens after each successive treatment, strategies such as maintenance therapy are needed to improve the degree and duration of response to previous therapies. Monoclonal antibodies, immunomodulatory agents, and targeted therapies are among the available options for maintenance therapy. People with CLL who achieve remission after previous therapy may choose to undergo medical observation or maintenance therapy to deepen the response. Even though there is widespread use of therapeutic maintenance agents, the benefits and harms of these treatments are still uncertain.

Objectives

To assess the effects and safety of maintenance therapy, including anti-CD20 monoclonal antibody, immunomodulatory drug therapy, anti-CD52 monoclonal antibody, Bruton tyrosine kinase inhibitor, and B-cell lymphoma-2 tyrosine kinase inhibitor, for individuals with CLL.

Search methods

We conducted a comprehensive literature search for randomised controlled trials (RCTs) with no language or publication status restrictions. We searched CENTRAL, MEDLINE, Embase, and three trials registers in January 2022 together with reference checking, citation searching, and contact with study authors to identify additional studies.

Selection criteria

We included RCTs with prospective identification of participants. We excluded cluster-randomised trials, cross-over trial designs, and non-randomised studies. We included studies comparing maintenance therapies with placebo/observation or head-to-head comparisons.

Data collection and analysis

We used standard Cochrane methodological procedures. We assessed risk of bias in the included studies using Cochrane's RoB 1 tool for RCTs. We rated the certainty of evidence for the following outcomes using the GRADE approach: overall survival (OS), health-related quality of life (HRQoL), grade 3 and 4 adverse events (AEs), progression-free survival (PFS), treatment-related mortality (TRM), treatment discontinuation (TD), and all adverse events (AEs).

Main results

We identified 11 RCTs (2393 participants) that met the inclusion criteria, including seven trials comparing anti-CD20 monoclonal antibodies (mAbs) (rituximab or ofatumumab) with observation in 1679 participants; three trials comparing immunomodulatory drug (lenalidomide) with placebo/observation in 693 participants; and one trial comparing anti-CD 52 mAbs (alemtuzumab) with observation in 21 participants. No comparisons of novel small molecular inhibitors were found.

The median age of participants was 54.1 to 71.7 years; 59.5% were males. The type of previous induction treatment, severity of disease, and baseline stage varied among the studies. Five trials included early-stage symptomatic patients, and three trials included advanced-stage patients (Rai stage III/IV or Binet stage B/C). Six trials reported a frequent occurrence of cytogenic aberrations at baseline (69.7% to 80.1%). The median follow-up duration was 12.4 to 73 months. The risk of selection bias in the included studies was unclear. We assessed overall risk of performance bias and detection bias as low risk for objective outcomes and high risk for subjective outcomes. Overall risk of attrition bias, reporting bias, and other bias was low.

Anti-CD20 monoclonal antibodies (mAbs): rituximab or ofatumumab maintenance versus observation

Anti-CD20 mAbs maintenance likely results in little to no difference in OS (hazard ratio (HR) 0.94, 95% confidence interval (CI) 0.73 to 1.20; 1152 participants; 3 studies; moderate-certainty evidence) and likely increases PFS significantly (HR 0.61, 95% CI 0.50 to 0.73; 1255 participants; 5 studies; moderate-certainty evidence) compared to observation alone.

Anti-CD20 mAbs may result in: an increase in grade 3/4 AEs (rate ratio 1.34, 95% CI 1.06 to 1.71; 1284 participants; 5 studies; low-certainty evidence); little to no difference in TRM (risk ratio 0.82, 95% CI 0.39 to 1.71; 1189 participants; 4 studies; low-certainty evidence); a slight reduction to no difference in TD (risk ratio 0.93, 95% CI 0.72 to 1.20; 1321 participants; 6 studies; low-certainty evidence); and an increase in all AEs (rate ratio 1.23, 95% CI 1.03 to 1.47; 1321 participants; 6 studies; low-certainty evidence) compared to the observation group.

One RCT reported that there may be no difference in HRQoL between the anti-CD20 mAbs (ofatumumab) maintenance and the observation group (mean difference −1.70, 95% CI −8.59 to 5.19; 480 participants; 1 study; low-certainty evidence).

Immunomodulatory drug (IMiD): lenalidomide maintenance versus placebo/observation

IMiD maintenance therapy likely results in little to no difference in OS (HR 0.91, 95% CI 0.61 to 1.35; 461 participants; 3 studies; moderate-certainty evidence) and likely results in a large increase in PFS (HR 0.37, 95% CI 0.19 to 0.73; 461 participants; 3 studies; moderate-certainty evidence) compared to placebo/observation.

Regarding harms, IMiD maintenance therapy may result in an increase in grade 3/4 AEs (rate ratio 1.82, 95% CI 1.38 to 2.38; 400 participants; 2 studies; low-certainty evidence) and may result in a slight increase in TRM (risk ratio 1.22, 95% CI 0.35 to 4.29; 458 participants; 3 studies; low-certainty evidence) compared to placebo/observation. The evidence for the effect on TD compared to placebo is very uncertain (risk ratio 0.71, 95% CI 0.47 to 1.05; 400 participants; 2 studies; very low-certainty evidence). IMiD maintenance therapy probably increases all AEs slightly (rate ratio 1.41, 95% CI 1.28 to 1.54; 458 participants; 3 studies; moderate-certainty evidence) compared to placebo/observation.

No studies assessed HRQoL.

Anti-CD52 monoclonal antibodies (mAbs): alemtuzumab maintenance versus observation

Maintenance with alemtuzumab may have little to no effect on PFS, but the evidence is very uncertain (HR 0.55, 95% CI 0.32 to 0.95; 21 participants; 1 study; very low-certainty evidence). We did not identify any study reporting the outcomes OS, HRQoL, grade 3/4 AEs, TRM, TD, or all AEs.

Authors' conclusions

There is currently moderate- to very low-certainty evidence available regarding the benefits and harms of maintenance therapy in people with CLL. Anti-CD20 mAbs maintenance improved PFS, but also increased grade 3/4 AEs and all AEs. IMiD maintenance had a large effect on PFS, but also increased grade 3/4 AEs. However, none of the above-mentioned maintenance interventions show differences in OS between the maintenance and control groups. The effects of alemtuzumab maintenance are uncertain, coupled with a warning for drug-related infectious toxicity. We found no studies evaluating other novel maintenance interventions, such as B-cell receptor inhibitors, B-cell leukaemia-2/lymphoma-2 inhibitors, or obinutuzumab.

PLAIN LANGUAGE SUMMARY

Is maintenance therapy following previous treatment better than observation or placebo for treating chronic lymphocytic leukaemia in adults?

What is the aim of this review?

We aimed to determine whether the currently available maintenance treatments for disease control in adults with chronic lymphocytic leukaemia (CLL) have an acceptable balance between benefits and harms following previous treatment.

Key messages

We found that maintenance therapy in adults with CLL after previous therapy probably has little to no effect on long-term survival. We also found that maintenance therapy probably increases the disease control period, but is associated with an increase in drug-related serious adverse events and all adverse events.

What was studied in this review?

CLL is a blood cancer that affects the lymphocytes, a group of white blood cells that help the body fight infection and disease. In the past 20 years, the control rate and survival of patients have significantly improved due to the use of chemotherapy plus immunotherapy, followed by the introduction of novel oral targeted therapy as the mainstream treatment for CLL. Despite these advancements, CLL patients may still develop disease that is resistant to treatment. Maintenance treatments aim to improve treatment response and extend survival, but may also increase the risk of adverse events. Available drugs for maintenance intervention include anti-CD20 monoclonal antibodies (mAbs), immunomodulatory drugs, and anti-CD52 mAbs. These drugs carry risks of serious side effects due to prolonged administration. It remains unclear whether the benefits of maintenance treatments outweigh the risks, as they may not achieve a satisfactory balance between disease control and prolonged survival without increasing treatment-related death or adverse events. CLL patients who achieve remission after the first phase of treatment must therefore weigh the benefits and risks of medical observation or maintenance therapy carefully.

What did we want to find out?

We wanted to know if maintenance therapy following previous treatment is effective in controlling disease for adults with CLL and if it causes any adverse effects.

What did we do?

We searched for studies that compared maintenance therapy with placebo (dummy treatment) or observation, or head-to-head comparisons of maintenance therapies. We compared and summarised the results and rated our confidence in the evidence based on factors such as study methods and sizes.

What did we find?

We identified 11 studies (2393 participants) for inclusion in the review. All studies were randomised controlled trials (RCTs), which are the most reliable clinical studies as they randomly allocate people to receive treatment (intervention group) or no treatment (control group). We synthesised the evidence from 10 studies (2341 participants) to assess the benefits and harms of maintenance therapy for CLL when compared with control groups receiving observation or placebo (dummy treatment).

Compared to observation, anti-CD20 mAbs probably have little to no effect on overall survival (evidence from 3 studies with 1152 participants), but probably increase the time in which the disease is controlled (progression-free survival) (evidence from 5 studies with 1255 participants).

Compared to placebo/observation, IMiD probably has little to no effect on overall survival (evidence from 3 studies with 461 participants), but probably results in a large increase in progression-free survival (evidence from 3 studies with 461 participants).

Compared to observation, anti-CD20 mAbs may result in increased serious side effects (evidence from 5 studies with 1284 participants), but may not affect the chance of dying from treatment (evidence from 4 studies with 1189 participants). Anti-CD20 mAbs may slightly reduce the number of people who stopped treatment (evidence from 6 studies with 1321 participants), but may slightly increase the overall number of side effects (evidence from 6 studies with 1321 participants).

Compared to placebo/observation, IMiD may result in increased serious side effects (evidence from 2 studies with 400 participants), and may slightly increase the number of people who died due to maintenance treatment (evidence from 3 studies with 458 participants). The evidence for the effect of IMiD on number of people who stopped treatment is very uncertain (evidence from 2 studies with 400 participants), but IMiD probably increases the overall number of side effects (evidence from 3 studies with 458 participants).

There was insufficient evidence to determine whether anti-CD20 mAbs maintenance therapy affected health-related quality of life, and our confidence in the evidence is low (evidence from 1 study with 480 participants).

No studies evaluated other new maintenance interventions, such as B-cell receptor inhibitors, B-cell leukaemia-2/lymphoma-2 inhibitors, or obinutuzumab.

What are the limitations of the evidence?

The overall certainty (confidence) of the evidence was mostly moderate to low due to issues within the trials such as the potential for bias, differences in baseline characteristics, and imprecise estimates of intervention effects.

How up-to-date is this review?

The evidence is current as of January 2022.

SUMMARY OF FINDINGS

Summary of findings 1. Anti-CD20 mAb maintenance therapy compared to observation for people with chronic lymphocytic leukaemia

Anti-CD20 mAb maintenance therapy compared to observation for people with chronic lymphocytic leukaemia

Patient or population: people with chronic lymphocytic leukaemia
Setting: inpatient and outpatient care
Intervention: anti-CD20 mAb maintenance therapy
Comparison: observation

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with observation	Risk with anti-CD20 mAb maintenance therapy				
Overall survival Follow-up: range 33.7 months to 47.7 months	Study population		HR 0.94 (0.73 to 1.20)	1152 (3 RCTs)	⊕⊕⊕⊝ Moderate [1]	Anti-CD20 mAb maintenance therapy probably results in little to no difference in overall survival. [2 3]
	240 per 1000	261 per 1000 (180 to 353)				
Health-related quality of life Follow-up: median 40.9 months	The mean health-related quality of life was 1.9.	MD 1.7 lower (8.59 lower to 5.19 higher)	-	480 (1 RCT)	⊕⊕⊝⊝ Low [1 4]	Anti-CD20 mAb maintenance therapy may not reduce health-related quality of life. [3]
Grade 3/4 adverse events Follow-up: range 14.4 months to 47.7 months	Study population		Rate ratio 1.34 (1.06 to 1.71)	1284 (5 RCTs)	⊕⊕⊝⊝ Low [4 5]	Anti-CD20 mAb maintenance therapy may result in an increase in grade 3/4 adverse events. [3 6]
	440 per 1000	590 per 1000 (466 to 752)				
Progression-free survival Follow-up: range 14.4 months to 47.7 months	Study population		HR 0.61 (0.50 to 0.73)	1255 (5 RCTs)	⊕⊕⊕⊝ Moderate [4]	Anti-CD20 mAb maintenance therapy probably increases progression-free survival. [2 3]
	480 per 1000	639 per 1000 (585 to 693)				
Treatment-related mortality Follow-up: range 33.7 months to 47.7 months	Study population		Risk ratio 0.82 (0.39 to 1.71)	1189 (4 RCTs)	⊕⊕⊝⊝ Low [1 7]	Anti-CD20 mAb maintenance therapy may result in little to no difference in treatment-related mortality. [3 6]
	68 per 1000	56 per 1000 (26 to 116)				

Treatment discontinuation Follow-up: range 14.4 months to 47.7 months	Study population		Risk ratio 0.93 (0.72 to 1.20)	1321 (6 RCTs)	⊕⊕⊝⊝ Low [1 8]	Anti-CD20 mAb maintenance therapy may result in a slight reduction in treatment discontinuation. [3 6]
	444 per 1000	409 per 1000 (320 to 533)				
All adverse events Follow-up: range 14.4 months to 47.7 months	Study population		Rate ratio 1.23 (1.03 to 1.47)	1321 (6 RCTs)	⊕⊕⊝⊝ Low [4 9]	Anti-CD20 mAb maintenance therapy may result in a slight increase in all adverse events. [3 6]
	800 per 1000	984 per 1000 (824 to 1176)				

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).
CI: confidence interval; **HR:** hazard ratio; **MD:** mean difference; **RCT:** randomised controlled trial

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[1]Downgraded one level due to serious imprecision (wide confidence intervals, which cross the line of no difference).
[2]Assumed risk was calculated based on large cohort study (Mato et al 2020, DOI: 10.1182/bloodadvances.2019001145).
[3]We could not address publication bias through funnel plots, as fewer than 10 studies were included in comparison. Given our comprehensive search for unpublished studies, including contacting the experts and researchers in the field, we believe suspicion about publication bias is reduced.
[4]Downgraded one level due to serious risk of bias (high risk of bias due to lack of blinding).
[5]Downgraded one level due to serious inconsistency (substantial heterogeneity, $I^2 = 56\%$).
[6]Assumed risk was calculated based on control group.
[7]Downgraded one level due to serious inconsistency (substantial heterogeneity, $I^2 = 47\%$, two studies suggest benefit and two suggest harm).
[8]Downgraded one level due to serious inconsistency (substantial heterogeneity, $I^2 = 64\%$).
[9]Downgraded one level due to serious inconsistency (substantial heterogeneity, $I^2 = 56\%$; inconsistency in the reporting criteria for adverse events (AEs): two studies reported all AEs occurring in more than 2% of participants, while another two studies reported all AEs occurring in more than 10% of participants. Additionally, one study reported all AEs, while another study reported AEs of grade 3, 4, and 5 occurring in more than 5% of participants).

Summary of findings 2. Immunomodulatory drug maintenance therapy compared to placebo/observation for people with chronic lymphocytic leukaemia

Immunomodulatory drug maintenance therapy compared to placebo/observation for people with chronic lymphocytic leukaemia

Patient or population: people with chronic lymphocytic leukaemia
Setting: inpatient and outpatient care
Intervention: immunomodulatory drug maintenance therapy

Comparison: placebo/observation

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo/observation	Risk with immunomodulatory drug maintenance therapy				
Overall survival Follow-up: range 17.9 months to 73 months	Study population		HR 0.91 (0.61 to 1.35)	461 (3 RCTs)	⊕⊕⊕⊝ Moderate [1]	Immunomodulatory drug maintenance therapy probably results in little to no difference in overall survival. [2,3]
	280 per 1000	314 per 1000 (179 to 460)				
Health-related quality of life – not reported	-	-	-	-	-	No study reporting the outcome was identified.
Grade 3/4 adverse events Follow-up: range 17.9 months to 31.5 months	Study population		Rate ratio 1.82 (1.38 to 2.38)	400 (2 RCTs)	⊕⊕⊝⊝ Low [4]	Immunomodulatory drug maintenance therapy may result in an increase in grade 3/4 adverse events. [3,5]
	420 per 1000	748 per 1000 (571 to 983)				
Progression-free survival Follow-up: range 17.9 months to 73 months	Study population		HR 0.37 (0.19 to 0.73)	461 (3 RCTs)	⊕⊕⊕⊝ Moderate [6]	Immunomodulatory drug maintenance therapy probably results in a large increase in progression-free survival. [2,3]
	480 per 1000	762 per 1000 (585 to 870)				
Treatment-related mortality Follow-up: range 17.9 months to 73 months	Study population		Risk ratio 1.22 (0.35 to 4.29)	458 (3 RCTs)	⊕⊕⊝⊝ Low [7]	Immunomodulatory drug maintenance therapy may result in a slight increase in treatment-related mortality. [3,5]
	19 per 1000	23 per 1000 (7 to 82)				
Treatment discontinuation Follow-up: range 17.9 months to 31.5 months	Study population		Risk ratio 0.71 (0.47 to 1.05)	400 (2 RCTs)	⊕⊝⊝⊝ Very low [8,9]	The evidence is very uncertain about the effect of immunomodulatory drug maintenance therapy on treatment discontinuation. [3,5]
	776 per 1000	551 per 1000 (365 to 815)				
All adverse events Follow-up: range 17.9 months to 73 months	Study population		Rate ratio 1.41 (1.28 to 1.54)	458 (3 RCTs)	⊕⊕⊕⊝ Moderate [10]	Immunomodulatory drug maintenance therapy probably increases all adverse events slightly. [3,5]
	900 per 1000	1242 per 1000 (1134 to 1368)				

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **HR:** hazard ratio; **RCT:** randomised controlled trial

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[1]Downgraded one level due to serious imprecision (wide confidence intervals, which cross the line of relative risk reduction threshold).
[2]Assumed risk was calculated based on large cohort study (Mato et al 2020, DOI: 10.1182/bloodadvances.2019001145).
[3]We could not address publication bias through funnel plots, as fewer than 10 studies were included in comparison. Given our comprehensive search for unpublished studies, including contacting the experts and researchers in the field, we believe suspicion about publication bias is reduced.

[4]Downgraded two levels due to very serious imprecision (wide confidence intervals, small number of participants)
[5]Assumed risk was calculated based on control group.
[6]Downgraded one level due to serious inconsistency (substantial heterogeneity, I^2 = 68%).
[7]Downgraded two levels due to very serious imprecision (very wide confidence intervals, which cross the line of relative risk reduction threshold, and low event rates).
[8]Downgraded one level due to serious inconsistency (substantial heterogeneity, I^2 = 75%).
[9]Downgraded two levels due to very serious imprecision (wide confidence intervals, which cross the line of relative risk reduction threshold, and small number of participants).
[10]Downgraded one level for serious inconsistency (inconsistencies in the reporting criteria for adverse events (AEs): one study reported all AEs occurring in more than 10% of participants, while another study reported AEs of grade 1 and 2 occurring in > 10% of participants, but only reported treatment-related AEs of grade 3, 4, and 5. Additionally, one study only reported grade 3, 4, and 5 events).

Summary of findings 3. Anti-CD52 mAb maintenance therapy compared to observation for people with chronic lymphocytic leukaemia

Anti-CD52 mAb maintenance therapy compared to observation for people with chronic lymphocytic leukaemia

Patient or population: people with chronic lymphocytic leukaemia
Setting: inpatient and outpatient care
Intervention: anti-CD52 mAb maintenance therapy
Comparison: observation

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with observation	Risk with Anti-CD52 mAb maintenance therapy				

Overall survival – not reported	-	-	-	-	-	No study reporting this outcome was identified.
Health-related quality of life – not reported	-	-	-	-	-	No study reporting this outcome was identified.
Grade 3/4 adverse events – not reported	-	-	-	-	-	No study reporting this outcome was identified.
Progression-free survival Follow-up: median 12.4 months	Study population		HR 0.55 (0.32 to 0.95)	21 (1 RCT)	⊕⊝⊝⊝ Very low [1,2]	Anti-CD52 mAb maintenance therapy may lead to an increase in progression-free survival, but the evidence is very uncertain. [3]
	400 per 1000	604 per 1000 (419 to 746)				
Treatment-related mortality – not reported	-	-	-	-	-	No study reporting this outcome was identified.
Treatment discontinuation – not reported	-	-	-	-	-	No study reporting this outcome was identified.
All adverse events – not reported	-	-	-	-	-	No study reporting this outcome was identified.

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).
CI: confidence interval; **HR:** hazard ratio; **RCT:** randomised controlled trial

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[1]Downgraded one level for serious risk of bias (single study that does not report methods of random sequence generation or allocation, high risk of bias due to lack of blinding).
[2]Downgraded two levels for very serious imprecision (single study, 21 participants, terminated early, likely underpowered to demonstrate an effect).
[3]Assumed risk was calculated based on large cohort study (Mato et al 2020, DOI: 10.1182/bloodadvances.2019001145).

BACKGROUND

Description of the condition

Chronic lymphocytic leukaemia (CLL) is the most common lymphoproliferative disease in adults (Siegel 2015). The median age of diagnosis is 72 years (Eichhorst 2015), and the age-adjusted incidence is five per 100,000 person-years (Howlader 2016; Statistics Canada 2016). CLL has a highly variable clinical course and prognosis (Hallek 2013a). Median overall survival (OS) varies from 18 months to over 20 years (Eichhorst 2015; Siddon 2013), and some individuals may lead a normal life with few or no symptoms for many years without requiring treatment. CLL is generally considered to be an indolent disease that progresses slowly, with most individuals needing treatment only after years of clinical medical observation (Hallek 2015). The severity of disease is reflected by the enlargement of the lymph nodes, liver, and spleen; present of B symptoms; an increased blood lymphocyte count; dysplastic haematopoiesis-induced cytopenias; and autoimmune complications (Hallek 2018). The two most widely used staging systems, proposed by Rai and colleagues, Rai 1975 and Binet and colleagues, Binet 1981, distinguish early (Rai 0; Binet A), intermediate (Rai I, II; Binet B), and advanced (Rai III/IV; Binet C) stages of disease, which are characterised by substantial differences in clinical course and long-term survival. The current clinical staging systems are often of limited prognostic value at diagnosis, which usually occurs at the early stage of the disease (Binet 1981; Hallek 2008; Rai 1975). Later, cytogenetic anomalies, including del(17p) deletions, somatic tumor protein p53 (*TP53*) gene, and immunoglobulin variable heavy-chain mutations (IGHV), have been found to be more effective in distinguishing more and less aggressive forms of CLL (Mertens 2014).

Most people with CLL do not require immediate treatment at the time of diagnosis. Treatment is usually initiated when the disease progresses, such as with progressive B symptoms, persistent cytopenias, hepatosplenomegaly, or autoimmune phenomena. However, most people with CLL will ultimately need treatment during their lifetime. Most individuals with CLL are treated when they are in an advanced stage of the disease and are experiencing symptoms or have haematopoietic insufficiency (Hallek 2008). Prognostic markers such as IGHV mutation status, fluorescence in situ hybridisation, zeta chain of T cell receptor associated protein kinase 70 (*ZAP70*), CD38 molecule (*CD38*), and β2-microglobulin can be used to predict the time to first CLL treatment. The standard of care for CLL has rapidly evolved from monotherapy with chlorambucil or purine analogues (fludarabine, pentostatine) to combination chemotherapy with cyclophosphamide, doxorubicin, vincristine, and prednisolone (CHOP); cyclophosphamide and vincristine with prednisolone (COP); or fludarabine with cyclophosphamide, to chemo-immunotherapy with fludarabine-based chemotherapy plus anti-CD20 monoclonal antibodies (mAbs) (rituximab) (Cramer 2016; Fischer 2012; Hallek 2010; Thompson 2016a). Recently, a number of novel drugs have been developed that target key pathogenic pathways in CLL, such as Bruton tyrosine kinase (*BTK*) and phosphoinositide 3-kinase (PI3K) inhibitors that disrupt B-cell receptor (BCR) signalling, and venetoclax, an antagonist of the anti-apoptotic protein B-cell leukaemia/lymphoma 2 (BCL-2). B-cell receptor kinase inhibitors (BCRi) like ibrutinib and idelalisib are initially beneficial for older individuals, and later show benefits for the entire population (Burger 2014; Byrd 2013; Byrd 2014; Furman 2014; Jones 2017; O'Brien 2016; Zelenetz 2017). Most people with CLL respond to first-line therapy. Although durable and sustained remissions can be achieved, the majority of people with CLL relapse within five years of initial treatment (Shindiapina 2016; Wierda 2005). Patients who relapse often become resistant to chemotherapy, develop fludarabine resistance, and experience poor outcomes due to limited treatment options and acquired genetic characteristics (unmutated immunoglobulin heavy chain gene, 17p13 deletion, and *TP53* mutations) in relapsed leukaemia clones (i.e. clonal evolution), with a poor prognosis (Dohner 2000; Grever 2007; Wierda 2017). The novel *BTK*-targeting agent ibrutinib has shown promising clinical responses for both naive and refractory CLL patients, but primary or acquired resistance is still common and often leads to poor clinical outcomes (George 2020). Such individuals usually have significantly reduced overall survival due to early progression and poor response to salvage therapy (Cramer 2013).

Previous National Comprehensive Cancer Network guidelines recommended retreatment with the chemo-immunotherapy used as first-line therapy for individuals who require second-line treatment if they achieved a duration of response of more than two years (Wierda 2017). However, fludarabine, cyclophosphamide, and rituximab (FCR) or fludarabine and bendamustine (RB) yield short remission times, and few previously treated individuals will have a complete response to treatment (Badoux 2011a; Robak 2010a; Wierda 2005); these types of retreatment modalities have become alternative treatment options due to the increasing number of treatment options. Second-line treatment with BCRi such as ibrutinib or idelalisib, or B-cell lymphoma 2 inhibitors (BCL-2i) such as venetoclax leads to partial and sustained responses in most individuals with refractory or relapsed disease, but complete responses with no minimal residual disease (MRD) are rare (Byrd 2014; Furman 2014; Thompson 2016b). As the progression-free period generally becomes shorter after each successive treatment period, strategies such as maintenance are needed to improve the depth and duration of the response to previous therapies. It has been proposed that maintenance therapy after chemo-immunotherapy might prolong the duration of remission by preventing the re-emergence of malignant CLL clones (Abrisqueta 2013; Wendtner 2004), and that maintenance regimens should be considered for individuals with high-risk CLL (Stilgenbauer 2010). In 2015, a study by van Oers and colleagues reported a significant improvement in progression-free survival (PFS) following ofatumumab maintenance in people with CLL in remission after retreatment compared to observation (Van Oers 2015). This pilot trial was followed by other trials of maintenance therapy, including the CLLM1 (Fink 2017), CONTINUUM (Chanan-Khan 2017), CLL-2007-SA (Dartigeas 2018), and CLL-8a (Greil 2016) trials, which reported good outcomes with lenalidomide and rituximab maintenance. The ongoing GALACTIC trial is evaluating the maintenance effect of the novel anti-CD20 mAb obinutuzumab since 2017 (Oughton 2017).

Despite the demonstrated ability of maintenance therapy to increase duration of response and deepen response in chemo-immunotherapy, there is ongoing debate about the long-term benefits and harms of this treatment modality. In the modern era of targeted therapy, the common treatment strategy is continuous treatment until progression, which combines induction and maintenance therapy to prevent refractory and resistant disease. The choice of maintenance therapy currently depends on the front-line therapy used, the presence of high-risk patient characteristics, and individual fitness (Grever 2007; Hallek 2017). There is a need for methods to determine the optimal maintenance treatment to maximise clinically relevant outcomes in people with CLL after previous therapy. To summarise the available data and discuss the role of maintenance in current treatment algorithms, we conducted a systematic review and pairwise meta-analysis of all phase-3 randomised clinical trials comparing maintenance therapy regimens in people with CLL.

Description of the intervention

Many available treatment methods can effectively induce disease remission and improve duration of response compared to observation. Allogeneic stem-cell transplantation is the only potentially curative treatment for CLL (Dreger 2014), but as it is feasible in only a small minority of individuals, prolonged PFS and OS are the realistic treatment goals for the majority of individuals. Before 2022, the standard of care for younger individuals with CLL without del 17p/TP53 mutation was front-line chemo-immunotherapy with regimens such as FCR (Hallek 2010; Thompson 2016b). BCRi such as ibrutinib are frequently used in older individuals and in those with relapsed disease (Byrd 2013). Intensifying FCR by adding drugs such as mitoxantrone results in additional toxicity without improving therapeutic efficacy (Faderl 2010). Chemo-immunotherapy approaches still do not result in a cure for the majority of individuals. Steady incidences of relapses are observed, and novel drugs have not demonstrated a complete reduction in relapse rates, particularly in refractory or relapsed CLL. This is evident from the need for treatment options after the exhaustion of BCR inhibitors, posing a persistent challenge (Jones 2018; Mato 2017; Thompson 2015). In the follow-up of front-line therapy trials using novel drugs, it remains unclear whether a cure may be attainable in such settings. The current standard of first-line treatment for CLL patients is small molecular inhibitors, which have shown better response rates compared to traditional chemo-immunotherapy. While targeting agents such as ibrutinib, acalabrutinib, zanubrutinib, and venetoclax have shown promising clinical responses in CLL patients, the presence of refractory or resistant disease after induction therapy remains an unsolved issue. In conclusion, with the exception of CLL with better cytogenetic prognosis (mutated IgVH and functional p53) treated with FCR, for which there may be a fraction that could be cured (Thompson 2016b), CLL remains an incurable disease.

The objective of maintenance therapy, which follows induction or salvage therapy, is to prevent recurrence and to strengthen the effect of previous treatment. It is not an integral part of the standard treatment for CLL. Early on, chlorambucil was used as a treatment-until-progression strategy with trials using up to 12 months of chlorambucil treatment (Keller 1986; Montserrat 1985; Vidal 2016). However, the benefit was very limited to achieve complete remissions even with prolonged treatment, and close to 50% of individuals were progressive within one year (i.e. during projected treatment), suggesting a relatively rapid development of resistance (Vidal 2016). Anti-CD52 antibody (alemtuzumab) maintenance was subsequently attempted, and long-term analysis of a phase 3 trial showed significant improvement of survival endpoints (Schweighofer 2009). However, severe drug-related toxicities (especially increased opportunistic infections) rendered these advantages moot (Byrd 2009; Jones 2013; Kaufman 2011; Lin 2010; Wendtner 2004).

As antibody and immunomodulatory drug (IMiD) strategies became available, their potential to counteract these problems was explored. Some small phase 2 trials showed that consolidation/maintenance treatment with lenalidomide, Chang 2016; Shanafelt 2013; Strati 2017, or rituximab, Abrisqueta 2013; Bo 2014; Del Poeta 2008; Foa 2014, is feasible. Following the development of a maintenance paradigm in other indolent lymphoma entities (Hainsworth 2002), a phase 2 trial which observed prolonged use of rituximab in CLL was launched by Hainsworth and colleagues (Hainsworth 2003). Rituximab induction follow-up of four weekly courses of rituximab in six-monthly intervals as maintenance for two years achieved a median PFS of 18.6 months. There are currently five significant completed randomised trials investigating the maintenance role of anti-CD20 antibodies in CLL (Dartigeas 2018; Greil 2016; Robak 2018; Van Oers 2015). Clinical trials of IMiD (lenalidomide) have reported prolongation of PFS (Chanan-Khan 2017; Fink 2017). Fink and colleagues recruited individuals with a high risk of cytogenetic aberrations after four cycles of front-line FCR induction by screening 468 individuals to randomise 89 individuals to either lenalidomide with ramp-up dosing design or placebo. After a median follow-up of 17.9 months, lenalidomide showed a significant benefit for prolongation of PFS (hazard ratio 0.168) (Fink 2017).

In recent years, significant progress has been made in understanding the biology of CLL, leading to the development of novel therapeutic strategies that have greatly enhanced patient outcomes (Yosifov 2019). Specifically, the identification of specific therapeutic targets within intracellular signalling pathways, such as the B-cell receptor (BCR) or BCL-2 (B-cell lymphoma), has transformed CLL treatment. While chemo-immunotherapy-based regimens were the mainstay of treatment for many years, they have been surpassed by target therapy-based regimens and their combinations, which have shown remarkable efficacy. Moreover, the introduction of second-generation anti-CD20 molecules in conjunction with targeted agents has further altered the treatment landscape. Although these advancements have provided benefits to the majority of CLL patients, the elderly and/or high-risk patients have likely benefited the most, experiencing significant improvements in their quality of life and life expectancy (Burger 2020). Nevertheless, the adoption of these new treatment approaches has posed challenges, such as the emergence of drug resistance, adverse side effects, and disease refractory and relapse.

The concepts of maintenance therapy are not only applied to conventional chemo-immunotherapy, but also to current targeted agent-based regimens. The CLL2-BXX phase 2 trials consist of four relevant trials that aimed to explore the effects of subsequent induction and maintenance treatment with an anti-CD20 antibody (obinutuzumab or ofatumumab) and a targeted oral agent (ibrutinib, idelalisib, or venetoclax) (Cramer 2018). The CLL2-BIG trial evaluated obinutuzumab plus ibrutinib as maintenance therapy, reporting a response rate of 27.1%, with six participants (10.2%) achieving a complete remission, or complete response (CR) with incomplete recovery of the bone marrow as their best response (Tresckow 2022). In addition, 42 of 59 (71.1%) participants had undetectable minimal residual disease in peripheral blood at the last staging during maintenance therapy. The CLL2-BIO trial evaluated ofatumumab plus ibrutinib as maintenance therapy for 24 months, showing good efficacy with an overall response rate of 92% and an improved CR/complete response with incomplete haematologic recovery (CRi) rate from 24.6% (end of induction) to 40.2% (post-maintenance therapy) (Cramer 2021).

Novel targeted agents such as ibrutinib, acabrutinib, and idelalisib are also treating CLL with a treatment-until-progression strategy. This treatment design is similar to the concept of maintenance therapy, which aims to deepen the response and prolong the duration of response, as well as prevent refractory and resistance.

Because maintenance therapy might preserve clinical or even molecular remission, omitting maintenance therapy could result in disease relapse, necessitating salvage therapy with high-dose chemotherapy, with or without autologous or allogeneic haematopoietic cell transplantation. Omitting maintenance therapy might ultimately result in a negative effect on the OS of individuals with CLL.

How the intervention might work

Maintenance therapy to improve the quality and duration of treatment response is practised in indolent lymphoma entities (Hainsworth 2002). CLL shares similar characteristics, and recent randomised phase 3 trials described below have reported significant benefits in PFS, but more evidence is needed regarding OS (Beauchemin 2015). Maintenance and consolidation both prolong treatment with the intent of strengthening the response after postremission treatment. Consolidation therapy is combined with induction treatment. Maintenance is a follow-up treatment to induction therapy intended to stabilise and deepen the response. In clinical practice, these two therapeutic interventions are difficult to distinguish in individuals where neither approach has a curative effect regardless of intent. In this review, all postremission interventions are considered 'maintenance therapy'.

Anti-CD52 monoclonal antibodies (anti-CD52 mAbs): alemtuzumab

Alemtuzumab (Campath, MabCampath) is a humanised antibody specific for CD52, a surface protein expressed on B cells in both CLL and normal B and T cells (Wierda 2005). In an early phase 3 study, individuals with CLL in at least partial remission after front-line chemotherapy with either fludarabine or fludarabine plus cyclophosphamide were randomised to either maintenance treatment with alemtuzumab 30 mg three times per week for a maximum of 12 weeks or observation (Wendtner 2004). At median follow-up of 21.4 months, individuals receiving alemtuzumab had significantly longer PFS (24.7 months) than controls, but the study was stopped because of unacceptable infectious toxicity. Follow-up surveillance reported that five individuals given alemtuzumab achieved molecular remission, but all individuals in the control group had evidence of MRD. At median follow-up of 48 months, individuals given alemtuzumab had significantly prolonged PFS compared with the observation group (Schweighofer 2009). Despite its toxicity, maintenance treatment with alemtuzumab induced molecular remission and reduced MRD, resulting in significantly improved long-term clinical outcome.

Immunomodulatory drug (IMiD): lenalidomide

Lenalidomide is a 4-amino-glutamyl analogue of thalidomide that has clinical activity in CLL (Riches 2016). Lenalidomide has a unique mechanism of action, not only targeting cancer cells, but also modulating or interrupting multiple interactions of CLL cells and elements in their microenvironment, Acebes-Huerta 2014; Ramsay 2008; Wu 2008, that promote leukaemia or influence survival (Badoux 2011b; Chanan-Khan 2006; Chen 2011; Ferrajoli 2008). Its multifaceted mechanism of action and diverse effects on CLL cells have been reflected in improved immuno-surveillance outcomes (Itchaki 2017; Kater 2014). Two large randomised controlled trials reported significant prolongation of PFS in individuals with CLL who were given lenalidomide maintenance (Chanan-Khan 2017; Fink 2017). The CLLM1 study (NCT01556776) randomly assigned 89 individuals with CCL 2:1 to receive lenalidomide 5 mg maintenance or placebo. The hazard ratio (HR) for PFS was 0.16 (95% confidence interval (CI) 0.07 to 0.37) with a median follow-up of 17.9 months. Median PFS was not reached (95% CI 32.3 to not evaluable) in the lenalidomide group, and was 13.3 months (95% CI 9.9 to 19.7) in the placebo group. However, study recruitment was stopped early because of poor accrual (Fink 2017). The global CONTINUUM study (NCT00774345) found that maintenance therapy with lenalidomide prolonged PFS without affecting potential subsequent lines of therapy (Chanan-Khan 2017). In the CONTINUUM trial, eligible individuals had achieved complete or partial responses to second-line therapy and were randomly assigned to maintenance therapy with lenalidomide (160 individuals) starting at 2.5 mg/day with escalation to 5 mg or 10 mg as tolerated or to a placebo group (154 individuals). Lenalidomide reduced the risk of progression by more than half compared with placebo (HR 0.40, 95% CI 0.29 to 0.55), but no difference was found in OS during follow-up (Chanan-Khan 2017).

Anti-CD20 monoclonal antibodies (anti-CD20 mAbs): rituximab, ofatumumab, obinutuzumab

The CD20 antigen is present in more than 90% of B-cell lymphoma cells and is neither shed nor internalised after antibody binding (Tedder 1994), which makes it a target for immunotherapy with monoclonal anti-CD20 antibody. Preclinical studies have shown that anti-CD20 activity includes complement-dependent cytotoxicity, antibody-dependent cellular cytotoxicity, and the induction of cell growth arrest and apoptosis (Li 2008; Li 2009; Teeling 2004). However, the increased occurrence of neutropenia and infections in individuals who received anti-CD20 has generated concern (Greil 2016; Hallek 2010; Robak 2010b; Van Oers 2015).

Rituximab

Rituximab is a genetically engineered chimeric murine/human monoclonal IgG1 kappa antibody directed against the CD20 antigen. It has a synergic effect when administered in combination with chemotherapeutic agents used in standard first-line treatment regimens (Nabhan 2014). The immuno-chemotherapeutic combination regimen increases treatment benefits compared with chemotherapy alone (Hallek 2017). Because of its benefit in induction therapy, the benefits of rituximab for maintenance therapy have been tested. Early data from small observational and uncontrolled, early-phase consolidation and maintenance trials suggest that maintenance therapy consisting of prolonged rituximab antibody therapy after use in induction can prolong disease-free survival (Abrisqueta 2013; Bo 2014; Del Poeta 2008; Foa 2014). The French Innovative Leukemia Organization conducted a multicentre phase 3 study that compared rituximab maintenance therapy versus observation to prolong PFS in individuals with CLL who were homogeneously treated with front-line FCR (Dartigeas 2018). After a median follow-up of 47.7 months, median PFS was longer in the rituximab group (59.3 months) than in the observation group (49.0 months; HR 0.55, 95% CI 0.40 to 0.75). Another phase 3 study randomly assigned 263 individuals in at least partial remission after first-line (80%) or second-line (20%) therapy to either two years of maintenance rituximab or to observation (Greil 2016). Rituximab maintenance significantly improved PFS (median 47.0 months versus 35.5 months; HR 0.50, 95% CI 0.33 to 0.75), with no significant difference in OS. The median PFS achieved by FCR is three to seven years (Fischer 2016; Thompson 2016b), and evidence of MRD-related relapse in the peripheral blood emerges very early at a median of 12.0 months after chemo-immunotherapy. However, it was delayed to 31.3 months by rituximab maintenance (Greil 2016).

Ofatumumab

Ofatumumab is a human type I CD20 (IgG1-κ) monoclonal antibody, with potent in vitro complement-dependent cytotoxicity even in rituximab-refractory cells (Barth 2012), a higher antibody-dependent cytotoxicity than rituximab (Rafiq 2013), and in vivo efficacy in rituximab-refractory CCL (Wierda 2010). The PROLONG trial was a phase 3 study of ofatumumab maintenance in individuals with CLL in complete or partial remission after second-line or subsequent therapy (Van Oers 2015). Participants were randomised to two years of either ofatumumab or observation. Ofatumumab maintenance resulted in an improvement in PFS (median 29.4 months versus 15.2 months; HR 0.50, 95% CI 0.38 to 0.66). There was no difference in OS.

Obinutuzumab

Obinutuzumab (formerly GA101) is a new-generation humanised anti-CD20 mAbs with a type II glycoengineered Fc portion selected to increase its affinity for FcγRIIIa receptors on immune effector cells. The increased affinity for neutrophils and macrophages is intended to elicit enhanced antibody-dependent cellular cytotoxicity (Bologna 2011; Dalle 2011; Niederfellner 2011). As the cells targeted by obinutuzumab are B cells rather than other lymphocytes including T cells, obinutuzumab is likely to be significantly less immunosuppressive than alemtuzumab (Cartron 2014). The GALACTIC study (ISRCTN64035629) is an ongoing phase 2/3, multicentre, randomised, open, parallel-group trial in previously treated individuals with CLL (Oughton 2017). Participants were randomised to either obinutuzumab maintenance or observation. The phase 2 study will assess safety and short-term efficacy to screen responses eligible for continuing to the phase 3 study. Planned enrolment was 188 participants at 40 study centres in the UK (Oughton 2017).

Small molecular inhibitors (ibrutinib, acalabrutinib, zanubrutinib, idelalisib, duvelisib, venetoclax)

Bruton tyrosine kinase inhibitors (BTKi): ibrutinib, acalabrutinib, zanubrutinib

Ibrutinib (Imbruvica), a covalent oral inhibitor of *BTK*, was the first BTKi to be approved by the US Food and Drug Administration (FDA) for first-line and refractory or relapsed treatment in individuals with CLL (Brown 2018; Byrd 2014; Roberts 2016). Second-generation BTK inhibitors, including acalabrutinib and zanubrutinib, have been formulated to reduce toxicity and improve tolerability compared to ibrutinib. Acalabrutinib has less potent inhibition of tyrosine-protein kinase compared to ibrutinib and does not inhibit epidermal growth factor receptor (*EGFR*) or IL2 inducible T cell kinase (*ITK*). Zanubrutinib is an orally administered BTK inhibitor with less potent inhibition of ITK and EGFR. All three BTK inhibitors – ibrutinib (Brown 2018; Byrd 2014), acalabrutinib (Ghia 2020; Sharman 2020), and zanubrutinib (Tam 2021) – are currently FDA approved for the treatment of CLL. The common toxicities of BTK inhibitors are generally well tolerated and usually resolve on their own for most individuals with CLL, including those who are frail. The notable adverse events of BTK inhibitors are increased bruising, hypertension, and incidence of atrial fibrillation.

Phosphatidylinositol 3-kinase inhibitors (PI3Ki): idelalisib, duvelisib

Idelalisib (GS-1101 or CAL-101) is an orally bioavailable and selective tyrosine kinase inhibitor of the phosphatidylinositol 3-kinase (PI3K). PI3K isoform is essential for antigen-induced BCR signalling. Idelalisib combined with rituximab has shown benefit in individuals with refractory or relapse CLL as induction or salvage treatment (Furman 2014), and is FDA approved. The common adverse events are colitis, pneumonitis, and liver function impairment. Further to their role as induction or salvage treatment, BCR inhibitors were also tested in subsequent maintenance therapy. However, most ibrutinib and idelalisib trials have used a treat-until-progression approach that makes distinguishing induction and maintenance phases difficult. Two trials have combined ibrutinib or idelalisib with bendamustin rituximab chemo-immunotherapy regimens in pretreated individuals (Chanan-Khan 2016; Zelenetz 2017), and recently the German CLL study group proposed a “sequential triple-T concept” of targeted, tailored treatment aiming for total eradication of CLL (Cramer 2018; Hallek 2013b). In that approach, individuals with CLL would receive two cycles of bendamustine followed by induction and maintenance treatment with CD20-antibody plus a BCR inhibitor. The CLL2-BIG trial (NCT02345863), evaluating ibrutinib and GA101 obinutuzumab; the CLL2-BCG trial (NCT02445131), evaluating idelalisib and obinutuzumab; and the CLL2-BIO trial (NCT02689141), evaluating ibrutinib and ofatumumab, have completed recruitment, and analysis of the primary endpoint outcomes are under way (Cramer 2018; Seymour 2018).

B-cell lymphoma-2 inhibitor (BCL-2i): venetoclax

BCL-2, encoded in humans by the BCL2 apoptosis regulator (*BCL2*) gene, is the founding member of the Bcl-2 family of apoptosis regulator proteins, by either inducing (pro-apoptotic) or inhibiting (anti-apoptotic) cell death (Tsujimoto 1984). Dysregulation of BCL-2 is associated with the development of many haematological malignancies and with treatment resistance, and these regulatory pathways can now be therapeutically targeted. The BCL2 inhibitor venetoclax was approved by the FDA in 2016 for second-line treatment of individuals with CLL associated with 17p deletion. The recent phase 3 MURANO trial reported the long-term outcome of venetoclax effect in individuals with refractory or relapsed CLL. While venetoclax plus rituximab combination demonstrated superior PFS than conventional bendamustine plus rituximab (HR 0.16, 95% CI 0.12 to 0.23 of three-year PFS), subsequent single maintenance venetoclax (400 mg once daily) also yielded efficacious free of disease progression (one-year PFS: 87.4%) (Seymour 2018).

Why it is important to do this review

Although published trials have supported the use of anti-CD20 monoclonal antibodies and IMiD (lenalidomide) as effective and well-tolerated maintenance therapy options for individuals with

CLL (Coiffier 2008; Hallek 2010; Robak 2010b), over the last decade the evidence has fluctuated, and new data for existing maintenance and novel target agents from recent randomised controlled trials have become available. A Cochrane Review was therefore necessary to provide up-to-date information for daily practice.

Our previous systematic review (Lee 2020), searched on 2019, used a network-meta analysis to compare the relative effectiveness of available maintenance therapies for CLL. The results showed that maintenance therapy has the potential to prolong the duration of response, but can also increase the risk of adverse events for people with CLL. It is important for physicians and patients to engage in shared decision-making to weigh the benefits and harms of maintenance therapy. Cochrane Reviews are a type of systematic review that aims to provide a readable and unbiased summary of the evidence for both physicians and patients. They adhere to strict methodological and reporting standards and use the GRADE approach to systematically assess the certainty of the evidence. Updating this topic with a Cochrane Review can provide physicians, other key decision-makers, and even patients with access to high-quality information on the evidence-based results of maintenance treatments for CLL such as anti-CD52 antibodies, anti-CD20 antibodies, IMiD, and novel small molecular inhibitors, and help them make informed clinical decisions.

OBJECTIVES

To assess the effects and safety of maintenance therapy, including anti-CD20 monoclonal antibody, immunomodulatory drug therapy, anti-CD52 monoclonal antibody, Bruton tyrosine kinase inhibitor, and B-cell lymphoma-2 tyrosine kinase inhibitor, for individuals with chronic lymphocytic leukaemia.

METHODS

Criteria for considering studies for this review

Types of studies

As prespecified in the protocol (Lee 2019), we included randomised controlled trials (RCTs), specifically RCTs that detailed prospective identification of participants. If studies had sufficient information and details outlined in their design, including participant characteristics, interventions, and outcomes, both the abstract and full-text publications were evaluated. We imposed no limitation on length of follow-up conducted in the studies or on language of publication. Any variants of randomised trials (e.g. cluster-randomised and cross-over trials) and non-randomised studies were ineligible. We excluded cluster-randomised trials because of the potential for bias based on how individual participants were identified and recruited within clusters, and cross-over trials because of the potential for bias from carry-over effects from one trial period to subsequent trial periods. Lastly, we excluded non-randomised studies based on the potential of confounders and the likelihood of increased heterogeneity resulting from residual confounding, as well as other biases that occur variedly across such studies.

Types of participants

We included individuals with histologically confirmed B-cell CLL. Included studies must have used the CLL diagnostic criteria specified by the 1989 and 2007 International Workshops on CCL or the World Health Organization (WHO) classification of lymphomas (Binet 1981; Cheson 1996; Hallek 2008; Swerdlow 2016). We included trials that enrolled individuals in at least partial haematological or molecular remission after previous treatment, with randomisation to maintenance performed either before induction or after the achievement of partial remission. Induction therapy, including chemotherapy alone, chemo-immunotherapy, and targeted therapy, was permitted. We excluded people with CLL who had received maintenance therapy without prior induction therapy or those who had undergone stem cell transplantation with curative intent. Trials that included mixed populations of individuals with different haematological malignancies were eligible if data were available separately for the CLL cohort. Trials for which subgroup data for individuals with CLL were not provided in the publication and for which no response was obtained from the original investigators were excluded from quantitative analysis.

Types of interventions

We included randomised trials of anti-CD52 mAb, anti-CD20 mAb, IMiD, BTKi, PI3Ki, or BCL-2i given as maintenance/consolidation treatment to individuals in at least partial remission after previous treatment. We excluded non-specific agents, such as interferon and cyclophosphamide, that were previously considered but are no longer eligible due to being of historical interest only. We also excluded treatment modalities that either could not clearly differentiate between induction and maintenance phases, or where treatment was continued until disease progression. We included studies comparing these interventions with placebo or observation controls, or head-to-head comparisons.

Experimental interventions

We included studies evaluating the following interventions.

- Anti-CD20 monoclonal antibodies
- Immunomodulatory drugs
- Anti-CD52 monoclonal antibodies
- BTK inhibitors
- PI3K inhibitors
- BCL-2 inhibitors

Comparator interventions

- Placebo/observation

Comparisons

We included studies using the following comparisons.

- Anti-CD20 antibodies versus placebo/observation
- Immunomodulatory drugs versus placebo/observation
- Anti-CD52 antibodies versus placebo/observation
- BTK inhibitors versus placebo/observation
- PI3K inhibitors versus placebo/observation
- BCL-2 inhibitors versus placebo/observation
- Head-to-head comparisons of active interventions

Types of outcome measures

Regardless of the reported outcomes of specific studies, we included trials if they met the criteria described above. Additionally, we considered trials regardless of any estimated relative differences in outcomes. We assessed outcome measurements based on the consensus guidelines published by the International Workshop on CLL for the design and conduct of clinical trials for individuals with CLL (Hallek 2018). For binary outcomes, if a study reported outcomes at multiple time points, we used the latest reported time point in the meta-analysis. For time-to-event outcomes, we used data with the longest follow-up if duplicate publications existed.

Primary outcomes

- Overall survival (OS): the time from study enrolment until death from any cause.
- Health-related quality of life (HRQoL): quality of life measures included the EORTC QLG Core Questionnaire (EORTC QLQ-C30) and QLQ-CLL17. We preferred to extract data from the EORTC QLQ-C30 first; if these data were unavailable, we would alternatively extract QLQ-CLL17 (Poll-Franse 2018).
- Grade 3 and 4 adverse events (AEs): defined using the common terminology criteria for adverse events (CTCAE 5.0), or as defined in the trial. Measured at any time after the initiation of treatment and up to 28 days after discontinuation of the study treatment.

Secondary outcomes

- Progression-free survival (PFS): as described in each trial, namely as the time from study entry until objective disease progression or death.
- Treatment-related mortality (TRM): measured at any time after the start of treatment and up to 28 days after discontinuation of the study treatment.
- Treatment discontinuation (TD): defined as treatment discontinuation from any cause at any time after participants were randomised into intervention/comparator groups.
- All adverse events (AEs): measured at any time after the start of treatment and up to 28 days after discontinuation of the study treatment.
- Complete response rate (CRR): measured at least two months after the last treatment.
- Overall response rate (ORR): measured at least two months after the last treatment.
- Minimal residual disease (MRD): the level of MRD negativity that is clinically important is 0.01%, which is lesser than 1 CLL cell per 10,000 leukocytes. Measured at least two months after the last treatment.

Search methods for identification of studies

The Systematic Review Initiative's Information Specialist, working in collaboration with the Cochrane Haematology Group, developed the search strategy. All qualified studies were included in the analysis, as outlined above.

Electronic searches

We searched databases and sources following the recommendations in Chapter 6 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Lefebvre 2011). We used the RCT filter strategy developed by Dickersin 1994. These platforms are listed in the following subsection. Search strategies are provided in the listed Appendices.

Databases of medical literature

- Cochrane Central Register of Controlled Trials (CENTRAL; 2022, Issue 1) in the Cochrane Library (Appendix 1)
- MEDLINE via Ovid (1946 to 30 January 2022) (Appendix 2)
- Embase (1988 to 30 January 2022) (Appendix 3)

Clinical trial registries

- ClinicalTrials.gov (www.clinicaltrials.gov/) (searched 30 January 2022) (Appendix 4)
- World Health Organization International Clinical Trials Registry Platform (WHO ICTRP) (apps.who.int/trialsearch/) (searched 30 January 2022) (Appendix 5)
- EU Clinical Trials Register (www.clinicaltrialsregister.eu) \(searched 30 January 2022) (Appendix 6)

Searching other resources

As prespecified in the protocol (Lee 2019), we searched the following additional sources.

- Manual reference search: review author(s) manually searched the references of all identified trials, relevant review articles, and current treatment guidelines for further literature.
- Personal contacts: review authors contacted the lead investigators of relevant studies.
- Conference proceedings:
    - American Society of Hematology (ASH) (from 2020 to 2022);
    - European Haematology Association (EHA) (from 2020 to 2022);
    - American Society of Clinical Oncology (ASCO) (from 2020 to 2022); and
    - European Society of Medical Oncology (ESMO) (from 2020 to 2022).

We included literature that matched our inclusion criteria whenever we were able to obtain adequate information from either the abstract or personal communication with the corresponding authors of the relevant studies. Additionally, we contacted the corresponding authors of relevant studies to identify any unpublished material, missing data, or information regarding ongoing studies.

The most recent set of searches was conducted on 30 January 2022, from which all records were screened for relevance. Any and all studies meeting the eligibility criteria were fully incorporated into this research and review.

Data collection and analysis

Selection of studies

Two review authors (CHL and YFZ) independently screened the abstracts of retrieved publications and decided whether to review the full text. In case of disagreement, the full text was obtained to aid in further discussion. Any studies that did not meet the inclusion criteria at this stage were considered ineligible and were excluded. We obtained the full-text copies of all publications that potentially met the inclusion criteria. Two review authors (CHL and YFZ) independently read the obtained studies and selected studies for inclusion in the review. In case of disagreement, a third review author (CLH) reviewed the full publication of the disputed study. We included a PRISMA flow chart in the review to illustrate the study selection process (Moher 2009). We used EndNote reference management software to identify and remove duplicate records (EndNote X8.1). We have documented the reasons for excluding studies that may have been reasonably expected to be included in the analysis in the Characteristics of excluded studies section.

Data extraction and management

Two review authors (CHL and YFZ) independently extracted data using a pre-piloted form based on the standardised Cochrane data extraction form. If necessary, we contacted study authors for supplementary information (Higgins 2011). If the review authors were unable to reach a consensus, a third review author (CLH) made the final decision. After agreement on the extracted data, the data were entered into Review Manager Web (RevMan Web 2022).

We extracted the information outlined below from the collated data.

- General information: author; title; source; publication date; country; language; and duplicate publications.
- Quality assessment: sequence generation; allocation concealment; blinding (participants, personnel, and outcome assessors); incomplete outcome data; selective outcome reporting; and other sources of bias.
- Study characteristics: trial design; study aims, setting, and dates; source of participants; inclusion/exclusion criteria; comparability of groups; subgroup analysis; statistical methods; power calculations; compliance with assigned treatment; length of follow-up; time of randomisation.
- Participant characteristics: eligibility and recruitment method; baseline demographics including age; number of participants recruited/allocated/evaluated; disease status after first-line or previous treatment; participants lost to follow-up; Rai/Binet stage.
- Interventions: drugs and dosages; administration route, frequency, and duration of maintenance and follow-up.
- Outcomes: definition and methods of measuring PFS; OS/mortality; complete/partial response rate; grades 3 to 4 adverse events requiring treatment discontinuation; withdrawal rates, including the number of individuals excluded from outcome assessment after randomisation and the reasons for their exclusion. If possible, we extracted data at the study-arm level, rather than the summary effects.
- Others: study sponsorship/funding; stated conflicts of interest of the investigators.

We extracted relevant outcome data on an intention-to-treat basis, as required, to calculate summary statistics and measures of variance. For dichotomous outcomes, we attempted to obtain the number of events and the population totals to create a 2 x 2 table. For continuous outcomes, means and standard deviations, or other data needed to calculate those estimates, were obtained. For count outcomes, we extracted the number of events, the number of the total population, and the median follow-up. For time-to-event outcomes, we extracted hazard ratios (HRs) and 95% confidence intervals (CIs).

Potential effect modifiers

We extracted the following potential effect modifiers from each study.

- Population characteristics (age)
- Timing of maintenance therapy (front-line or above second-line)
- Intervention (type of drug)
- Comparison (placebo or observation)

Duplicate and companion publications

All articles were mapped to unique trials, with all relevant data juxtaposed for identifying duplicate publications, companion records, or multiple published reports of an identical trial.

Assessment of risk of bias in included studies

We assessed risk of bias in the included studies using the Cochrane RoB 1 tool, according to Chapter 8 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011). Based on this evaluation, we compiled a risk of bias table for each included study using Review Manager Web (RevMan Web 2022). Two review authors (CHL and YFZ) independently assessed risk of bias in each included study, with any disagreements addressed by group consensus or by consulting a third review author (CL) for the final decision. Risk of bias criteria are outlined below (Higgins 2011).

- Random sequence generation (selection bias): was the random sequence appropriately generated?
- Allocation concealment (selection bias): was allocation adequately concealed?
- Blinding of participants and personnel (performance bias): where possible, were the study participants and personnel adequately blinded?
- Blinding of outcome assessment (detection bias): was blinding of the outcome assessors effective in preventing systematic differences such that the outcomes could be determined?
- Incomplete outcome data (attrition bias): did any missing data (drop-out during the study or exclusions from the analysis) introduce bias?
- Selective reporting (reporting bias): were reports of the study free of selective outcome reporting?
- Other sources of bias: was the study apparently free of other problems that could put it at risk of bias?

We judged the risk of bias domains for each included study as 'low risk', 'high risk', or 'unclear risk' (defined below), as described in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011).

- Low risk of bias: if the criterion is adequately fulfilled in the study, the study is at a low risk of bias for the given domain.
- High risk of bias: if the criterion is not fulfilled in the study, the study is at a high risk of bias for the given domain.
- Unclear risk of bias: if the study report provides insufficient information to permit a clear judgement of low risk or high risk, or if the risk of bias is unknown for the given domain.

We evaluated the risk of performance bias (blinding of participants and personnel) and detection bias (blinding of outcome assessment) separately for each outcome. We grouped outcomes by subjective or objective measurement when reporting the findings in the risk of bias tables.

Objective measurements (endpoints are not considered to be influenced by blinding)

- Overall survival
- Treatment-related mortality
- Treatment discontinuation
- Minimal residual disease

Subjective measurements (endpoints are considered to be potentially influenced by blinding)

- Health-related quality of life
- Grade 3/4 adverse events
- Progression-free survival
- All adverse events
- Complete response rate
- Overall response rate

Measures of treatment effect

We evaluated the intention-to-treat results, and calculated risk ratios with 95% confidence intervals (CIs) for dichotomous data. We used hazard ratios (HRs) with 95% CIs for the analysis of time-to-event outcomes. If HRs were not available, we made every effort to estimate the HR as accurately as possible using the available data and a purpose-built method based on the Parmar and Tierney approach (Parmar 1998; Tierney 2007). If data were only available that could be extracted from figures, we used GetData Graph Digitizer software (GetData Graph Digitizer). We calculated mean differences (MD) with 95% CIs for continuous outcomes. If different instruments were used to assess continuous treatment outcomes, we estimated standardised mean differences (SMD) with 95% CIs. For count data, we calculated rate ratios with 95% CIs. In the case of duplicate publications, we used data with the longest follow-up.

Unit of analysis issues

The individual participant was the unit of analysis. All cross-over and cluster-randomised trials were excluded.

Dealing with missing data

We accounted for missing data as recommended in Chapter 16 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011). We analysed data by the effect of assignment to intervention (the intention-to-treat effect). We contacted authors of the included trials to request relevant missing data and additional information not reported in the trial publications.

If the number of trial participants assessed for a given outcome was not reported, we used the number of participants randomised per treatment arm as the denominator. We calculated numerators using percentages if only percentages but no absolute number of events were reported for binary outcomes. If estimates for mean and standard deviations were missing, we calculated these statistics from reported data, using the approaches described in Chapter 6 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2019), and imputed as previously described (Furukawa 2006). If results were reported graphically but not numerically, we estimated the missing data from the figures. If not all participants (i.e. randomised participants) were included in the evaluated data, we assessed the missing information on a case-by-case basis to determine whether the loss was random or non-random or could not be categorised. Apart from the data acquisition methods mentioned above, we did not input any missing data.

Assessment of heterogeneity

We estimated heterogeneity using the Chi^2 test and the I^2 statistic among the trials in each analysis. We used P = 0.10 for the Chi^2 test for statistical significance. We interpreted the I^2 statistic as described below, in accordance with Section 9.5.2 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Deeks 2011):

- 0% to 40%: may not be important;
- 30% to 60%: indicates moderate heterogeneity;
- 50% to 90%: indicates substantial heterogeneity;
- 75% to 100%: indicates considerable heterogeneity.

If we observed substantial heterogeneity ($I^2 > 50\%$), we reported it and explored all possible causes through subgroup analyses (Deeks 2011). The Tau^2 estimate of between-study variance was calculated in a random-effects meta-analysis. If excessive heterogeneity was not explained by subgroup analysis, the outcome results were not pooled in the meta-analysis, and a narrative description of the study results was presented instead.

Assessment of reporting biases

Potential reporting biases were reported graphically through funnel plots, which were statistically tested by conducting a linear regression test for all outcomes if at least 10 trials were included. Funnel plots of treatment effect were compared with trial precision to illustrate asymmetry that indicated the potential for selection bias (e.g. the selective publication of trials with positive findings; Egger 1997). Funnel plot asymmetry was tested using a linear regression analysis (Egger 1997). We considered P < 0.1 as significant for this test (Sterne 2011).

Data synthesis

We conducted all data analysis as recommended in Chapter 9 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Deeks 2011). Data were entered using the Cochrane Review Manager 5 statistical package (RevMan Web 2022), and summarised using both fixed-effect and random-effects models. We used random-effects models in the primary analyses and the fixed-effect model for sensitivity analyses. We interpreted random-effects meta-analyses with due consideration of the complete distribution of effects. In addition, we performed statistical analyses based on the statistical guidelines in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011).

We calculated dichotomous outcomes using the Mantel–Haenszel method. We used the inverse variance method to measure continuous outcomes. We used the generic inverse variance method to measure count and time-to-event outcomes.

Multi-arm studies were eligible for inclusion in the review. Our approach to performing data synthesis of such studies was either to omit groups irrelevant to the comparison or to combine multiple groups considered eligible as the experimental and/or comparator intervention. This was performed to create a single pair-wise comparison. Only one multi-arm trial was included in the review (Byrd 2018).

Subgroup analysis and investigation of heterogeneity

We planned to explore potential causes of heterogeneity by performing subgroup analyses. This was done by stratifying the outcomes to assess heterogeneity for the following parameters.

- Type of drug
- Age of participants (adults < 70 versus adults ≥ 70 years of age)
- When maintenance therapy was given (first-line or treatment of previously treated individuals)

We assessed differences between subgroups using the Chi^2 test (Deeks 2011).

If the number of studies was low, we did not perform subgroup analyses. Instead, we presented the results of stratified analyses for different anti-CD20 drugs without testing for differences between subgroups.

Sensitivity analysis

We planned to test the robustness of the results by conducting a fixed-effect pair-wise meta-analysis. Additionally, we planned a sensitivity analysis to compare studies at high risk of bias for random sequence generation, allocation concealment, and blinding with those at low risk of bias (Higgins 2011).

We did not perform a sensitivity analysis comparing anti-CD20 mAbs at high risk of bias for random sequence generation, allocation concealment, and blinding to those at low risk of bias for two reasons: (i) there was no high risk of bias in the domains of random sequence generation and allocation concealment; and (ii) blinding for objective outcomes was all low risk of bias, while blinding for subjective outcomes was all high risk of bias.

Regarding the IMiD comparison, we were also unable to conduct a sensitivity analysis as there was no high risk of bias in the domains of random sequence generation and allocation concealment.

Regarding anti-CD52 mAb, we did not perform sensitivity analysis as only one trial was included.

Summary of findings and assessment of the certainty of the evidence

Certainty of the evidence

We used GRADEpro GDT software to assess the certainty of the body of evidence for each outcome (GRADEpro GDT), as described in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011; Schünemann 2017). The GRADE approach considers problems associated with not only internal validity (risk of bias, inconsistency, imprecision, and publication bias) but also those associated with external validity (directness of results). Three review authors (CHL, YFZ, and CL) independently rated the certainty of the evidence for each outcome, with any disagreements resolved by discussion or by a senior review author (CLH).

We downgraded evidence rated as high certainty by one level for serious, or by two levels for extremely serious concerns for each limitation. We explained our decisions to downgrade the certainty of the evidence in footnotes, and added comments to aid the reader's understanding of the review when necessary.

The levels of certainty of the evidence are as follows.

- High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
- Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
- Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
- Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Tabulated summary of findings

We used the GRADE approach to tabulate research findings and judgement summaries, as described by the GRADE Working Group. For each comparison, a summary of the evidence was demonstrated for the main outcomes in a summary of findings table, as follows.

- Overall survival
- Health-related quality of life
- Grade 3 and 4 adverse events
- Progression-free survival
- Treatment-related mortality
- Treatment discontinuation
- All adverse events

This table provides key information related to the best estimate of the magnitude of the effect in relative terms. It also includes the absolute differences for each relevant comparison of alternative management strategies, numbers of participants and studies addressing each important outcome, and the rating of the overall confidence in effect estimates for each outcome (Guyatt 2011; Schünemann 2017).

RESULTS

Description of studies

For details, see Characteristics of included studies, Characteristics of excluded studies, Characteristics of studies awaiting classification, and Characteristics of ongoing studies.

Results of the search

A study flow diagram is presented in Figure 1.

Figure 1. PRISMA study flow diagram.

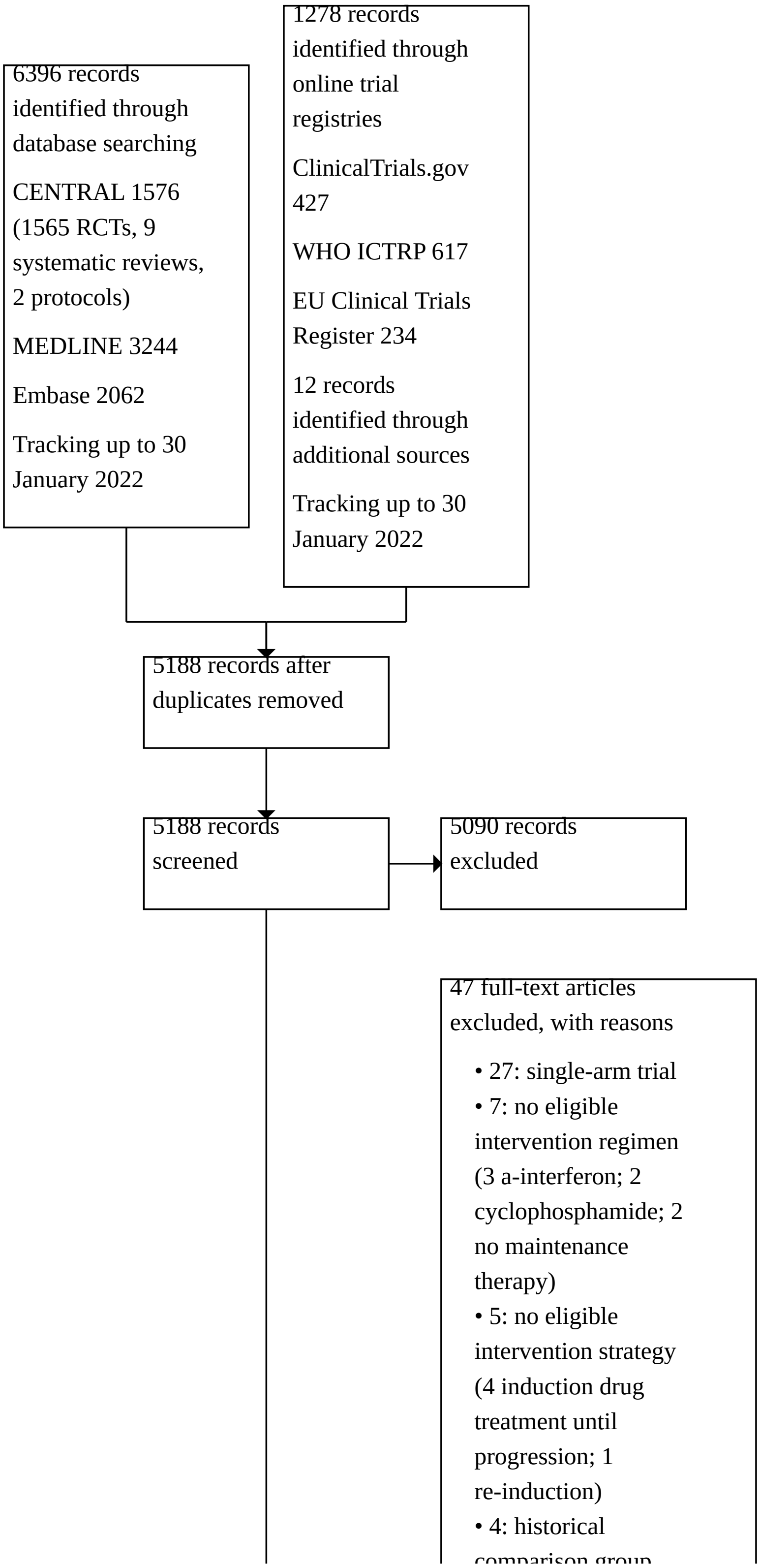
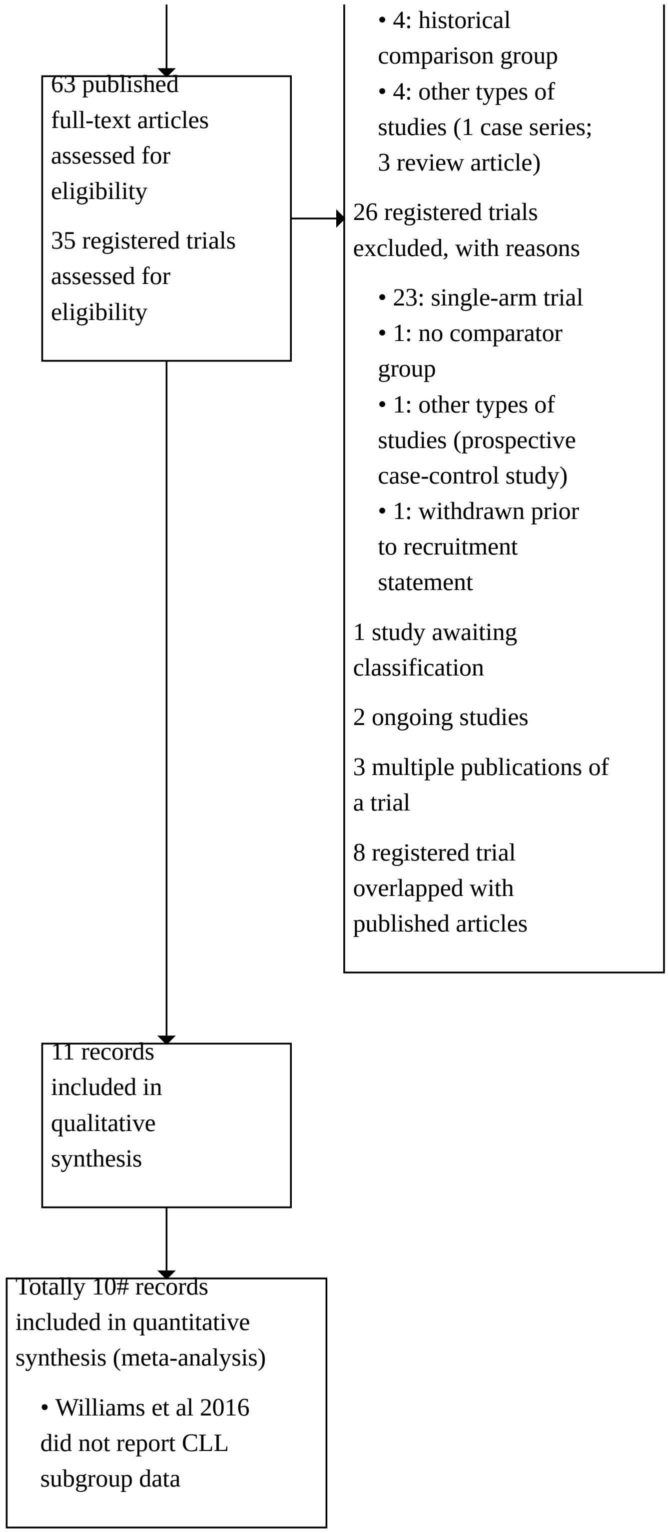


Figure 1. (Continued)



As of 30 January 2022, we conducted a literature search of the electronic databases and trial registries and identified a total of 6396 references. After collating these references in a library (EndNote X8.1), and removing 2252 duplicates, we screened the remaining 5188 articles based on title and abstract and excluded 5090 articles that did not meet our inclusion criteria.

We retrieved 63 published articles and 35 registered trials for full-text screening, of which 11 records met our inclusion criteria for qualified synthesis. One report (conference abstracts) is currently awaiting classification (Jaksic 1988). We excluded 47 articles for the following reasons:

- single-arm trials (27 articles);
- no eligible intervention regimen (7 articles);
- no eligible intervention strategy (5 articles);
- non-randomised historical comparison group (4 articles);
- other types of studies (4 articles).

We also excluded 26 registered trials, for the following reasons: 23 were single-arm trials; one was a prospective case-control study; one had no comparator group; and one was withdrawn prior to recruitment. Notably, of the registered trials, eight were included in our qualified review (all of which overlapped with published articles), and two prospectively registered trials are still ongoing (NCT03280160; Oughton 2017).

We also identified 12 additional records through manual searches and our searches of conference proceedings (ASCO, ASA, ESMO, and EHA). Most of these records involved individuals with cancer or haematological malignancies, which were not identified by our search strategy due to the lack of CLL indexing. None of these records were included after full-text screening. We contacted the corresponding or first authors, or both, of the record awaiting classification, as well as the two ongoing trials, because we were unsure if full publications existed for these studies (see Characteristics of studies awaiting classification; Characteristics of ongoing studies). We finally included 11 records in our qualified review.

Included studies

For details of each study, see Characteristics of included studies.

We included 11 RCTs involving a total of 2393 participants (range 21 to 480 per trial) in the review (Byrd 2018; Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Foa 2014; Greil 2016; Osterborg 2016; Robak 2018; Schweighofer 2009; Van Oers 2019; Williams 2016). Data from one included trial (52 participants) were not analysed in the meta-analysis because they did not report the prespecified outcomes for the CLL group (E4402 trial Williams 2016 did not report the results for the CLL subgroup).

Trial design and setting

The included trials were grouped by types of intervention. Seven trials compared anti-CD20 mAb with observation. Among them, five trials compared rituximab maintenance therapy with observation (Dartigeas 2018; Foa 2014; Greil 2016; Robak 2018; Williams 2016), and two trials compared ofatumumab maintenance therapy with observation (Osterborg 2016; Van Oers 2019). Three trials compared lenalidomide maintenance therapy with placebo/observation (Byrd 2018; Chanan-Khan 2017; Fink 2017), and one trial compared alemtuzumab maintenance therapy with observation (Schweighofer 2009).

One study reported no periods for trial recruitment (Schweighofer 2009). Recruitment for the earliest trial began in November 2003, Williams 2016, and lasted until March 2016, Fink 2017. All included studies were published as full-text publications.

Ten of the included trials were two-armed RCTs, while the remaining trial was a four-armed RCT (Byrd 2018). In order to include this last trial in the review, we divided it into two subgroups and analysed it correspondingly. One subgroup included participants with induction chemotherapy of fludarabine plus rituximab followed by with/without lenalidomide maintenance therapy. The other subgroup evaluated participants with induction chemotherapy of fludarabine, cyclophosphamide, and rituximab (FCR) followed by with or without lenalidomide maintenance therapy.

Nine studies were phase 3 trials (Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Greil 2016; Osterborg 2016; Robak 2018; Schweighofer 2009; Van Oers 2019; Williams 2016), and two were phase 2 trials (Byrd 2018; Foa 2014).

Six multicentre trials were conducted across different countries (Chanan-Khan 2017; Fink 2017; Greil 2016; Osterborg 2016; Robak 2018; Schweighofer 2009; Van Oers 2019). Two trials were conducted in Italy with a multicentre design (Byrd 2018; Foa 2014). One multicentre trial was conducted in the USA (Williams 2016), and another multicentre trial was conducted in France (Dartigeas 2018).

The median duration of follow-up in five trials that administered rituximab maintenance therapy ranged from 14.4 months, Robak 2018, to 73 months, Byrd 2018. The median duration of follow-up in two trials that administered ofatumumab maintenance therapy was 40.9 months, Van Oers 2019, and unavailable, Osterborg 2016, respectively. In three trials of lenalidomide as maintenance therapy, the median duration of follow-up ranged from 17.9 months, Fink 2017, to 31.5 months, Chanan-Khan 2017. The median duration of follow-up in one trial of alemtuzumab maintenance therapy was 12.4 months (Schweighofer 2009).

Study participants

A total of 2393 adult patients with CLL post-previous therapy were randomly allocated to a maintenance therapy group or an observation/placebo group. Participants of the male gender accounted for 60.5% and 58.5%, and median age ranged from 57.8 to 71.7 years old and from 54.1 to 71.1 years old in the maintenance group and control group, respectively. The type of previous induction and disease severity varied between studies. In rituximab maintenance trials, one trial used rituximab for induction (Williams 2016); one trial used rituximab plus cladribine plus cyclophosphamide (RCC) (Robak 2018); one trial used rituximab plus chlorambucil (RC) (Foa 2014); and two trials used fludarabine plus cyclophosphamide plus rituximab (FCR) (Dartigeas 2018; Greil 2016).

In the ofatumumab maintenance trials, one trial used ofatumumab for induction (Osterborg 2016), and another used either FCR or rituximab and bendamustine (RB) (Van Oers 2019). All three lenalidomide maintenance trials used FCR or RB for induction therapy (Byrd 2018; Chanan-Khan 2017; Fink 2017). The induction chemotherapy of alemtuzumab utilised either fludarabine or fludarabine/cyclophosphamide (Schweighofer 2009).

There were variations in baseline disease severity. Eight trials reported baseline Binet or Rai stage status (Byrd 2018; Dartigeas 2018; Foa 2014; Greil 2016; Osterborg 2016; Robak 2018; Schweighofer 2009; Van Oers 2019), whereas the remaining three trials did not (Williams 2016 included non-Hodgkin's lymphoma reported ISS (international staging system) stage; Chanan-Khan 2017 and Fink 2017 did not report baseline stage). Among the eight trials that reported baseline stage, five had a higher proportion of participants in the early stage with Rai stage I/II or Binet stage A with symptoms or high cytogenetic aberrations (Byrd 2018; Dartigeas 2018; Greil 2016; Robak 2018; Van Oers 2019). The other three trials had a higher proportion of participants in Rai stage III/IV or Binet stage B/C (Foa 2014; Osterborg 2016; Schweighofer 2009). Regarding cytogenic evaluation, only three trials did not present these data (Chanan-Khan 2017; Robak 2018; Williams 2016); six trials showed the presence of cytogenic aberrations more frequently, ranging from 69.7% to 80.1% (Byrd 2018; Fink 2017; Foa 2014; Greil 2016; Osterborg 2016; Schweighofer 2009), whereas the remaining two trials were opposites (71.9% and 67.7% without cytogenic abnormalities in Dartigeas 2018 and Van Oers 2019, respectively). Most trials performed the analysis for immunoglobulin heavy-chain variable region gene (IGHV) mutation status, with unmutated IGHV rate of approximately 41.4% to 91% (Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Foa 2014; Greil 2016; Osterborg 2016; Schweighofer 2009; Van Oers 2019), with the exception of three trials for which baseline IGHV status was not reported (Byrd 2018; Robak 2018; Williams 2016).

Study interventions

In seven trials, participants received intravenous anti-CD20 mAbs as maintenance therapy (n = 847) versus observation (n = 832). In five trials, participants received intravenous rituximab as maintenance therapy (n = 583) versus observation (n = 579): four trials administered 375 mg/m^2 every 12 weeks (Foa 2014; Greil 2016; Robak 2018; Williams 2016), and one trial administered 500 mg/m^2 every eight weeks for up to two years or until disease progression (Dartigeas 2018). In two trials, participants received intravenous ofatumumab as maintenance therapy (n = 264) for up to two years with a target dosage of 1000 mg, Van Oers 2019, or 2000 mg versus observation (n = 253), Osterborg 2016.

In another three trials, participants received oral IMiD (lenalidomide) as maintenance therapy (n = 360) with a targeted dosage of 5 mg, Byrd 2018; Fink 2017, or 15 mg, Chanan-Khan 2017, daily until progression versus placebo/observation (n = 333). In another trial, participants received 30 mg intravenous anti-CD52 mAb (alemtuzumab, n = 11) three times per week for up to 12 weeks versus observation (n = 10) (Schweighofer 2009).

Anti-CD20 mAb – rituximab maintenance trials

Foa 2014: participants with previously untreated CLL (Binet stage A or B with active disease or stage C) who received induction therapy of RC and achieved complete response, complete response with incomplete bone marrow recovery, and partial response were randomised to maintenance rituximab 375 mg/m^2 intravenously once every 8 weeks for 12 doses or to observation.

Greil 2016 and Egle 2019: participants who achieved complete response, complete response with incomplete bone marrow recovery, or partial response following previous first- or second-line rituximab-containing chemo-immunotherapy were randomised to either rituximab or observation. Participants in the maintenance group were offered intravenous rituximab 375 mg/m^2 with a three-month interval up to two years or until disease progression.

Williams 2016: participants who achieved a partial or complete response after induction therapy were randomised to maintenance rituximab 375 mg/m^2 intravenously once every three months until treatment failure or observation group rechallenged rituximab until disease progression.

Greil 2016: participants with CLL who achieved clinical response following previous first- or second-line rituximab-containing chemo-immunotherapy were randomly assigned to maintenance rituximab 375 mg/m^2 intravenously with a three-month interval and up to two years or until disease progression or to observation.

Robak 2018: participants with CLL with an immunologically confirmed diagnosis (CD5-, CD19-, CD23-, and CD20-positive clone), at stages I to IV according to the Rai classification, achieved a partial response or complete response after six cycles. RCC induction was randomly assigned to either rituximab 375 mg/m^2 intravenously, up to eight cycles every 12 weeks, or observation.

Anti-CD20 mAb – ofatumumab maintenance trials

Osterborg 2016: participants with bulky fludarabine-refractory CLL who achieved at least stable disease or better after ofatumumab induction therapy underwent randomisation (2:1) to either intravenous ofatumumab fixed 2000 mg once every 4 weeks for up to an additional 24 weeks or no further treatment.

Van Oers 2019: participants who achieved either partial response or complete response after second- or third-line therapy were randomised to either intravenous ofatumumab 1000 mg once every eight weeks for up to two years or to observation.

IMiD – lenalidomide maintenance trials

Fink 2017: participants with CLL who responded to chemo-immunotherapy two to five months after completion of first-line therapy and who were assessed to be at high risk for early progression, defined by minimal residual disease (MRD) levels of 10^{-2} or higher or MRD levels of 10^{-4} to < 10^{-2} after completion of first-line treatment combined with either an unmutated IGHV status of del(17p) or *TP53* mutation at baseline, were randomly assigned to either oral lenalidomide or placebo. Lenalidomide was initiated at 5 mg daily in the first 28 days' cycle; if well-tolerated, the dose was increased to 10 mg daily in each 28-day cycle in cycles 2 to 6. A target dose of 15 mg daily was given starting with the seventh cycle up to progression of the disease. Further increase in dosage (to 20 mg starting with the 13th cycle and to 25 mg starting with the 19th cycle) was guided by MRD assessments and allowed in participants with MRD levels of 10^{-4} or higher in peripheral blood who could tolerate previous dose levels. The maximal daily dose of lenalidomide was 25 mg.

Chanan-Khan 2017: participants who received bendamustine, antiCD20 antibody, chlorambucil, or alemtuzumab as first- or second-line treatment and achieved at least achieved a partial response were randomised to lenalidomide maintenance or placebo. Lenalidomide was started at 2 or 5 mg, once per day during days 1 to 28. If lenalidomide was well-tolerated, an increase in dose to 5 mg/day was permitted cycle 2 onward. If the participant completed five continuous cycles of 5 mg/day lenalidomide, which was tolerated well and an MRD-negative complete response was not achieved, the dose could be increased to 10 mg/day. Treatment continued until disease progression or unacceptable toxic effects.

Byrd 2018: participants with naive CLL who had an asymptomatic intermediate (Rai I or II) or high-risk (Rai III or IV) stage CLL and a performance status of 2 or more and achieved clinical response were blindly randomised to receive either oral lenalidomide or observation. Lenalidomide was started at 5 mg initially for days 1 to 21 and increased to 10 mg if tolerated for six months.

Anti-CD52 mAb – alemtuzumab maintenance trials

Schweighofer 2009: participants who responded to induction treatment with either fludarabine or fludarabine plus cyclophosphamide were randomised to either 30 mg intravenous alemtuzumab three times per week for up to 12 weeks or to observation. The starting dose of alemtuzumab was 3 mg on day 1; if well tolerated, it was increased to 10 mg on day 2 and the target dose of 30 mg on day 3. The 30 mg dose was subsequently given three times a week for a maximum of 12 weeks.

Study outcomes

Primary outcome measure

Our primary outcomes were overall survival (OS), health-related quality of life (HRQoL), and grade 3/4 adverse events (AEs). A total of 10 trials were included in the meta-analyses for these outcomes.

Six trials reported OS, which was presented as median hazard ratios (HRs) and 95% confidence intervals (CIs). Three trials compared anti-CD20 mAbs maintenance with observation (two with rituximab (Dartigeas 2018; Greil 2016), and one with ofatumumab (Van Oers 2019)), and three trials compared lenalidomide maintenance versus placebo/observation (Byrd 2018; Chanan-Khan 2017; Fink 2017). OS data for Byrd 2018 were extracted from survival curves.

Only one trial reported HRQoL, which compared ofatumumab with observation presented as mean difference (MD) and standard deviation (SD) (Van Oers 2019).

Seven trials reported grade 3 and 4 AEs, which were calculated from tables: five trials compared anti-CD20 mAbs maintenance with observation (four with rituximab (Dartigeas 2018; Foa 2014; Greil 2016; Robak 2018), and one with ofatumumab (Van Oers 2019)), and two trials compared lenalidomide maintenance versus placebo (Chanan-Khan 2017; Fink 2017).

Secondary outcome measures

Nine trials reported progression-free survival (PFS), with a total of five trials comparing anti-CD20 mAbs maintenance with observation (three trials with rituximab (Dartigeas 2018; Greil 2016; Robak 2018), and two trials with ofatumumab (Osterborg 2016;Van Oers 2019)); two trials comparing lenalidomide with placebo (Chanan-Khan 2017; Fink 2017), and one trial comparing lenalidomide with observation (Byrd 2018); and one trial comparing alemtuzumab with observation (Schweighofer 2009). Foa 2014 only reported a three-year PFS rate. We extracted PFS data for Schweighofer 2009, Osterborg 2016, Byrd 2018, and Robak 2018 from survival curves due to the lack of actual values.

Seven trials reported treatment-related mortality (TRM) outcomes: four anti-CD20 mAbs trials (two with rituximab (Dartigeas 2018; Greil 2016), and two with ofatumumab (Osterborg 2016;Van Oers 2019)), and three lenalidomide maintenance trials (Byrd 2018; Chanan-Khan 2017; Fink 2017).

Eight trials reported treatment discontinuation (TD): six anti-CD20 mAbs trials (four with rituximab maintenance (Dartigeas 2018; Foa 2014; Greil 2016; Robak 2018), and two with ofatumumab maintenance (Osterborg 2016; Van Oers 2019)), and two lenalidomide maintenance trials (Chanan-Khan 2017; Fink 2017).

Three trials reported complete response rate (CRR) and overall response rate (ORR): two rituximab maintenance trials (Foa 2014; Robak 2018), and one alemtuzumab maintenance trial (Schweighofer 2009).

Nine trials reported all adverse events: six anti-CD20 mAbs trials (four with rituximab maintenance (Dartigeas 2018; Foa 2014; Greil 2016; Robak 2018), and two with ofatumumab maintenance (Osterborg 2016; Van Oers 2019)), and three lenalidomide trials (Byrd 2018; Chanan-Khan 2017; Fink 2017).

No trials reported the outcome of MRD.

Study funding

All trials reported their sources of funding. Seven trials received commercial sponsorship from the pharmaceutical industry; one trial received funding from Sanofi Genzyme (Schweighofer 2009); four trials from Roche (Dartigeas 2018; Foa 2014; Greil 2016; Robak 2018); three trials from Novartis AG, Inc (Osterborg 2016; Schweighofer 2009; Van Oers 2019); and three trials from GlaxoSmithKline and Genmab A/S, Inc. (Osterborg 2016; Schweighofer 2009; Van Oers 2019). Seven trials were supported by the national government or academic research funding (Byrd 2018; Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Greil 2016; Robak 2018; Williams 2016).

Ongoing studies

We identified two ongoing trials (for details, see Characteristics of ongoing studies) (NCT03280160; Oughton 2017).

The planned enrolment for these two trials included a total of 272 participants. We will be monitoring their progress, and when published, they will be included in future updates of this review if deemed eligible.

Excluded studies

For details, see Characteristics of excluded studies.

We excluded a total of 73 studies merged from 47 publications and 26 registered trials. We excluded publications for the following reasons: single-arm trial (Abrisqueta 2013; Chang 2016; Egle 2018; Flinn 2016; Hainsworth 2003; Kaufman 2011; Keller 1986; Kempin 2019; Lamanna 2009; Lampson 2019; Lin 2010; Mato 2015; Montillo 2002; Montillo 2006; Noy 2001; O'Brien 2003; Osterborg 2015; Pleyer 2020; Rai 2002; Rosenthal 2017; Schey 1999; Sehn 2012; Shadman 2016; Srock 2007; Strati 2017; von Tresckow 2019; Weiss 2000); no eligible intervention regimen (Khouri 2017;Mauro 2003; O'Brien 1995; Pan 2003;Walewski 2020; Zinzani 1994; Zinzani 1997); no eligible intervention strategy (Kater 2019; Moreno 2019; Shanafelt 2019; Sharman 2019; Sharman 2020); non-randomised historical comparison group (Huang 2014; Shanafelt 2013; Strati 2016; Thieblemont 2004), or other types of studies (Scaramucci 2004 with a case series study design; Moccia 2008, O'Brien 2011, and Ruppert 2019 with a review article study design).

We excluded registered trials for the following reasons: single-arm trial (EUCTR2005-001569-33-ES; EUCTR2008-001823-71-IT; EUCTR2010-018519-14-IT; EUCTR2014-000569-35-DE; EUCTR2014-000580-40-DE; EUCTR2014-000582-47-DE; EUCTR2014-000590-39-DE; EUCTR2015-004985-27-NL; NCT00336206; NCT00587847; NCT00860457; NCT00974233; NCT01258933; NCT01392079; NCT01403246; NCT01465230; NCT01496976; NCT01521689; NCT01600053; NCT01754857; NCT01754870; NCT01809847; NCT02013817); prospective case-control design (NCT03847727); no comparator group (NCT00771602; only included one participant); withdrawn prior to recruitment (ISRCTN63375144; with alemtuzumab licence withdrawn by Genyme Sanofi on 8 August 2012).

Risk of bias in included studies

We assessed risk of bias in the included studies (for details, see Included studies) using Cochrane's RoB 1 tool, given that all 11 studies were RCTs. Overall, risk of bias was unclear; detailed assessments for each study are described in Characteristics of included studies and summarised in Figure 2; Figure 3, according to the methods presented in Assessment of risk of bias in included studies.

Figure 2. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

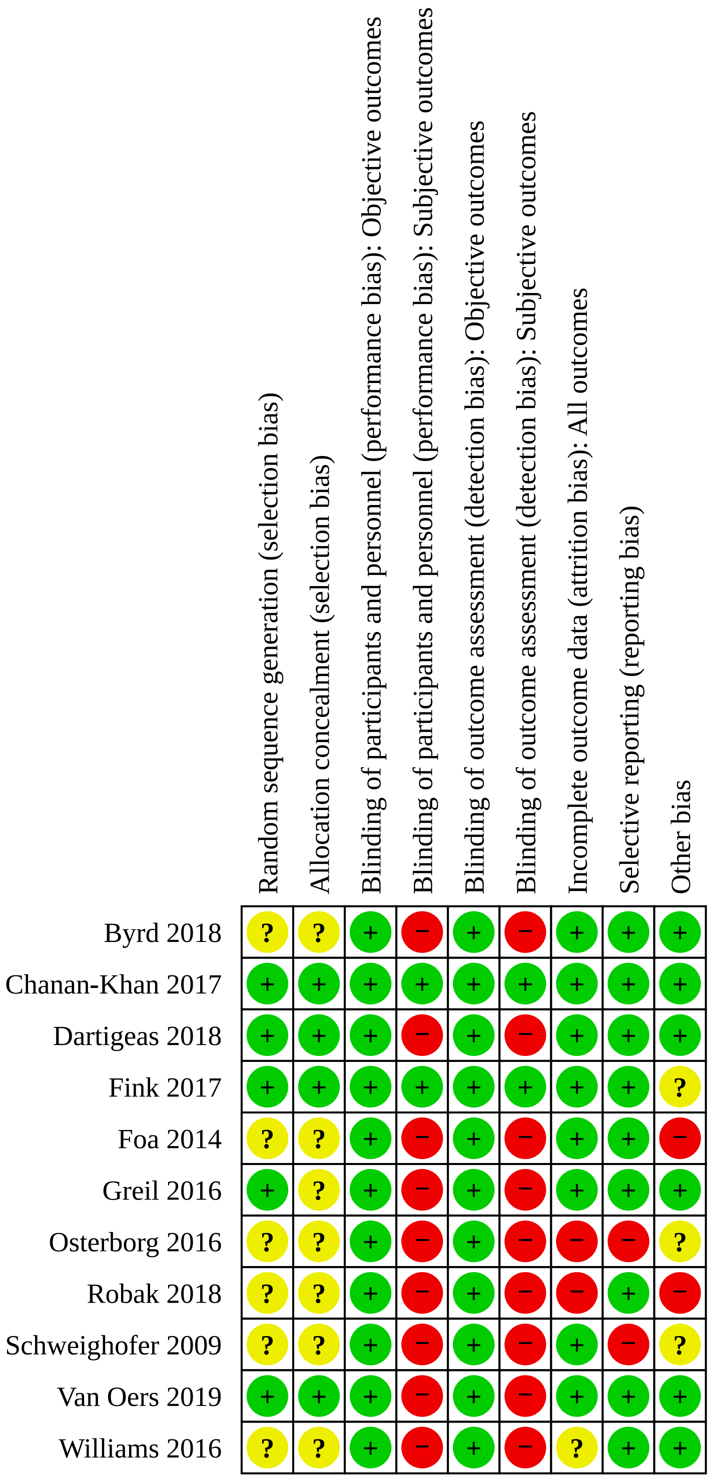
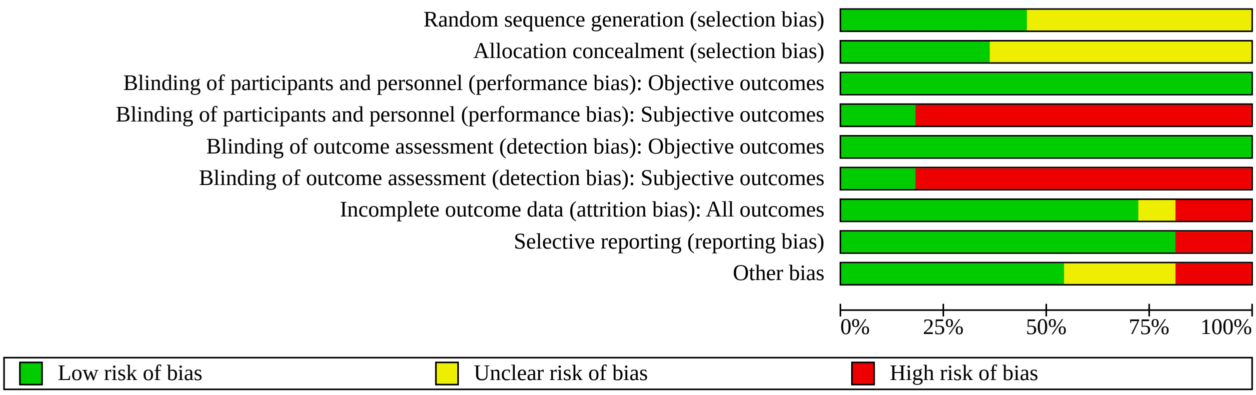


Figure 3. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.



Allocation

Overall, the included studies were at unclear risk of selection bias. We evaluated the random sequence generation as adequate in five trials (Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Greil 2016; Van Oers 2019), which we categorised as at low risk of bias. Six studies reported randomisation procedures without sufficient detail to permit a judgement, thus we considered their risk of bias as unclear (Byrd 2018; Foa 2014; Osterborg 2016; Robak 2018; Schweighofer 2009; Williams 2016).

Allocation concealment was appropriate for four trials (Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Van Oers 2019), and unclear for eight studies (Byrd 2018; Foa 2014; Greil 2016; Osterborg 2016; Robak 2018; Schweighofer 2009; Williams 2016).

Blinding

Overall, we judged the risk of performance and detection bias to be low for objective outcomes and high for subjective outcomes.

Performance bias

While exploring the effect of maintenance therapy intervention on people with CLL, we found that only two RCTs had double-blind and identical placebo designs (Chanan-Khan 2017; Fink 2017). The other nine RCTs had an open-label design; the performance of objective outcomes may not be affected, but that of other outcomes may be affected because of lack of blinding. Consequently, we judged all studies to have a low risk of bias for objective outcomes, and the risk of bias for blinding of participants and physicians as high for subjective outcomes (Byrd 2018; Dartigeas 2018; Foa 2014; Greil 2016; Osterborg 2016; Robak 2018; Schweighofer 2009; Van Oers 2019; Williams 2016).

Detection bias

We rated nine trials at low risk of detection bias for the assessment of objective outcomes.

Given that the judgements of unblinded outcome assessors for subjective outcomes may be influenced, we assessed the risk of bias for blinding of outcome assessment in nine trials as high (Byrd 2018; Dartigeas 2018; Foa 2014; Greil 2016; Osterborg 2016; Robak 2018; Schweighofer 2009; Van Oers 2019; Williams 2016). The other two trials had a triple-blind design (Chanan-Khan 2017; Fink 2017), thus we judged the risk of bias as low.

Incomplete outcome data

Overall, we judged risk of attrition bias as low. No risk of attrition bias could be detected, with all randomised participants analysed in the arms to which had been assigned; we therefore judged the risk of attrition bias to be low for nine trials (Byrd 2018; Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Foa 2014; Greil 2016; Schweighofer 2009; Van Oers 2019). We assessed risk of attrition bias for the other two trials as high: one trial had lost a follow-up ratio of up to 23.7% (Robak 2018), and the other trial enrolled 66 participants; however, only 39 participants were included in the final analysis (Osterborg 2016). One trial provided insufficient detail to permit a judgement, thus we considered risk of bias as unclear (Williams 2016).

Selective reporting

Overall, we considered there to be no obvious risk of reporting bias. For 9 of the 11 included trials, for which registered protocols existed, we judged the risk of reporting bias as low (Byrd 2018; Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Foa 2014; Greil 2016; Robak 2018; Van Oers 2019; Williams 2016).

Two trials had a registered protocol but did not fully report the prespecified outcomes. In the GCLLSG trial (Schweighofer 2009), results for patient quality of life were not reported. In a trial performed by Osterborg 2016, results were not reported for the following outcomes: duration of response, adverse event of interest (infusion reactions, infections, mucocutaneous reactions, tumour lysis syndrome, cardiovascular events, and small bowel obstruction), mean immunoglobulin (Ig) antibodies IgA, IgG and IgM over time, and number of participants who were positive or negative for human anti-human antibodies post-ofatumumab therapy.

Other potential sources of bias

We considered the overall risk of other sources of bias as low.

We considered two trials to be at high risk of other potential sources of bias: Robak 2018 had a prolonged randomisation process due to slow enrolment and included fewer participants than planned, leading to premature termination. Foa 2014 had an imbalance in the number of participants with a history of disease severity between the treatment arms (20.6% of participants in the rituximab arm had Binet stage C, compared to only 9.4% in the observation arm).

We considered three trials to be at unclear risk of other potential sources of bias: Fink 2017 had premature closure of recruitment due to poor accrual (enrolling only 89 of the estimated 200 planned participants). Osterborg 2016 did not report detailed baseline characteristics of the ofatumumab and observation arms after the second randomisation (the first randomisation was for induction therapy). Schweighofer 2009 was stopped prematurely due to severe infection.

We considered six trials to be at low risk of other potential sources of bias because no obvious sources of bias were detected (Byrd 2018; Chanan-Khan 2017; Dartigeas 2018; Greil 2016; Van Oers 2019; Williams 2016).

Effects of interventions

See: **Summary of findings 1** Anti-CD20 mAb maintenance therapy compared to observation for people with chronic lymphocytic leukaemia; **Summary of findings 2** Immunomodulatory drug maintenance therapy compared to placebo/observation for people with chronic lymphocytic leukaemia; **Summary of findings 3** Anti-CD52 mAb maintenance therapy compared to observation for people with chronic lymphocytic leukaemia

A total of 11 trials were included for the main descriptive analysis (Byrd 2018; Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Foa 2014; Greil 2016; Osterborg 2016; Robak 2018; Schweighofer 2009; Van Oers 2019; Williams 2016). We excluded one trial from the quantitative analysis because it grouped non-Hodgkin's lymphoma together and did not report prespecified outcomes specifically for participants with CLL (Williams 2016). Given the lack of data regarding the included RCTs, MRD outcomes were not analysed. The effect estimates of all meta-analyses were based on the intention-to-treat design principle if possible and presented with a random-effects model of forest plots.

Anti-CD20 monoclonal antibodies (mAbs) maintenance therapy versus observation

Anti-CD20 mAbs: rituximab, ofatumumab

Primary outcomes

Overall survival (OS)

Three trials (N = 1152) that reported OS were analysed (Dartigeas 2018; Greil 2016; Van Oers 2019). Maintenance treatment with anti-CD20 mAbs probably results in little to no difference in OS compared with observation alone (hazard ratio (HR) 0.94, 95% confidence interval (CI) 0.73 to 1.20; P = 0.60; Analysis 1.1). Heterogeneity was small (I^2 = 0%). We assessed the certainty of the evidence as moderate because the CI includes both clinically relevant benefits and harms. We downgraded one level for serious imprecision. This result remained consistent in sensitivity analyses using both fixed-effect and random-effects models (Table 1).

Overall survival (OS): type of drug

A stratified analysis by drug type (rituximab versus ofatumumab) demonstrated that rituximab maintenance therapy had an HR of 0.86 (95% CI 0.58 to 1.28; participants = 672; studies = 2) (Dartigeas 2018; Greil 2016), while ofatumumab maintenance therapy had an HR of 0.99 (95% CI 0.72 to 1.36; participants = 480; studies = 1) (Van Oers 2019) compared with observation (Analysis 1.2).

Health-related quality of life (HRQoL)

Only one trial reported HRQoL (Van Oers 2019), with 480 participants. Anti-CD20 mAbs may not reduce HRQoL compared with observation (mean difference −1.70, 95% CI −8.59 to 5.19; P = 0.63; Analysis 1.3). We assessed the certainty of the evidence as low, downgrading one level for serious risk of bias due to lack of blinding and one level for imprecision based on the wide CIs. We did not perform sensitivity analysis due to there being only one trial.

Grade 3 and 4 adverse events (AEs)

Five trials assessed grade 3/4 AEs (Dartigeas 2018; Foa 2014; Greil 2016; Robak 2018; Van Oers 2019), including a total of 1284 participants. We reported that maintenance therapy of anti-CD20 mAbs may result in an increase in grade 3/4 adverse events compared with observation alone (rate ratio 1.34, 95% CI 1.06 to 1.71; P = 0.02; Analysis 1.4). Heterogeneity was considerable (I^2 = 56%). We assessed the certainty of the evidence as low, downgrading one level for serious risk of bias due to most trials having an open-label design and one level for serious inconsistency based on substantial heterogeneity. This result remained consistent in sensitivity analyses using both fixed-effect and random-effects models (Table 1).

Grade 3 and 4 adverse events (AEs): type of drug

There was no statistically significant increase of grade 3/4 AEs reported with rituximab maintenance (rate ratio 1.30, 95% CI 0.93 to 1.82; participants = 804; studies = 4) (Dartigeas 2018; Foa 2014; Greil 2016; Robak 2018), but there was a significantly increasing risk of grade 3/4 AEs with ofatumumab maintenance (rate ratio 1.43, 95% CI 1.06 to 1.93; participants = 480; studies = 1) (Van Oers 2019) compared with observation (Analysis 1.5).

Secondary outcomes

Progression-free survival (PFS)

Five trials assessed PFS (Dartigeas 2018; Greil 2016; Osterborg 2016; Robak 2018; Van Oers 2019), including a total of 1255 participants. Our meta-analysis showed that maintenance treatment with anti-CD20 mAbs probably significantly increases PFS compared with observation alone (HR 0.61, 95% CI 0.50 to 0.73; P < 0.001; Analysis 1.6). Heterogeneity was small (I^2 = 38%). We assessed the certainty of the evidence as moderate due to lack of blinding. We downgraded one level for serious risk of bias. This result remained consistent in sensitivity analyses using both fixed-effect and random-effects models (Table 1).

We used GetData Graph Digitizer software to estimate the HR from the survival curve provided by Osterborg 2016 (GetData Graph Digitizer).

Progression-free survival (PFS): type of drug

A stratified analysis by drug type (rituximab versus ofatumumab) demonstrated that rituximab maintenance therapy had an HR of 0.50 (95% CI 0.37 to 0.68; participants = 738; studies = 3) (Dartigeas 2018; Foa 2014; Greil 2016; Robak 2018), while ofatumumab maintenance therapy had an HR of 0.66 (95% CI 0.50 to 0.86;

participants = 517; studies = 2) (Osterborg 2016; Van Oers 2019) compared with observation (Analysis 1.7).

Treatment-related mortality (TRM)

Four trials assessed TRM (Dartigeas 2018; Greil 2016; Osterborg 2016; Van Oers 2019), including 1189 participants. Maintenance treatment with anti-CD20 mAbs may result in little to no difference in TRM compared with observation alone (risk ratio 0.82, 95% CI 0.39 to 1.71; P = 0.60; Analysis 1.8). We assessed the certainty of the evidence as low, downgrading one level for serious imprecision because of the wide CIs, which crossed the line of no difference, and a further level due to substantial heterogeneity (I^2 = 47%; and two studies suggest benefit and two suggest harm). This result remained consistent in sensitivity analyses using both fixed-effect and random-effects models (Table 1).

Treatment-related mortality (TRM): type of drug

A stratified analysis by drug type (rituximab versus ofatumumab) demonstrated that rituximab maintenance therapy had a risk ratio of 1.12 (95% CI 0.44 to 2.88; participants = 672; studies = 2) (Dartigeas 2018; Greil 2016), while ofatumumab maintenance therapy had a risk ratio of 0.52 (95% CI 0.25 to 1.08; participants = 517; studies = 2) (Osterborg 2016; Van Oers 2019) compared with observation (Analysis 1.9).

Treatment discontinuation (TD)

Six trials (N = 1321) reporting TD were analysed (Dartigeas 2018; Foa 2014; Greil 2016; Osterborg 2016; Robak 2018; Van Oers 2019). The evidence suggested that anti-CD20 mAbs maintenance may result in a slight reduction in treatment discontinuation compared with observation (risk ratio 0.93, 95% CI 0.72 to 1.20; P = 0.57; Analysis 1.10). We assessed the certainty of the evidence as low, downgrading one level for serious imprecision because of the wide CI, which crossed the line of no difference, and a further level due to substantial heterogeneity (I^2 = 64%). This result remained consistent in sensitivity analyses using both fixed-effect and random-effects models (Table 1).

Treatment discontinuation (TD): type of drug

A stratified analysis by drug type (rituximab versus ofatumumab) demonstrated that rituximab maintenance therapy had a risk ratio of 0.87 (95% CI 0.58 to 1.31; participants = 804; studies = 4) (Dartigeas 2018; Foa 2014; Greil 2016; Robak 2018), while ofatumumab maintenance therapy had a risk ratio of 0.93 (95% CI 0.77 to 1.12; participants = 517; studies = 2) (Osterborg 2016; Van Oers 2019) compared with observation (Analysis 1.11).

All adverse events (AEs)

Six trials assessed grade 3/4 AEs (Dartigeas 2018; Foa 2014; Greil 2016; Osterborg 2016; Robak 2018; Van Oers 2019), including a total of 1321 participants. Anti-CD20 mAbs maintenance therapy may result in a slight increase in all AEs compared with observation alone (rate ratio 1.23, 95% CI 1.03 to 1.47; P = 0.02; Analysis 1.12). Heterogeneity was considerable (I^2 = 56%). We assessed the certainty of the evidence as low, downgrading by one level for serious risk of bias, primarily due to most trials having an open-label design, and by another level for serious inconsistency based on substantial heterogeneity (I^2 = 56%) and inconsistencies in the reporting criteria for AEs. This result remained consistent in sensitivity analyses using both fixed-effect and random-effects models (Table 1).

All adverse events (AEs): type of drug

A stratified analysis by drug type (rituximab versus ofatumumab) demonstrated that rituximab maintenance therapy had a rate ratio of 1.26 (95% CI 0.97 to 1.64; participants = 804; studies = 4) (Dartigeas 2018; Foa 2014; Greil 2016; Robak 2018), while ofatumumab maintenance therapy had a rate ratio of 1.16 (95% CI 0.96 to 1.40; participants = 517; studies = 2) (Osterborg 2016; Van Oers 2019) compared with observation (Analysis 1.13).

Complete response rate (CRR)

Two trials (N = 132) that reported CRR were analysed (Foa 2014; Robak 2018). Both trials suggested that rituximab maintenance therapy may increase CRR slightly (risk ratio 1.42, 95% CI 0.81 to 2.48; P = 0.22; Analysis 1.14). Heterogeneity was small (I^2 = 0%). We assessed the certainty of the evidence as low because of the wide CIs, which crossed the line of no difference; the small number of participants; and high risk of bias for Robak 2018 owing to incomplete outcome data and prolonged enrolment/low recruitment rate. We downgraded one level for serious imprecision and one level for serious risk of bias. This result remained consistent in sensitivity analyses using both fixed-effect and random-effects models (Table 1).

Overall response rate (ORR)

Only two trials reported ORR (Foa 2014; Robak 2018), including 132 participants. The evidence suggested that maintenance therapy with rituximab may result in an increase in ORR compared with observation alone (risk ratio 1.53, 95% CI 1.05 to 2.24; P = 0.03; Analysis 1.15). Heterogeneity was small (I^2 = 0%). We assessed the certainty of the evidence as low because of the wide CIs, which crossed the line of no difference; the small number of participants; and high risk of bias for Robak 2018 owing to incomplete outcome data and prolonged enrolment/low recruitment rate. We downgraded one level for serious imprecision and one level for serious risk of bias. This result remained consistent in sensitivity analyses using both fixed-effect and random-effects models (Table 1).

Minimal residual disease (MRD)

No study reporting this outcome was identified.

Immunomodulatory drug (IMiD) maintenance therapy versus placebo/observation

IMiD: lenalidomide

Two trials compared an IMiD with placebo (Chanan-Khan 2017; Fink 2017), while another trial compared an IMiD with observation (Byrd 2018).

Primary outcomes

Overall survival (OS)

Three trials (N = 461) that reported OS were analysed (Byrd 2018; Chanan-Khan 2017; Fink 2017). We found evidence suggesting that maintenance treatment with lenalidomide probably results in little to no difference in overall survival compared with observation alone (HR 0.91, 95% CI 0.61 to 1.35; P = 0.63; Analysis 2.1). Heterogeneity was small (I^2 = 0%). We assessed the certainty of the evidence as moderate because of the wide CIs, which crossed the line of relative risk reduction threshold, and the small number of participants. We downgraded one level for serious imprecision.

The consistency of this result was maintained in sensitivity analyses using both fixed-effect and random-effects models, as well as when assessing the impact of high versus low risk of bias for blinding (Table 1).

We used GetData Graph Digitizer software to estimate the HR from the survival curve provided by Byrd 2018 (GetData Graph Digitizer).

Health-related quality of life (HRQoL)

No study reporting this outcome was identified.

Grade 3 and 4 adverse events (AEs)

Two trials reported grade 3 and 4 AEs (Chanan-Khan 2017; Fink 2017). A total of 400 participants were included in the meta-analysis. The evidence suggested that maintenance therapy of lenalidomide may result in an increase in grade 3/4 AEs compared to placebo (rate ratio 1.82, 95% CI 1.38 to 2.38; $P < 0.001$; Analysis 2.2). Heterogeneity was small ($I^2 = 0\%$). We assessed the certainty of the evidence as low because of wide CIs and the small number of participants. We downgraded two levels for very serious imprecision. This result remained consistent in sensitivity analyses using both fixed-effect and random-effects models (Table 1).

Secondary outcomes

Progression-free survival (PFS)

Three trials assessed PFS (Byrd 2018; Chanan-Khan 2017; Fink 2017), including a total of 461 participants. Meta-analysis showed that maintenance treatment with lenalidomide probably results in a large increase in PFS compared to observation alone (HR 0.37, 95% CI 0.19 to 0.73; $P = 0.004$; Analysis 2.3). Heterogeneity was substantial ($I^2 = 68\%$). We assessed the certainty of the evidence as moderate because of the substantial heterogeneity with visualised dissimilarity of point estimates. We downgraded one level for serious inconsistency. The consistency of this result was maintained in sensitivity analyses using both fixed-effect and random-effects models, as well as when assessing the impact of high versus low risk of bias for blinding (Table 1).

We used GetData Graph Digitizer software to estimate the HR from the survival curve provided by Byrd 2018 (GetData Graph Digitizer).

Treatment-related mortality (TRM)

Three trials assessed TRM (Byrd 2018; Chanan-Khan 2017; Fink 2017), including a total of 458 participants. Meta-analysis suggested that maintenance treatment with lenalidomide may result in a slight increase in TRM compared to observation alone (risk ratio 1.22, 95% CI 0.35 to 4.29; $P = 0.76$; Analysis 2.4). Heterogeneity was small ($I^2 = 0\%$). We assessed the certainty of the evidence as low because of the very wide CIs, which crossed the line of relative risk reduction threshold, and low event rates. We downgraded two levels for very serious imprecision. The consistency of this result was maintained in sensitivity analyses using both fixed-effect and random-effects models, as well as when assessing the impact of high versus low risk of bias for blinding (Table 1).

Treatment discontinuation (TD)

Two trials (N = 400) that reported our secondary outcome TD were analysed (Chanan-Khan 2017; Fink 2017). The evidence is very uncertain about the effect of IMiD maintenance therapy on TD (risk ratio 0.71, 95% CI 0.47 to 1.05; $P = 0.09$; Analysis 2.5). Heterogeneity was substantial ($I^2 = 75\%$). We assessed the certainty of the evidence as very low. We downgraded two levels for very serious imprecision due to wide CIs, which crossed the line of relative risk reduction threshold, and the small number of participants. We downgraded one level for serious inconsistency due to substantial heterogeneity. This result was found not to be robust in sensitivity analysis. However, when a fixed-effect model was applied, meta-analysis indicated that the proportion of TD may be reduced with IMiD maintenance (risk ratio 0.78, 95% CI 0.68 to 0.89; Table 1).

All adverse events (AEs)

Three trials reported all AEs (Byrd 2018; Chanan-Khan 2017; Fink 2017). A total of 458 participants were included in the meta-analysis. IMiD maintenance therapy probably increases all AEs slightly compared to placebo (rate ratio 1.41, 95% CI 1.28 to 1.54; $P < 0.001$; Analysis 2.6). Heterogeneity was small ($I^2 = 0\%$). We assessed the certainty of the evidence as moderate. We downgraded the study by one level for serious inconsistency, specifically due to inconsistencies in the reporting criteria for AEs. The consistency of this result was maintained in sensitivity analyses using both fixed-effect and random-effects models, as well as when assessing the impact of high versus low risk of bias for blinding (Table 1).

Complete response rate (CRR)

No study reporting this outcome was identified.

Overall response rate (ORR)

No study reporting this outcome was identified.

Minimal residual disease (MRD)

No study reporting this outcome was identified.

Anti-CD52 mAbs maintenance therapy versus observation

Anti-CD52 mAb: alemtuzumab

Primary outcomes

Overall survival (OS)

No study reporting this outcome was identified.

Health-related quality of life (HRQoL)

No study reporting this outcome was identified.

Grade 3 and 4 adverse events (AEs)

No study reporting this outcome was identified.

Secondary outcomes

Progression-free survival (PFS)

Only one trial reported PFS (Schweighofer 2009), with 21 participants. We reported the trial result, finding that maintenance with alemtuzumab may lead to an increase in PFS, but the evidence is very uncertain (HR 0.55, 95% CI 0.32 to 0.95; $P = 0.003$; Analysis 3.1). Heterogeneity was not applicable. We evaluated the certainty of the evidence as very low due to the trial's lack of reporting on random sequence generation and allocation methods, lack of blinding, potential for selective reporting, and the inclusion of 21 participants with early termination, suggesting the study may be underpowered to demonstrate an effect. We downgraded two levels for very serious imprecision and one level for serious risk of bias. We did not perform sensitivity analysis due to there being only one trial.

We used GetData Graph Digitizer software to estimate the HR from the survival curve provided by Schweighofer 2009 (GetData Graph Digitizer).

Treatment-related mortality (TRM)

No study reporting this outcome was identified.

Treatment discontinuation (TD)

No study reporting this outcome was identified.

All adverse events (AEs)

Schweighofer 2009 (reporting on a total of 21 participants) reported 51 AEs in the alemtuzumab arm, but did not report any events in the control arm.

Complete response rate (CRR)

Only one trial reported CRR (Schweighofer 2009), with 21 participants. We reported the trial result, finding that the effect of maintenance with alemtuzumab on CRR is very uncertain (risk ratio 1.36, 95% CI 0.28 to 6.56; P = 0.70; Analysis 3.2). Heterogeneity was not applicable. We evaluated the certainty of the evidence as very low due to the trial's lack of reporting on random sequence generation and allocation methods, lack of blinding, potential for selective reporting, and the inclusion of 21 participants with early termination, suggesting the study may be underpowered to demonstrate an effect. We downgraded two levels for very serious imprecision and one level for serious risk of bias. We did not perform sensitivity analysis due to there being only one trial.

Overall response rate (ORR)

Only one trial reported ORR (Schweighofer 2009), with 21 participants. We reported the trial result, finding that the effect of maintenance of alemtuzumab on ORR is very uncertain (risk ratio 1.41, 95% CI 0.92 to 2.14; P = 0.11; Analysis 3.3). Heterogeneity was not applicable. We evaluated the certainty of the evidence as very low due to the trial's lack of reporting on random sequence generation and allocation methods, lack of blinding, potential for selective reporting, and the inclusion of 21 participants with early termination, suggesting the study may be underpowered to demonstrate an effect. We downgraded two levels for very serious imprecision and one level for serious risk of bias. We did not perform sensitivity analysis due to there being only one trial.

Minimal residual disease (MRD)

No study reporting this outcome was identified.

DISCUSSION

The main objective of this review was to examine the evidence for the clinical effectiveness and safety of maintenance therapy for the treatment of CLL. A total of 11 trials were eligible for inclusion (Byrd 2018; Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Foa 2014; Greil 2016; Osterborg 2016; Robak 2018; Schweighofer 2009; Van Oers 2019; Williams 2016). For the purposes of quantitative analysis, seven trials compared anti-CD20 mAbs versus observation: five trials used rituximab (Dartigeas 2018; Foa 2014; Greil 2016; Robak 2018), and two trials used ofatumumab (Osterborg 2016; Van Oers 2019) as maintenance therapy. Three trials compared IMiD (lenalidomide) maintenance therapy versus placebo/observation (Byrd 2018; Chanan-Khan 2017; Fink 2017), and another trial compared anti-CD52 mAb (alemtuzumab) maintenance therapy versus observation (Schweighofer 2009). One trial lumped non-Hodgkin's lymphoma together and did not report prespecified outcomes for this review specifically for participants with CLL (Williams 2016); no results can be ascertained from this study.

Summary of main results

The available evidence from these trials showed the following.

Overall survival (OS) (time to event)

We included three trials (1152 participants) comparing anti-CD20 mAbs with observation and three trials (461 participants) comparing IMiD with placebo/observation. Both types of maintenance therapy probably result in little to no difference in OS (moderate-certainty evidence).

No trial comparing anti-CD52 mAb reported OS.

Health-related quality of life (HRQoL)

Only one study (480 participants), comparing anti-CD20 mAbs versus observation, reported this outcome. Anti-CD20 mAbs may not reduce HRQoL (low-certainty evidence).

No studies evaluating IMiD or anti-CD52 mAbs reported HRQoL.

Grade 3 and 4 adverse events (AEs)

Five trials (1284 participants) evaluating anti-CD20 mAbs maintenance therapy assessed this outcome. Anti-CD20 mAbs maintenance therapy may result in an increase in grade 3/4 AEs (low-certainty evidence).

Two trials (400 participants) evaluating IMiD maintenance therapy assessed this outcome. IMiD maintenance therapy may result in an increase in grade 3/4 AEs (low-certainty evidence).

No trial comparing anti-CD52 mAb reported grade 3 and 4 AEs.

Progression-free survival (PFS) (time to event)

We included five trials (1255 participants) comparing anti-CD20 mAbs with observation; three trials (461 participants) comparing IMiD with placebo/observation; and one trial (21 participants) comparing anti-CD52 mAbs with observation.

Anti-CD20 mAbs maintenance therapy probably increases PFS, and IMiD maintenance therapy probably results in a large increase in PFS (moderate-certainty evidence). We are very uncertain whether anti-CD52 mAb has any effect on PFS (very low-certainty evidence).

Treatment-related mortality (TRM)

We included four trials (1189 participants) evaluating anti-CD20 mAbs and three trials (458 participants) evaluating IMiD. Treatment with anti-CD20 mAbs maintenance therapy may result in little to no difference in TRM (low-certainty evidence). Treatment with IMiD maintenance therapy may result in a slight increase in TRM (low-certainty evidence).

No trial evaluating anti-CD52 mAb reported TRM.

Treatment discontinuation (TD)

Six trials (1321 participants) evaluating anti-CD20 mAbs and two trials (400 participants) evaluating IMiD assessed this outcome.

Anti-CD20 mAbs maintenance therapy may result in a slight reduction in TD (low-certainty evidence). The evidence for the effect of IMiD maintenance therapy on TD is very uncertain (very low-certainty evidence).

No trial evaluating anti-CD52 mAb reported TD.

All adverse events (AEs)

Of the 11 trials included in this review, 9 reported safety outcomes; one trial only reported safety outcomes during induction therapy, and one trial only reported safety outcomes in the experimental arm, with no information on the control arm. The reporting criteria for AEs varied across the included studies: one trial reported any AE; two trials reported AEs with an incidence of ≥ 2% of total adverse events in all cycles; four trials reported any incidence of AEs in ≥ 10%; and two trials only reported grade 3, 4, and 5 AEs. Furthermore, the duration of follow-up for surveillance of AEs also varied across studies.

A total of nine trials were included to assess this quantitative outcome: six trials (1321 participants) evaluating anti-CD20 mAbs and three trials (458 participants) evaluating IMiD.

Anti-CD20 mAbs maintenance therapy may result in a slight increase in all AEs (low-certainty evidence). IMiD maintenance therapy probably slightly increases all AEs (moderate-certainty evidence).

No trial evaluating anti-CD52 mAb reported all AEs.

Response rate (complete response rate (CRR) and overall response rate (ORR))

Two trials (132 participants) evaluating anti-CD20 mAbs and one trial (21 participants) evaluating anti-CD52 mAb assessed CRR and ORR.

The evidence suggests that anti-CD20 mAbs maintenance therapy may result in a slight increase in both CRR and ORR (low-certainty evidence). We are very uncertain whether there is a difference in CRR or ORR between anti-CD52 mAbs and observation (very low-certainty evidence).

No trial comparing IMiD maintenance therapy reported CRR or ORR.

Minimal residual disease

None of the included studies reported this outcome.

Overall completeness and applicability of evidence

This review provides the most up-to-date evidence on the role of maintenance therapy in individuals diagnosed with CLL. To facilitate the development of comprehensive inclusion criteria, we included studies that examined either consolidation or maintenance therapy.

Eleven randomised trials met the inclusion criteria for this review: seven trials compared anti-CD20 mAbs with observation (N = 1679, with five trials including 856 participants evaluating rituximab and two trials with 517 participants evaluating ofatumumab); three trials compared lenalidomide with placebo/observation (N = 693); and one trial compared alemtuzumab with observation (N = 21). Nine trials were phase 3, and two were phase 2 trials.

One trial enrolled participants with non-Hodgkin's lymphoma, while the remaining 10 trials enrolled those with CLL; the corresponding author of Williams 2016 was unable to provide us with more detailed information on the CLL subgroup due to the inability to obtain permissions from other centres' principal investigators. We attempted to contact the original author of one study for further information required to determine eligibility for inclusion in the review, but did not receive a response (Jaksic 1988). Two ongoing trials are still active (NCT03280160; Oughton 2017), one of which is evaluating ibrutinib in combination with ofatumumab as a maintenance agent, and the other which is evaluating obinutuzumab as a maintenance agent.

There are several limitations that may affect the strength and applicability of our review. These include moderate clinical heterogeneity among participants, different regimens of active interventions, uneven follow-up durations, and previous interventions across trials. Additionally, the landscape of CLL is evolving, with novel tyrosine kinase inhibitors (TKI) treating CLL with a treatment-until-progression strategy that includes induction and maintenance phases. However, we did not report this comparison due to a lack of available data.

In nine trials participants received previous treatment with rituximab-based chemo-immunotherapy, while those in one trial received ofatumumab-based therapy, and in the remaining trial received chemotherapy.

Eight trials reported baseline Binet/Rai disease stage status, with five trials predominantly involving early-stage CLL patients but with symptoms or high cytogenetic aberrations, while the other three trials predominantly involved patients in later stages. Eleven trials reported cytogenic abnormality, with six trials having a higher frequency of cytogenic abnormality ranging from 69.7% to 80.1%. Unmutated IGHV status was reported in eight trials, ranging between 41.4% and 91%.

The intravenous rituximab dosage was 375 mg/m^2 every 12 weeks in four trials and 500 mg/m^2 every 8 weeks in one trial. The intravenous ofatumumab maintenance dosage was 1000 mg in one trial and 2000 mg in the second trial. The lenalidomide maintenance dosage was administered with a targeted daily dosage of 5 mg or 15 mg until progression in three trials. Approximately 30 mg of alemtuzumab was administered intravenously three times per week.

The median duration of follow-up in seven trials that administered anti-CD20 mAbs (rituximab and ofatumumab) ranged from 14.4 months to 73 months. In three trials of IMiD (lenalidomide) maintenance, the mean duration of follow-up ranged from 17.9 months to 31.5 months. For anti-CD52 mAb (alemtuzumab) maintenance therapy, the median duration of follow-up was 12.4 months.

There was a moderate degree of clinical heterogeneity among participant characteristics and previous interventions across trials. In particular, differences in the underlying disease severity (baseline Rai/Binet status, cytogenetic aberration, and IGHV status), as well as the previous treatment regimen, post-therapy response, timing of maintenance therapy, and the regimens of maintenance therapy administered, may all have contributed to the expected clinical heterogeneity across these studies. Notably, for many outcomes, the values of I^2 were 0%, but I^2 values may be underestimated when there are few events or a limited number of studies included in a meta-analysis (Huedo-Medina 2006; Ioannidis 2007).

Two trials compared ofatumumab to observation, and only one trial reported the comparison of efficacy between alemtuzumab and observation. Therefore, the evidence is limited in regards to the efficacy of ofatumumab in people with CLL and the role of alemtuzumab in CLL maintenance.

Across all outcomes, maintenance therapy showed efficacy in prolonging PFS, but there were insufficient data to determine if maintenance therapy has a beneficial effect on OS. Nonetheless, maintenance therapy was associated with an increased incidence of grade 3/4 AEs and all AEs.

No trial reported minimal residual disease as an outcome.

Quality of the evidence

All 11 included studies were parallel-group RCTs. Four trials reported random sequence generation and allocation concealment (Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Van Oers 2019). Adequate blinding was implemented in only two trials, utilising a placebo arm (Dartigeas 2018; Fink 2017). Nine studies were open-label trials, with high risk of both performance bias and detection bias for subjective outcomes (Byrd 2018; Dartigeas 2018; Foa 2014; Greil 2016; Osterborg 2016; Robak 2018; Schweighofer 2009; Van Oers 2019; Williams 2016). Eleven trials received financial support from pharmaceutical companies, including Roche, GlaxoSmithKline, and Novartis AG (Byrd 2018; Chanan-Khan 2017; Dartigeas 2018; Fink 2017; Foa 2014; Greil 2016; NCT00771602; Osterborg 2016; Robak 2018; Schweighofer 2009; Van Oers 2019; Williams 2016).

We did not address publication bias through funnel plots because fewer than 10 studies were included in all comparisons.

Our risk of bias judgements for the included studies are presented in the risk of bias tables in Characteristics of included studies, and risk of bias summaries are shown in Figure 2 and Figure 3.

- Two trials were at high risk of bias in four out of seven domains (Osterborg 2016; Robak 2018).
- Two trials were at high risk of bias in three out of seven domains (Foa 2014; Schweighofer 2009).
- Five trials were at high risk of bias in two out of seven domains (Byrd 2018; Dartigeas 2018; Greil 2016; Van Oers 2019; Williams 2016).

Our GRADE assessment for most outcomes was moderate or low. The certainty of evidence for the main comparisons of each maintenance therapy versus observation or placebo is summarised in Summary of findings 1; Summary of findings 2; Summary of findings 3.

We evaluated the certainty of evidence as moderate for the following major outcomes.

- Anti-CD20 mAbs maintenance therapy: OS and PFS. We downgraded one level due to serious risk of bias (open-label design) or serious imprecision (wide CIs).
- IMiD maintenance therapy: OS, PFS, and all AEs. We downgraded one level due to serious imprecision (wide CIs) or serious inconsistency (significant heterogeneity or inconsistencies in the reporting criteria for AEs).
- Anti-CD52 mAbs maintenance therapy: nil

We assessed the certainty of evidence as low for the following major outcomes.

- Anti-CD20 mAbs maintenance therapy: HRQoL, grade 3 and 4 AEs, TRM, TD, and all AEs. We downgraded the certainty of the evidence due to serious risk of bias (lack of blinding), serious imprecision (wide CIs), and inconsistency (substantial heterogeneity or inconsistencies in the reporting criteria for AEs).
- IMiD maintenance therapy: grade 3 and 4 AEs and TRM. We downgraded the certainty of the evidence based on very serious imprecision (very wide CIs, or wide CIs with small number of participants).
- Anti-CD52 mAbs maintenance therapy: nil

We assessed the certainty of evidence as very low for the following major outcomes.

- Anti-CD20 mAbs maintenance therapy: nil
- IMiD maintenance therapy: TD. We downgraded two levels for very serious imprecision (wide CIs with small number of participants) and one level for serious inconsistency (substantial heterogeneity).
- Anti-CD52 mAbs maintenance therapy: PFS. We downgraded one level for serious risk of bias (open-label design) and two levels for very serious bias (single study terminated early, likely underpowered to demonstrate an effect).

There was notable clinical heterogeneity in the types and stages of the malignant disease being treated, previous induction chemotherapy regimens, maintenance therapy dosage and frequency, definitions of outcomes, and differences in study quality. Unfortunately, we could not perform subgroup analyses due to the lack of suitable studies. However, these will be addressed in future review updates as further RCTs are published.

Potential biases in the review process

We are not aware of any apparent biases in the review process. We used a sensitive search strategy that included ongoing trial databases and handsearching of conference proceedings. We also contacted researchers in the field to identify any further as yet unpublished trials. We placed no restrictions on the language in which the paper was originally published or publication status. We tried to avoid bias by having all relevant processes duplicated by two review authors and resolving any discrepancies through discussion. However, as the included studies are limited to performing tests for publication bias, we cannot confidently claim that we obtained all relevant studies.

None of the review authors was involved in the included or excluded studies.

Agreements and disagreements with other studies or reviews

We found moderate- to low-certainty evidence on the effectiveness and safety of maintenance therapy for people with CLL. To our knowledge, there are few systematic reviews and meta-analyses that fully evaluate the benefits and harms associated with the use of maintenance therapy in adults with CLL using anti-CD20 mAbs (rituximab and ofatumumab), IMiDs (lenalidomide), anti-CD52 mAb (aletuzumab), or other regimens.

Lee 2020 regularly evaluate the outcomes of people with maintenance therapy compared to standard care. Their last update was on 6 March 2019, and it reported on six included RCTs. Lee and colleagues compared the interventions for each maintenance drug, finding that lenalidomide, rituximab, and ofatumumab resulted in the longest PFS compared to no intervention (HR 0.37, 95% CI 0.27 to 0.50; HR 0.50, 95% CI 0.38 to 0.66; HR 0.52, 95% CI 0.41 to 0.66, respectively), with lenalidomide showing the greatest efficacy in preventing disease progression (HR 0.37, 95% CI 0.27 to 0.50). The study also assessed serious adverse events and provided relative risk estimates. Our assessments of clinical benefits for OS and PFS were similar, but the study did not report on HRQoL, TRM, TD, and all AEs. Additionally, the study did not include six RCTs included in our review (Byrd 2018; Foa 2014; Osterborg 2016; Robak 2018; Schweighofer 2009; Williams 2016).

Mkhwanazi 2022 included a total of four studies that reported improved PFS following maintenance therapy with anti-CD20 therapy compared to an observation group. The results showed a pooled effect estimate with a non-significant improvement in PFS (HR 0.51, 95% CI 0.42 to 0.60; P = 0.93). There were no differences in OS between participants receiving maintenance therapy and those who were not on treatment. Mkhwanazi 2022 only reported on survival outcomes and did not include outcomes for clinical harms. Additionally, the interpretation of the effects on PFS was incorrect. The CI of the pooled effect estimate was 0.42 to 0.60, which is statistically significant. The study also did not include three RCTs included in our review (Foa 2014; Osterborg 2016; Williams 2016).

The concept of maintenance therapy is not new in haematological malignancy. For instance, a review published on maintenance therapy for B-chronic lymphocytic leukaemia announced that maintenance therapy of rituximab and lenalidomide had a potential role in deep treatment response (O'Brien 2011).

Robak 2013 commented on the role of maintenance in CLL, concluding that maintenance therapy had PFS benefit and deepened the treatment response, but at the expense of severe haematological and immunologic AEs, and that maintenance or consolidation therapy with rituximab, alemtuzumab, or lenalidomide should not be used outside of clinical trials. Tam 2016 also commented that anti-CD20 maintenance after chemo-immunotherapy for CLL should depend on the risk of the individual, emphasising that the clinician must balance the costs and risks of two years of antibody maintenance, as there is no evident advantage in terms of the time to the next treatment and overall survival.

After the PROLONG trial was published, ofatumumab was approved by the US Food and Drug Administration (FDA) for the treatment of patients with recurrent or progressive CLL who are in complete or partial remission following two or more lines of prior therapy since 2016.

Before 2021, in the chemo-immunotherapy era, the National Comprehensive Cancer Network (NCCN) guidelines in the US recommended lenalidomide maintenance therapy following first-line chemo-immunotherapy for high-risk patients, and lenalidomide or ofatumumab maintenance therapy following second-line chemo-immunotherapy for people with CLL who had a complete or partial response, based on lower-level evidence with NCCN consensus that the intervention was appropriate. However, the CLL treatment landscape is changing rapidly, and TKI have become the mainstream induction therapy in CLL. Chemotherapy-immunotherapy has become a choice rather than a preferred regimen for induction. Therefore, because there is no evidence supporting maintenance therapy followed by TKI induction, maintenance therapies with lenalidomide and ofatumumab have been removed from later versions of the NCCN guidelines (NCCN 2021).

Even in the TKI era of CLL, maintenance therapy remains an important intervention to prolong and deepen the treatment response, in addition to the higher response rate of TKI induction. Mark-David Levin and colleagues launched a phase 2 randomised trial (HOVON 139/GIVE trial) in which people with CLL received a fixed-duration regimen of venetoclax plus obinutuzumab followed by consolidation with venetoclax, which resulted in high rates of undetectable measurable residual disease. This pilot trial demonstrated the potential benefit of a measurable residual disease-guided treatment strategy and the role of maintenance therapy. We will continue to monitor available randomised trials and update this review accordingly.

AUTHORS' CONCLUSIONS

Implications for practice

There is currently moderate- to very low-certainty evidence available on the benefits and harms of maintenance therapy for people with chronic lymphocytic leukaemia (CLL). Anti-CD20 monoclonal antibodies (rituximab, ofatumumab) or immunomodulatory drug (lenalidomide) maintenance therapy, when compared to observation or placebo, probably improves progression-free survival, but possibly at the cost of increasing grade 3 and 4 and all adverse events. However, none of the above maintenance interventions currently show a difference between maintenance and control group in overall survival.

Implications for research

Further trials using tyrosine kinase inhibitors for induction therapy, rather than chemo-immunotherapy, are needed to determine the potential effectiveness of maintenance therapy in people with CLL. Head-to-head comparisons of different maintenance therapies are also needed. In addition, we did not include patients who received continuous therapy with small molecular inhibitor and responded, but continued or discontinued treatment beyond response, due to inconsistency with the conventional definition of maintenance therapy. We may consider including this category of trials in future updated reviews, as it becomes a major treatment modality for CLL patients. Finally, because minimal residual disease-guided treatment strategies have only recently become mainstream in haematological care, trials with minimal residual disease-guided maintenance therapy designs should also be included and studied in future research.

ACKNOWLEDGEMENTS

We would like to thank all the following individuals for their comments and suggestions that have helped to improve the review.

- Cochrane Haematology and Cochrane Central Editorial Service supported the authors in the development of this intervention review.
- The following people conducted the editorial process for this article:
  - Sign-off Editor (final editorial decision): Nicole Skoetz, University of Cologne and University Hospital Cologne, Germany; Co-ordinating Editor of the Cochrane Haematology Group
  - Managing Editor (selected peer reviewers, provided comments, collated peer-reviewer comments, provided editorial guidance to authors, edited the article): Lara Kahale, Cochrane Central Editorial Service
  - Editorial Assistant (conducted editorial policy checks and supported editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service
  - Copy Editor (copy-editing and production): Lisa Winer, Cochrane Central Production Service

Peer reviewers (provided comments and recommended an editorial decision): Samir Agrawal, Immunobiology, Blizard Institute, Queen Mary University of London (clinical), Liat Vidal (clinical), Brian Duncan (consumer review), Rachel Richardson, Associate Editor, Cochrane (methods review), Emma Axon, Cochrane (methods review), Jo Platt, Information Specialist Cochrane GNOC (search review), Ina Monsef, Information Specialist Cochrane CHG (search review)

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* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies *[ordered by study ID]*

Byrd 2018

Study characteristics	
Methods	**CALGB 10404 (Alliance)**

Byrd 2018 *(Continued)*

	Study design: Open-label, randomised, parallel-group phase 2 study, multicentre Language of publication: English Major induction regimen: induction with FR, fludarabine 25 mg/m^2 IV days 1 to 5 + rituximab 50 mg/m^2 IVPB day 1 and rituximab 375 mg/m^2 IVPB day 5 for 6 months or FCR, fludarabine 25 mg/m^2 IV days 1 to 3 + cyclophosphamide 250 mg/m^2 days 1 to 3 + rituximab 500 mg/m^2 IVPB day 1 for 6 months Randomisation: 4 arms • FR induction with lenalidomide consolidation versus FR induction with observation • FCR induction with lenalidomide consolidation versus FCR induction with observation Recruitment period: March 2008 and August 2012 Median follow-up time: median 73 months Sample size calculation: with a targeted 103 participants per arm with non-del(11q), there was at least 84% power to detect an improvement in 2-year PFS rate from 60% to 73%, using a critical value of 69% and constraining the one-sided type I error rate to 4%.
Participants	Inclusion criteria: patients enrolled in this study were 18 years of age and older, had CLL with an absolute lymphocytosis of > 5 × 10^9/L, immunophenotype typical of CLL, and features defined by the International Workshop on CLL (IWCLL) 2008 guidelines. All participants had symptomatic intermediate- (Rai I or II) or high-risk (Rai III or IV) stage CLL and a performance status of 2 or better. Participants had no prior therapy for CLL and a creatinine ≤ 1.5 × the upper limit of normal. Number of included participants (overall): • FR+L versus FR: 232 • FCR+L versus FCR: 58 Numbers of SLL/CLL: • FR+L versus FR: 232 • FCR+L versus FCR: 58 Study sites: multicentre, 364 centres Stage/type of disease (experimental): • FR+L: Rai III/IV: 50.4% • FCR+L: Rai III/IV: 29.3% Stage/type of disease (control): • FR: Rai III/IV: 46.3% • FCR: Rai III/IV: 55.6% Total cytogenetic aberrations rate: 80.1% Median age (experimental): • FR+L: median 62 (36 to 79) • FCR+L: median 59 (30 to 75) Median age (control): • FR: median 61 (28 to 81) • FCR: median 60 (41 to 78) Sex of male percentage (experimental):

Byrd 2018 *(Continued)*

	• FR+L: 63% • FCR+L: 74% Sex of male percentage (control): • FR: 67% • FCR: 59%
Interventions	Numbers of experimental group (lenalidomide): • FR+L: 109 • FCR+L: 31 Numbers of control group (observation group): • FR: 123 • FCR: 27 Experimental intervention: lenalidomide 5 mg, orally, days 1 to 21 and titrated to 10 mg if tolerated for 6 months Control intervention: observation after induction either FR or FCR
Outcomes	Primary outcome: 2-year progression-free survival rate Secondary outcomes: • Progression-free survival • Overall survival • Grade 3/4 adverse events • All adverse events • Fetal adverse events • Overall response rate • Complete response rate • PFS rate of patients with Del(11q22.3) [Time Frame: 2 years] • Time-to-progression in patients with Del(11q22.3) • Time-to-progression in patients without Del(11q22.3) Reported and relevant outcomes for this review: nil Final numbers of primary outcome analysis (experimental): • FR+L: 109 • FCR+L: 31 Final numbers of primary outcome analysis (control): • FR: 123 • FCR: 27 Outcome assessment design: intention-to-treat
Notes	Conflict of interest: the authors declare no competing financial interests. Funding: supported by the National Cancer Institute of the National Institutes of Health under award numbers U10CA180821 and U10CA180882 (to the Alliance for Clinical Trials in Oncology), U10CA180833, U10CA180836, U10CA180850, U10CA180867; Canadian Cancer Trials Group: National Clinical Trials Network U10CA180863 and Canadian Cancer Society Research Institute 704970; Eastern Cooperative Oncology Group/American College of Radiology Imaging Network: U10CA180791, U10CA180799, and U10CA180802; Southwest Oncology Group U10CA180828 and U10CA180888. This work was also supported in part by funding from the Four Winds Foundation, the Leukemia & Lymphoma Society, Vysis

Byrd 2018 *(Continued)*

Inc (part of Abbott Molecular), and the National Cancer Institute (P01 CA095426 and R35 CA197734) (J.C.B.).

Summary conclusion: this phase 2 study randomised 342 untreated patients with non-del(11q) CLL requiring therapy of fludarabine plus rituximab (FR; n = 123), FR plus lenalidomide consolidation (FR+L; n = 109), or FR plus cyclophosphamide (FCR; n = 110) and compared 2-year progression-free survival (PFS) rates of each to the historical control rate with FC (60%). Patients with del(11q) in at least 20% of pretreatment cells continued with FCR (n = 27) or were reassigned to FCR+L (n = 31) and excluded from the primary analysis. Among non-del(11q) patients, 2-year PFS rates were 64% (90% confidence interval (CI) 57 to 71; FR), 72% (90% CI 65 to 79; FR+L), and 74% (90% CI 66 to 80; FCR); FR+L and FCR had rates significantly greater than historical control. Median PFS was significantly shorter with FR compared with FR+L (P = 0.04) and FCR (P < 0.001): 43 (95% CI 33 to 50), 61 (95% CI 45 to 71), and 97 (95% CI 61 to not reached) months, respectively. Median follow-up was 73 months, and median overall survival was only reached with FCR (101 months; 95% CI 96 to not reached). With FR+L, the risk of death decreased over time and was lower than with FR at later time points (P = 0.01), but not significantly different from FCR (P = 0.21). Future studies incorporating short courses of lenalidomide into other novel treatment regimens are justified.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote"Within strata defined by Rai stage at diagnosis, patients were then randomized to 1 of 3 treatment regimens: arm A, FR; arm B, FR with consolidative lenalidomide (FR1L); and arm C, FCR" Comments: unkown random sequence generation.
Allocation concealment (selection bias)	Unclear risk	Unkown allocation concealment.
Blinding of participants and personnel (performance bias): Objective outcomes	Low risk	Participants and personnel were not blinded to treatment, no effect on subjective outcomes expected.
Blinding of participants and personnel (performance bias): Subjective outcomes	High risk	This was an open-label trial. The study is at risk of performance bias.
Blinding of outcome assessment (detection bias): Objective outcomes	Low risk	Investigator assessment of total population, subjective outcomes assumed reliable.
Blinding of outcome assessment (detection bias): Subjective outcomes	High risk	This was an open-label trial. No blinding of outcome assessors was reported. Potential risk of detection bias is due to knowledge of allocated interventions by outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	A total of 418 patients enrolled on this study, 400 patients are included in this analysis. lost follow up ratio: 4.3%. Analysis with intention-to-treat. Comments: ITT, relative low lost of follow up rate.
Selective reporting (reporting bias)	Low risk	All outcomes mentioned in the protocol are reported
Other bias	Low risk	No other sources of bias were detected

Chanan-Khan 2017

Study characteristics	
Methods	**CONTINUUM Trial** Study design: double-blind, placebo-controlled, randomised phase 3 trial, international, 111 hospitals, medical centres, and clinics in 21 countries Language of publication: English Major induction regimen: previous treatment at first-line therapy: fludarabine, cyclophosphamide, and rituximab (27.7%), bendamustine and rituximab (1%); previous treatment at second-line therapy: fludarabine, cyclophosphamide, and rituximab (48%), bendamustine and rituximab (22.6%) Randomisation: 2 arms: lenalidomide maintenance vs placebo Recruitment period: 16 February 2009 and 29 September 2015 Median follow-up time: median 31.5 months (IQR 18.9 to 50.8) Sample size calculation: planned enrolment of 320 participants, 160 in each group. 160 events were calculated to be necessary for 80% power to detect a clinically significant difference of 61.3% in median progression-free survival and overall survival between the 2 groups, using a two-sided logrank test with P = 0.025. For the study to be sufficiently powered, α was split between progression-free survival and overall survival, with one--sided α of 0.0125 for both progression-free survival and overall survival.
Participants	Inclusion criteria: eligible patients had received a purine analogue, bendamustine, antiCD20 antibody, chlorambucil, or alemtuzumab as first-line or second-line treatment, were aged 18 years or older, and had an Eastern Cooperative Oncology Group (ECOG) performance score of 0 to 2. Patients were randomised if they had been treated with 2 lines of therapy, achieving at least a partial response to the second-line treatment. Number of included participants (overall): 314 Numbers of SLL/CLL: 314 Study sites: 111 hospitals, medical centres, and clinics in 21 countries Stage/type of disease (experimental): no Rai or Binet stage profile. IGHV mutational status: mutated 23 (14%); unmutated 65 (41%); unable to interpret 54 (34%); unable to interpret because of MRD negativity 12 (8%) Stage/type of disease (control): IGHV mutational status: mutated 25 (16%); unmutated 65 (42%); unable to interpret 50 (32%); unable to interpret because of MRD negativity 8 (5%) Median age (experimental): median 63 (58 to 69) Median age (control): median 63 (58 to 68) Sex of male percentage (experimental): 72% Sex of male percentage (control): 72%
Interventions	Numbers of experimental group (lenalidomide): 160 Numbers of control group (observation): 154 Experimental intervention: lenalidomide 2.5 mg/day orally, on days 1 to 28 of the first 28-day cycle. If lenalidomide was well tolerated, escalation to 5 mg/day was permitted from cycle 2. If the participant completed 5 continuous cycles of 5 mg/day lenalidomide, and this was well tolerated and they had not achieved an MRD-negative complete response, escalation to 10 mg/day was permitted. Treatment continued until disease progression or unacceptable toxic effects. Participants who could not tolerate 2.5 mg/day discontinued treatment and were followed up for survival. Participants could discontinue

Chanan-Khan 2017 *(Continued)*

	study drug because of serious adverse events, disease progression, death, loss to follow-up, withdrawal of consent, or a protocol violation. Control intervention: oral placebo capsules (2.5 mg/day). Placebo doses matched lenalidomide doses.
Outcomes	Primary outcomes: • Progression-free survival • Overall survival Secondary outcomes: • PFS2 (time from randomisation to second objective disease progression, death from any cause, or start of CLL therapy after next-line treatment, whichever occurred first) • Tumour response (improvement in response and duration of response) • Overall adverse events • Grade 3/4 adverse events • Fetal adverse events • Health-related quality of life Reported and relevant outcomes for this review: PFS, OS, grade 3/4 AE, TD Final numbers of primary outcome analysis (experimental): 160 Final numbers of primary outcome analysis (control): 154 Outcome assessment design: intention-to-treat
Notes	Conflict of interest: AAC-K, ME, and JK declare no competing interests. AZ reports consultancy with Novartis, research funding from Bristol-Myers Squibb and Astellas, and is on the speakers bureau of Novartis. SV reports consultancy with Roche and Takeda, and honoraria from Roche and Takeda. SS reports research funding from Celgene Corporation, Bayer Healthcare, and Astellas, and being on the speakers bureau of Celgene Corporation, Sandoz, Novartis, Astellas, Janssen Pharmaceuticals, and Bristol-Myers Squibb. AS reports consultancy with Gilead, Roche, Janssen, Novartis, Celgene Corporation, and AbbVie, research funding from Gilead, and honoraria from Gilead, Roche, Janssen, Novartis, Celgene Corporation, and AbbVie. DS reports research funding from Amgen Pharmaceuticals, and honoraria from Celgene Corporation, Roche, and Janssen. JZ and BP both report employment and equity in Celgene Corporation. RF reports consultancy with Roche, Genentech, Janssen, Gilead, Amgen, Celgene Corporation, and Bristol-Myers Squibb, and is on the speakers bureau of Roche, Janssen, Gilead, Amgen, Celgene Corporation, Pfizer, and Ariad. Funding: funded by Celgene Corporation. Summary conclusion: between 16 February 2009 and 29 September 2015, 314 patients with CLL were enrolled and randomly assigned to receive either lenalidomide (n = 160) or placebo (n = 154). With a median follow-up of 31.5 months (IQR 18.9 to 50.8), there was no significant difference in overall survival between the lenalidomide and the placebo groups (median 70.4 months, 95% CI 57.5 to not estimable (NE) vs NE, 95% confidence interval (CI) 62.8 to NE; hazard ratio (HR) 0.96, 95% CI 0.63 to 1.48; P = 0.86). Progression-free survival was significantly longer in the lenalidomide group (median 33.9 months, 95% CI 25.5 to 52.5) than in the placebo group (9.2 months, 7.4 to 13.6; HR 0.40, 95% CI 0.29 to 0.55; P < 0.001). PFS2 was significantly longer in the lenalidomide group than in the placebo group (median 57.5 months (47.7 to NE) vs 32.7 months (26.4 to 49.0); HR 0.46, 95% CI 0.29 to 0.70; P < 0.01). Improved responses from baseline were observed in 10 (6%) of 160 lenalidomide-treated participants versus 4 (3%) of 154 placebo-treated participants (P = 0.12). Median time to improved response was 12.2 weeks (IQR 7.2 to 22.5) in the lenalidomide group versus 76.3 weeks (20.2 to 182.6) in the placebo group. Duration of improved response was not estimable in either group (95% CI 22.9 to NE in the lenalidomide group vs NE to NE for the placebo group). There were no clinically meaningful differences in HRQoL between lenalidomide-treated participants and placebo-treated participants, as measured by FACT-Leu and EQ-5D, during maintenance treatment. In the safety population, the most common grade 3 or 4 adverse events included neutropenia (94 (60%) of 157 participants in the lenalidomide group vs 35 (23%) of 154 participants in the placebo group), thrombocytopenia (26 (17%) vs 10

Chanan-Khan 2017 *(Continued)*

	(6%)), and diarrhoea (13 (8%) vs 1 (< 1%)). There were 5 fatal adverse events (3 (2%) participants in the lenalidomide group and 2 (1%) participants in the placebo group).

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Subjects were randomized using interactive voice-response system
Allocation concealment (selection bias)	Low risk	Subjects were randomized using interactive voice-response system
Blinding of participants and personnel (performance bias): Objective outcomes	Low risk	Both lenalidomide and placebo were identical in appearance.
Blinding of participants and personnel (performance bias): Subjective outcomes	Low risk	Both lenalidomide and placebo were identical in appearance.
Blinding of outcome assessment (detection bias): Objective outcomes	Low risk	All study participants, including patients, investigators, and those completing data analysis, were masked to the treatment assignments.
Blinding of outcome assessment (detection bias): Subjective outcomes	Low risk	All study participants, including patients, investigators, and those completing data analysis, were masked to the treatment assignments.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote:"All 314 randomized patients represented the intent-to-treat population"
Selective reporting (reporting bias)	Low risk	All outcomes mentioned in the protocol are reported
Other bias	Low risk	No other sources of bias were detected

Dartigeas 2018

Study characteristics

Methods	**CLL 2007 SA Trial** Study design: open-label, randomised, parallel-group phase 3 study, multicentre, 89 centres in France Language of publication: English Major induction regimen: induction of 4 monthly courses of full-dose FCR with 2 interim rituximab doses on day 14 of cycles 1 and 2. This oral fludarabine plus cyclophosphamide regimen combined with intravenous rituximab has been fully described (oral fludarabine (40 mg/m^2 per day) and oral cyclophosphamide (250 mg/m^2 per day) for the first 3 days of each cycle, rituximab at 375 mg/m^2 intravenously on day 0 of cycle 1 and subsequently at 500 mg/m^2 on day 14 of cycle 1, days 1 and 14 of cycle 2, and day 1 of cycles 3 and 4).

Dartigeas 2018 *(Continued)*

	Randomisation: 2 arms: rituximab vs observation Recruitment period: 14 December 2007 and 18 February 2014 Median follow-up time: median 47.7 months (30.4 to 65.8) Sample size calculation: accordingly, it was calculated that 406 participants needed to be randomised. Thus, assuming a dropout rate of 25% during induction, the enrolment of 542 participants was required. The study was designed to detect a hazard ratio (HR) of 0.6 for the rituximab group compared with the observation group, corresponding to a 32% relative improvement in 3-year progression-free survival (66% with rituximab maintenance treatment vs 50% for observation), with 90% power and a two-sided α error of 0.05. One interim analysis was planned after observing 121 (75%) of the required 161 progression-free survival events. To maintain an overall two-sided α error rate of 0.05 using the O'Brien and Fleming method, the interim analysis was done at a two-sided α error of 0.0052 and the final analysis at a two-sided α error of 0.048. The interim analysis was done after 150 events, and the prespecified statistical boundary was crossed (P = 0.0009 for the primary endpoint, progression-free survival).
Participants	Inclusion criteria: eligible participants were fit patients aged 65 years or older, treatment-naive, and with active Binet stage B or C chronic lymphocytic leukaemia requiring treatment and completed 4 monthly FCR induction therapy Number of included participants (overall): 409 Numbers of SLL/CLL: 409 Study sites: 89 centres in France Stage/type of disease (experimental): Binet B: 65%; Binet C: 20.6%; Rai stage: III/IV: 40% Stage/type of disease (control): Binet B: 67%; Binet C 33%; Rai stage III/IV: 35% Total cytogenetic aberrations rate: 71.9% Median age (experimental): median 71.7 (68.5 to 75.9) Median age (control): median 71.1 (67.4 to 74.8) Sex of male percentage (experimental): 62% Sex of male percentage (control): 70%
Interventions	Numbers of experimental group (rituximab): 202 Numbers of control group (observation): 207 Experimental intervention: rituximab, 500 mg/m^2, intravenously, every 8 weeks for 2 years Control intervention: observation every 8 weeks for 2 years
Outcomes	Primary outcome: progression-free survival Secondary outcomes: • Event-free survival • Disease-free survival • Overall survival • Time to next treatment • Overall response rate • Overall adverse events • Grade 3/4 adverse events • Fetal adverse events • Secondary cancers

Dartigeas 2018 *(Continued)*

	• Phenotypic response rate • Rates of treatment-related adverse events • Pharmacokinetics of rituximab • Quality of life Reported and relevant outcomes for this review: PFS, OS, TRM, CRR, grade 3/4 AE, TD Final numbers of primary outcome analysis (experimental): 202 Final numbers of primary outcome analysis (control): 207 Outcome assessment design: intention-to-treat
Notes	Conflict of interest: CD received honoraria from Roche and Janssen for consultancy, travel support from Mundipharma, Gilead, AbbVie, and Roche, and declares that the University Hospital of Tours received research funds from the French National Cancer Institute INCa for this clinical trial. EVDN received honoraria from Celgene and Servier for speaker's bureau and travel grants from Roche. M-SD reports personal fees and non-financial support from Janssen, AbbVie, and Gilead, outside the submitted work. MCB received honoraria from Bristol-Myers Squibb and Janssen, research funding from Beckman Coulter, and support for congress visit from Chugai, outside the submitted work. FC received personal fees from Gilead, Janssen, and AbbVie, non-financial support from Gilead and Roche, and grants from Janssen, outside the submitted work. OT received grants and personal fees from Roche, Janssen, and AbbVie and grants from Gilead, outside the submitted work. KL declares grants and personal fees from Takeda, personal fees from Amgen, and received grants from Mundipharma, Hospira, Teva, and Roche, outside the submitted work. AD received honoraria from Janssen, Gilead, AbbVie, and Roche, served on advisory boards of Gilead, AbbVie, and Roche, and received travel support from Novartis. PF served as a consultant for Roche. VLév received personal fees from AbbVie, Janssen, Gilead, and Roche. VLeb received personal fees and non-financial support from Roche, Janssen, and Gilead and personal fees from Servier and AbbVie. JL, HM, CB, SDG, SL, FN-K, RL, PR, TA-S, J-PV, BM, A-SM, RD, and PC declare no competing interests. Funding: funded by the French National Cancer Institute INCa (Hospital Clinical Research Programs, application number INCa 10-05 and INCa 11-200), Roche Pharma AG, and Chugai. Summary conclusion: between 14 December 2007 and 18 February 2014, 542 participants were enrolled, of whom 525 started FCR induction. Between 10 June 2008 and 14 August 2014, 409 (78%) participants were randomly assigned to rituximab maintenance (n = 202) or observation (n = 207). 4 (2%) participants in the rituximab group did not receive the allocated treatment (progressive disease (n = 1), adverse events (n = 3)). After a median follow-up of 47.7 months (IQR 30.4 to 65.8), median progression-free survival in the rituximab group (59.3 months, 95% confidence interval (CI) 49.6 to not estimable) was improved compared with the observation group (49.0 months, 39.9 to 60.5; hazard ratio 0.55, 95% CI 0.40 to 0.75; P = 0.0002). Neutropenia and grade 3 to 4 infections were more common with rituximab maintenance (105 (53%) of 198 participants vs 74 (36%) of 207 participants and 38 (19%) vs 21 (10%), respectively) during the study. The most common grade 3 to 4 infection was lower respiratory tract infection (24 (12%) vs 8 (4%)). The incidence of second cancers, except basal cell carcinoma, was similar in both groups (29 (15%) vs 23 (11%)). Deaths were related to adverse events for 23 (11%) participants in the rituximab group and 16 (8%) in the observation group.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Subjects were randomized using computer-generated randomisation list, balanced by the use of randomly permuted blocks of variable sizes and was stratified according to del(11q), IGHV mutational status and response to FCR treatment
Allocation concealment (selection bias)	Low risk	Quote" centrally allocated registration code and communicated by the FILO central study office. Neither investigators nor patients were masked to group allocation. Crossover between the study groups was not permitted."

Dartigeas 2018 *(Continued)*

Blinding of participants and personnel (performance bias): Objective outcomes	Low risk	Participants and personnel were not blinded to treatment, no effect on subjective outcomes expected.
Blinding of participants and personnel (performance bias): Subjective outcomes	High risk	This was an open label trial. The study is at risk of performance bias.
Blinding of outcome assessment (detection bias): Objective outcomes	Low risk	Investigator assessment of total population, subjective outcomes assumed reliable
Blinding of outcome assessment (detection bias): Subjective outcomes	High risk	Quote: "Response and disease progression were assessed in accordance with the International Workshop on Chronic Lymphocytic Leukaemia updated National Cancer Institute-sponsored Working Group guidelines and confirmed by a central medical review group that was masked to treatment group assignment." Comments: This was an open-label trial. No blinding of outcome assessors was reported. Potential risk of detection bias in all objective outcomes apart from response rate and progression free survival (centrally reviewed) is due to knowledge of allocated interventions by outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Intention to treat analysis for efficacy evaluation.Per protocol analysis for safety outcome.
Selective reporting (reporting bias)	Low risk	All outcomes mentioned in the protocol are reported
Other bias	Low risk	No other sources of bias were detected

Fink 2017

Study characteristics	
Methods	**CLLM1 Trial** Study design: double-blind, placebo-controlled, randomised parallel-group phase 3 study, international, 62 centres in 5 countries Language of publication: English Major induction regimen: first-line treatment: fludarabine, cyclophosphamide, rituximab (38.2%), bendamustine plus rituximab (60.7%), fludarabine plus cyclophosphamide (1.1%) Randomisation: 2-arm study: lenalidomide maintenance vs placebo Recruitment period: 5 July 2012 and 15 March 2016 Median follow-up time: median 17.9 months (IQR 9.1 to 28.1) Sample size calculation: data published before the initiation of the study indicated that median progression-free survival could be expected at 22.4 months in the placebo group; the intervention was assumed to lead to an improvement of the median progression-free survival by 75%, which should result in a median progression-free survival of 39.2 months and a corresponding hazard ratio of 0.571. 118

Fink 2017 *(Continued)*

	progression-free survival events were required to have 80% power at a 5% significance level yielding 200 participants to be recruited. In the course of the study, it was realised that the recruitment goal of 200 participants might not be reached. Therefore, 2 interim analyses allowing stopping for efficacy or futility (non-binding) after 24 (20%) and 48 (41%) progression-free survival events were implemented subsequently using a group sequential testing design. The corresponding P values for early stopping for futility and efficacy were 0.1889 and 0.0006 as lower and upper boundaries, respectively.
Participants	Inclusion criteria: patients older than 18 years and diagnosed with immunophenotypically confirmed CLL with active disease, who responded to chemoimmunotherapy 2 to 5 months after completion of first-line therapy and who were assessed as having a high risk for an early progression, defined by minimal residual disease levels of 10^{-2} or higher or minimal residual disease levels of 10^{-4} to less than 10^{-2} after completion of first-line treatment combined with either an unmutated IGHV gene status, del(17p) or *TP53* mutation at baseline, were eligible. Number of included participants (overall): 89 Numbers of SLL/CLL: 89 Study sites: 62 centres across 5 countries (Austria, Germany, Italy, the Netherlands, and Spain) Stage/type of disease (experimental): No Rai or Binet stage profile Total cytogenetic aberrations rate: 74.2% Cytogenetic abnormalities • del(17p): 7 (14%) • del(11q): 16 (31%) • *TP53* mutational status (n = 83): 10/56 (18%) • IGHV unmutated (n = 81): 50/55 (91%) • Complex karyotype (n = 43): 5/29 (17%) Stage/type of disease (control): • del(17p): 2 (8%) • del(11q): 7 (27%) • *TP53* mutational status (n = 83): 7/27 (26%) • IGHV unmutated (n = 81): 24/26 (92%) • Complex karyotype (n = 43): 4/14 (29%) Median age (experimental): median 64 (57.3 to 69.8) Median age (control): median 64 (58.0 to 69.5) Sex of male percentage (experimental): 88% Sex of male percentage (control): 79%
Interventions	Numbers of experimental group (lenalidomide): 60 Numbers of control group (observation): 29 Experimental intervention: lenalidomide starting with 5 mg daily in the first 28-day cycle. If the 5 mg dose level was well tolerated, dose was escalated to 10 mg daily in each 28-day cycle in cycles 2 to 6; the target dose of 15 mg daily was given starting with the seventh cycle up to progression of disease. Further escalations (to 20 mg starting with the 13th cycle and to 25 mg starting with the 19th cycle, respectively) were guided by minimal residual disease assessments and were allowed in participants with minimal residual disease levels of 10^{-4} or higher in peripheral blood that tolerated previous dose levels. The maximal daily dose of lenalidomide was 25 mg.

Fink 2017 *(Continued)*

	Control intervention: placebo administration as lenalidomide dosage
Outcomes	Primary outcome: progression-free survival based on independent review committee Secondary outcomes: • Progression-free survival based on investigator's assessment • Progression-free survival based on investigator's assessment • Time to next treatment censoring progression • Event-free survival • Overall survival • Grade 3/4 adverse events • Overall adverse events • Fetal adverse events • Treatment discontinuation • MRD levels per central lab (evaluation by flow cytometry and comparison of MRD levels immediately after the completion of first-line therapy to levels 6, 12, 18, and 24 months and subsequently annually) • Health-related quality of life by EORTC QLQ-C30 and EQ-5D • Treatment-free survival after second-line treatment Reported and relevant outcomes for this review: PFS, OS, grade 3/4 AE, TD Final numbers of primary outcome analysis (experimental): 60 Final numbers of primary outcome analysis (control): 29 Outcome assessment design: intention-to-treat
Notes	Conflict of interest: AMF received research grants and travel grants from Celgene and grants and personal fees from Hoffmann-La Roche, Mundipharma, and AbbVie. OA-S received personal fees from Gilead, Roche, and AbbVie. KF received grants from Hoffmann-La Roche. PG received grants and personal fees from AbbVie, Janssen, and Gilead; personal fees from Roche; and grants from Novartis and Celgene. FB received personal fees from Celgene. APK received grants from Celgene. MK received honoraria for participating in advisory boards and research funding from Celgene and Roche. ET received non-financial support from Celgene, GSK, and Amgen; personal fees and non-financial support from Gilead; and grants from Novartis. SS received grants and personal fees from Celgene. MR received grants from the University of Cologne; grants and research support from Hoffmann-La Roche, Celgene, and AbbVie; and consultancy from BMS and Gilead. SB received grants from Celgene and grants and personal fees from AbbVie and Roche. BE received grants and personal fees from Celgene, Roche, Janssen, AbbVie, and Gilead. MH received research funding from Celgene. JB, SR, AA, HH, KJ-U, SD, C-MW, TN, HD, and K-AK declare no competing interests. Funding: sponsor University of Cologne Summary conclusion: between 5 July 2012 and 15 March 2016, 468 previously untreated patients with CLL were screened for the study, of which 379 (81%) were found not to be eligible. Recruitment was closed prematurely due to poor accrual after 89 of 200 planned patients were randomly assigned: 60 (67%) enrolled patients were assigned to the lenalidomide group and 29 (33%) to the placebo group, of whom 56 (63%) received lenalidomide and 29 (33%) placebo, with a median of 11.0 (IQR 4.5 to 20.5) treatment cycles at data cutoff. After a median observation time of 17.9 months (IQR 9.1 to 28.1), the hazard ratio for progression-free survival assessed by an independent review was 0.168 (95% confidence interval (CI) 0.074 to 0.379). Median progression-free survival was 13.3 months (95% CI 9.9 to 19.7) in the placebo group and not reached (95% CI 32.3 to not evaluable) in the lenalidomide group. The most frequent adverse events were skin disorders (35 participants (63%) in the lenalidomide group vs 8 participants (28%) in the placebo group), gastrointestinal disorders (34 (61%) vs 8 (28%)), infections (30 (54%) vs 19 (66%)), haematological toxicity (28 (50%) vs 5 (17%)), and general disorders (28 (50%) vs 9 (31%)). 1 fatal adverse event was reported in each of the treatment groups (1 participant (2%) with fatal acute lymphocytic leukaemia in the lenalidomide group and 1 participant (3%) with fatal multifocal leukoencephalopathy in the placebo group).

Fink 2017 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Subjects were randomized using a computer-generated randomisation schedule prepared by ICON
Allocation concealment (selection bias)	Low risk	Subjects were randomized using a unique treatment code was sent automatically by the IWRS to the investigators
Blinding of participants and personnel (performance bias): Objective outcomes	Low risk	placebo capsules that were identical in appearance were provided.
Blinding of participants and personnel (performance bias): Subjective outcomes	Low risk	placebo capsules that were identical in appearance were provided.
Blinding of outcome assessment (detection bias): Objective outcomes	Low risk	Results based on the assessment of an independent review committee
Blinding of outcome assessment (detection bias): Subjective outcomes	Low risk	Results based on the assessment of an independent review committee
Incomplete outcome data (attrition bias) All outcomes	Low risk	89 participants were randomly assigned to both study arms. In study analysis only 81 participants.
Selective reporting (reporting bias)	Low risk	All outcomes mentioned in the protocol are reported
Other bias	Unclear risk	Quote"118 progression-free survival events were required to have 80% power at a 5% significance level yielding 200 patients to be recruited" and "Recruitment was closed prematurely due to poor accrual after 89 of 200 planned patients were randomly assigned" Comment: terminated early due to poor recruitment.

Foa 2014

Study characteristics

Methods	Study design: open-label, randomised phase 2 study, multicentre, 19 centres in Italy Language of publication: English Major induction regimen: induction treatment of eight 28-day cycles of chlorambucil (8 mg/m^2/day, days 1 to 7) and rituximab (day 1 of cycle 3, 375 mg/m^2; cycles 4 to 8, 500 mg/m^2). Randomisation: 2 arms: rituximab maintenance versus observation Recruitment period: October 2008 and January 2013

Foa 2014 *(Continued)*

	Median follow-up time: median 34.9 months Sample size calculation: based on an a-level of 0.05 to show a significant difference with respect to the primary endpoint of OR rate after the induction phase, the planned sample size was estimated from 72 to 94 participants to obtain a power of at least 80% and a maximum of 90%. The calculation of the sample size was based on the following assumptions: • 0.65 OR proportion in null hypothesis; • 0.80 OR proportion expected under alternative hypothesis. Considering 20% of non-evaluable participants, from 90 to 118 participants should have been included in the study.
Participants	Inclusion criteria: patients aged > 65 years (or 60 to 65 years not eligible for fludarabine-based regimens) with previously untreated CLL (Binet stage A or B with active disease or stage C) received induction therapy of chlorambucil plus rituximab achieved complete response, complete response with incomplete bone marrow recovery, and partial response were randomised to rituximab maintenance and observation group. Number of included participants (overall): 66 Numbers of SLL/CLL: 66 Study sites: multicentre, 19 centres in Italy Stage/type of disease (experimental): Binet B: 61.8%; Binet C: 20.6% Stage/type of disease (control): Binet B: 53.1%; Binet C: 9.4% Total cytogenetic aberrations rate: 74.7% Median age (experimental): mean 69.8 (SD: 5.4) Median age (control): mean 70.2 (SD: 5.1) Sex of male percentage (experimental): 70.60% Sex of male percentage (control): 71.90%
Interventions	Numbers of experimental group (rituximab): 34 Numbers of control group (observation): 32 Experimental intervention: rituximab 375 mg/m^2, intravenously, every 8 weeks for 12 doses Control intervention: observation
Outcomes	Primary outcome: progression-free survival Secondary outcomes: • Overall adverse events • Grade 3/4 adverse events • Overall survival • 3-year progression-free survival • Conversion rate • Event-free survival • Time to new CLL treatment Reported and relevant outcomes for this review: CRR, ORR, grade 3/4 AE, TD Final numbers of primary outcome analysis (experimental): 34

Foa 2014 *(Continued)*

	Final numbers of primary outcome analysis (control): 32 Outcome assessment design: intention-to-treat
Notes	Conflict of interest: RF received compensation from Roche and Celgene as expert testimony and by Roche, BMS, Janssen, and ARIAD as member of advisory boards. AC received honoraria from Roche. AA is employed by Roche as Hematology Medical Manager. EJR is employed by Roche as Local Safety Responsible. EG is employed by Roche as Medical Manager. All other authors declare no potential conflict of interest. Funding: Hoffmann-La Roche Summary conclusion: rituximab maintenance tended towards a better PFS than observation and was safe and most beneficial for participants in partial response and for unmutated IGHV cases. Moreover, the study authors suggest that rituximab maintenance is doable in elderly CLL patients and tends to improve PFS in responders to a CLB-R induction, not increasing toxicity compared to observation in terms of AE, SAE, neutropenia, or infections. It seems beneficial mostly in PR/nPR patients after induction in terms of response improvement and in unmutated IGHV patients.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unkown allocation sequence generation and randomization.
Allocation concealment (selection bias)	Unclear risk	Unkown allocation sequence concealment.
Blinding of participants and personnel (performance bias): Objective outcomes	Low risk	Participants and personnel were not blinded to treatment, no effect on subjective outcomes expected.
Blinding of participants and personnel (performance bias): Subjective outcomes	High risk	This was an open-label trial. The study is at risk of performance bias
Blinding of outcome assessment (detection bias): Objective outcomes	Low risk	Investigator assessment of total population, subjective outcomes assumed reliable.
Blinding of outcome assessment (detection bias): Subjective outcomes	High risk	This was an open-label trial. No blinding of outcome assessors was reported. Potential risk of detection bias in all objective outcomes is due to knowledge of allocated interventions by outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote:"Ninety-seven patients entered the induction phase (safety population), and 85 received 1 dose of R (intention-to-treat, ITT population). Secondary randomized to rituximab maintenance therapy (66 patients, analysis with ITT)" Comment: clear CONSORT flow diagram, all randomised participants included in ITT analysis. Attrition bias likely to be low
Selective reporting (reporting bias)	Low risk	All outcomes mentioned in the protocol are reported
Other bias	High risk	There was an imbalance in the number of participants with a history of disease severity between treatment arms. 20.6% (7/34) of participants had Binet stage

Foa 2014 *(Continued)*

	C in the rituximab arm, whereas only 9.4% (3/32) had Binet stage C in the observation arm. Comments: imbalance of basic characteristics, at high risk of bias

Greil 2016

Study characteristics	
Methods	**AGMT CLL-8a Mabtenance randomised phase 3 trial** Study design: open-label, randomised phase 3 trial, international, 29 centres in 5 countries Language of publication: English Major induction regimen: induction therapy of fludarabine, cyclophosphamide, and rituximab (73.4%) and bendamustine and rituximab (20%) Randomisation: 2 arms: rituximab maintenance versus observation Recruitment period: 1 April 2010 and 23 December 2013 Median follow-up time: median 33.7 months Sample size calculation: a hazard ratio (HR) of 0.55 was estimated from published maintenance data in follicular lymphoma. Thus, the study authors expected to observe progression-free survival in 72% of participants (rituximab) compared with 55% of participants (observation) after 3 years, corresponding to an HR of 0.55 (assuming exponential distribution). With a recruitment ratio of 1:1, a two-sided significance level of 5%, and an estimated accrual rate of 128 participants per year, the study authors estimated that a maximum number of 92 events would be required to achieve 80% power with 1 interim analysis after the occurrence of 50% of events.
Participants	Inclusion criteria: CLL patients aged 18 years or older; Eastern Cooperative Oncology Group performance score (ECOG PS) of 0 to 2; complete response (CR), complete response with incomplete bone marrow recovery (CRi), or partial response, following previous first-line or second-line rituximab-containing chemo-immunotherapy Number of included participants (overall): 263 Numbers of SLL/CLL: 263 Study sites: 29 centres in Austria, the Czech Republic, Slovakia, Bulgaria, and Romania Stage/type of disease (experimental): Rai stage: III/IV 38%; cytogenetic aberrations: 74% Stage/type of disease (control): Rai stage: III/IV 37%; cytogenetic aberrations: 81% Median age (experimental): median 63 (55 to 69) Median age (control): median 63 (58 to 71) Sex of male percentage (experimental): 75% Sex of male percentage (control): 67%
Interventions	Numbers of experimental group (rituximab): 134 Numbers of control group (observation): 129 Experimental intervention: rituximab 375 mg/m^2, intravenously, 3-monthly intervals, for a period of 2 years or until disease progression

Greil 2016 *(Continued)*

	Control intervention: observation at 3-monthly intervals, for a period of 2 years or until disease progression
Outcomes	Primary outcomes: progression-free survival Secondary outcomes: • Time to next treatment • Grade 3/4 adverse events • Overall survival • Event-free survival • Treatment discontinuation • Conversion rate to CR or CRi and to MRD negativity; median MRD levels in peripheral blood • The effect of baseline risk factors on treatment outcome • Influence of depth of clinical remission and MRD status after induction therapy on clinical progression-free and overall survival • Benefit of rituximab treatment according to cytogenetic risk group, IGVH mutation status, and CD38 expression Reported and relevant outcomes for this review: PFS, OS, TRM, CRR, grade 3/4 AE, TD Final numbers of primary outcome analysis (experimental): 134 Final numbers of primary outcome analysis (control): 129 Outcome assessment design: intention-to-treat
Notes	**Egle 2019 reported "Subgroup analysis of AGMT CLL-8a mabtenance randomized phase III trial"** Conflict of interest: SB reports support from Roche. AE reports grants and personal fees from Roche. MF reports personal fees from Amgen, BMS, Boehringer Ingelheim, Gilead, Janssen, Novartis, Roche, and AstraZeneca. RG reports grants and personal fees from Roche and Takeda, grants from Celgene and Novartis, and personal fees from BMS and Amgen. MH reports personal fees from the Czech Lymphoma Study Group, and other support from Roche. TK reports other from Roche. UJ reports grants and personal fees from Roche. AK reports personal fees from AGMT, Hoffmann-La Roche, the Central European Society for Anticancer Drug Research, and Arbeitsgemeinschaft Internistische Onkologie. JM reports other support from Roche, GlaxoSmithKline, and Novartis. HO reports personal fees from Sanofi-Aventis, Amgen, and Celgene. PO reports grants from Roche, non-financial support from Janssen and Roche, and personal fees from GlaxoSmithKline and Gilead. TP reports personal fees from Roche. AP reports personal fees from Roche. LS reports personal fees from AbbVie, and personal fees and non-financial support from Gilead, Janssen, and Roche. MS reports grants and personal fees from Roche. JA, PA, CB, MD, EF, LG, MG, EK, ML, EM, GM, TM, TN, LP, MP, SP, JR, ES, OS, and AZ declare no competing interests. Funding: supported by Arbeitsgemeinschaft Medikamentöse Tumortherapie gemeinnützige GmbH (AGMT), Roche. Summary conclusion: median observation times were 33.4 months (IQR 25.7 to 42.8) for the rituximab group and 34.0 months (25.4 to 41.9) for the observation group. Progression-free survival was significantly longer in the rituximab maintenance group (47.0 months, IQR 28.5 to incalculable) than with observation alone (35.5 months, 95% CI 25.7 to 46.3; hazard ratio (HR) 0.50, 95% confidence interval (CI) 0.33 to 0.75, P = 0.00077). The incidence of grade 3/4 haematological toxicities other than neutropenia was similar between groups. Grade 3/4 neutropenia occurred in 28 (21%) participants in the rituximab group and 14 (11%) participants in the observation group. Apart from neutropenia, the most common grade 3/4 adverse events were upper (5 (4%) participants in the rituximab group vs 1 (1%) participant in the observation group) and lower (3 (2%) vs 1 (1%)) respiratory tract infection, pneumonia (9 (7%) vs 2 (2%)), thrombocytopenia (4 (3%) vs 4 (3%)), neoplasms (5 (4%) vs 4 (3%)), and eye disorders (4 (3%) vs 2 (2%)). The overall incidence of infections of all grades was higher among rituximab recipients (88 (66%) vs 65 (50%)).

Risk of bias

Greil 2016 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "patients were block randomised in a 1:1 ratio centrally in the electronic case report form system into the two treatment groups, and stratification applied" Comment:low risk of random sequence generation
Allocation concealment (selection bias)	Unclear risk	Quote"patients were block randomised in a 1:1 ratio centrally in the electronic case report form system" and "No masking was performed in this open-label trial." Comment:Unkown allocation sequence concealment.
Blinding of participants and personnel (performance bias): Objective outcomes	Low risk	Participants and personnel were not blinded to treatment, no effect on subjective outcomes expected
Blinding of participants and personnel (performance bias): Subjective outcomes	High risk	This was an open-label trial. The study is at risk of performance bias.
Blinding of outcome assessment (detection bias): Objective outcomes	Low risk	subjective outcomes outcome assumed reliable
Blinding of outcome assessment (detection bias): Subjective outcomes	High risk	This was an open-label trial. No blinding of outcome assessors was reported. Potential risk of detection bias in all objective outcomes is due to knowledge of allocated interventions by outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All analyses were done in the intention-totreat (ITT) population (ie, all participants who were enrolled and randomly assigned).
Selective reporting (reporting bias)	Low risk	All outcomes mentioned in the protocol are reported
Other bias	Low risk	No other sources of bias were detected

Osterborg 2016

Study characteristics	
Methods	Study design: international, multicentre, 80 centres in 19 countries, open-label, randomised phase 3 trial Language of publication: English Major induction regimen: induction with ofatumumab treatment consisted of 8 weekly infusions followed by 4 monthly infusions (dose 1: 300 mg; dose 2 to 12: 2000 mg) Randomisation: 2 arms: ofatumumab versus observation Recruitment period: 10 March 2011 and 14 April 2011 Median follow-up time: not available

Osterborg 2016 *(Continued)*

	Sample size calculation: a total of 120 participants were planned for enrolment to observe 95 events of progressive disease or death. Approximately 95 events were needed to achieve at least 90% power to demonstrate a treatment-arm difference in median PFS of 3 months in the primary comparison (ofatumumab versus physicians' choice treatment arm, 6 months versus 3 months) at a 2-sided alpha level of 5%.
Participants	Inclusion criteria: patients with bulky fludarabine-refractory CLL received induction with ofatumumab intravenously, initial dose 300 mg, followed 1 week later with 2000 mg once weekly for 7 weeks, followed 4 weeks later by 1 infusion of 2000 mg every 4 weeks for 4 infusions, for a total of 12 infusions over 24 weeks. After 24 weeks of ofatumumab treatment, participants who have achieved at least stable disease or better, and whom the investigator would deem appropriate for the therapy to continue, would undergo a second randomisation (2:1) to either 1) an additional ofatumumab dose regimen of 2000 mg once every 4 weeks for up to an additional 24 weeks, or 2) no further therapy. Number of included participants (overall): 37 (122 first randomisation, then 79 in ofatumumab arm, then only 37 in ofatumumab maintenance secondary randomisation) Numbers of SLL/CLL: 37 Study sites: international, 19 countries Stage/type of disease (overall): Binet B: 43%; Binet C: 48% Median age (overall): median 61.5 (46 to 82) Sex of male percentage (overall): 70%
Interventions	Numbers of experimental group (ofatumumab): 24 Numbers of control group (observation): 13 Experimental intervention: ofatumumab 2000 mg, intravenously, once every 4 weeks for up to an additional 24 weeks Control intervention: observation
Outcomes	Primary outcome: progression-free survival Secondary outcomes: • Overall survival • Overall response rate • Grade 3 and 4 adverse events • All adverse events • Time to next treatment Reported and relevant outcomes for this review: TD Final numbers of primary outcome analysis (experimental): 24 Final numbers of primary outcome analysis (control): 13 Outcome assessment design: intention-to-treat
Notes	Conflict of interest: nil Funding: sponsored by GlaxoSmithKline and Genmab; ofatumumab is an asset of Novartis AG as of 2 March 2015. GlaxoSmithKline and Genmab provided the drug and worked closely together with the investigators in the development of study design and interpretation of the data. GlaxoSmithKline and Genmab funded the study and were also responsible for collection and analysis of the data. Of the sponsor, only Chai-Ni Chang had access to the raw data after the official treatment unblinding.

Osterborg 2016 *(Continued)*

	Summary conclusion: extending ofatumumab treatment was associated with an improvement in PFS from time of second randomisation (n = 24; median 5.6 months) compared with observation (n = 13; median 3.5 months; P = 0.026). Time to next treatment was also significantly longer in the ofatumumab extended treatment arm than in the observation arm (median 9.2 versus 2.8 months, P = 0.002). The median OS times from second randomisation were 25.4 months and 17.8 months, respectively (P = 0.65). PFS and time to next therapy were significantly longer with ofatumumab extended treatment than observation (P = 0.026 and P = 0.002, respectively; n = 37). The adverse event profile of long-term ofatumumab administration showed no unexpected findings.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote"Patients in the ofatumumab arm who did not have disease progression after 6 months of treat- ment, as assessed by the investigator, were eligible for a second 2:1 randomization to receive either six additional infusions (2000 mg every 4 weeks) (ofatumumab extended) or no further treatment (observation). Patients in the PC arm who developed progressive disease (PD) could receive ofatumumab salvage therapy for up to 12 months." Comment: Unkown random sequence generation and randomization.
Allocation concealment (selection bias)	Unclear risk	Unkown allocation sequence concealment.
Blinding of participants and personnel (performance bias): Objective outcomes	Low risk	Participants and personnel were not blinded to treatment, no effect on subjective outcomes expected.
Blinding of participants and personnel (performance bias): Subjective outcomes	High risk	This was an open-label trial. The study is at risk of performance bias.
Blinding of outcome assessment (detection bias): Objective outcomes	Low risk	Investigator assessment of total population, subjective outcomes assumed reliable.
Blinding of outcome assessment (detection bias): Subjective outcomes	High risk	Quote: "Disease progression was evaluate through Independent Review Committee (IRC)." Comment:This was an open-label trial. No blinding of outcome assessors was reported. Potential risk of detection bias in all objective outcomes apart from response rate and progression free survival (reviewed by IRC) is due to knowledge of allocated interventions by outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	Enrolled 66 participants, but only 39 participants were included in the final analysis; and relative higher rate of treatment discontinuation, 12.5% (3/24) at ofatumumab arm and 7.6%(1/13) at observation arm.
Selective reporting (reporting bias)	High risk	Duration of Response, AE of interest (infusion reactions, infections, mucocutaneous reactions, Tumour Lysis Syndrome (TLS), cardiovascular events, and small bowel obstruction), mean Immunoglobulin (Ig) Antibodies IgA, IgG, and IgM Over Time, Number of Participants Who Were Positive or Negative for Human Anti-Human Antibodies (HAHA) Post-OFA Therapy was not reported
Other bias	Unclear risk	detailed baseline characteristics of ofatuzumab and observation arms were not reported

Robak 2018

Study characteristics	
Methods	**PALG – CLL4 (ML21283) trial** Study design: international, multicentre, open-label, randomised phase 3b trial Language of publication: English Major induction regimen: induction treatment consisted of 6 cycles of an RCC regimen administered every 4 weeks. In cycle 1, participants received rituximab: 375 mg/m^2 IV on day 1, cladribine: 0.12 mg/kg/day IV days 2 to 4, cyclophosphamide: 250 mg/m^2/day (IV, over 15 to 30 min) on days 2 to 4. In cycles 2 to 6, participants were treated with rituximab: 500 mg/m^2 IV on day 1, cladribine: 0.12 mg/kg/day IV days 2 to 4, cyclophosphamide: 250 mg/m^2/day IV, over 15 to 30 min, days 2 to 4. Randomisation: 2 arms: rituximab versus observation Recruitment period: July 2009 and December 2011 Median follow-up time: median 14.4 months (IQR 8.7 to 19.8) Sample size calculation: under the assumptions made, 63 events were expected if the 148 participants included in the maintenance/observation period (74 per year over 2 years) were followed for 3.5 years (total time, maintenance/observation + follow-up 5.5 years). A dropout rate of 10% was assumed both during induction therapy and maintenance/observation period, a 90% OR rate (CR, PR) after induction therapy and 10% withdrawal for CR/PR patients. Based on these assumptions, 228 participants should have been recruited for induction therapy; 205 (90%) of them should be available for treatment assessment after induction therapy; 184 were expected to achieve CR/PR (90%); and 90% of these participants (165 participants) should have been eligible for randomisation.
Participants	Inclusion criteria: previously untreated, progressive CLL with immunologically confirmed diagnosis (CD5, CD19, CD23, and CD20-positive clone), at stages I-IV according to the Rai classification; achieved with a partial response (PR) or complete response (CR) after 6 cycles of RCC induction treatment. Number of included participants (overall): 66 Numbers of SLL/CLL: 66 Study sites: Europe, Poland, Belarus Stage/type of disease (experimental): Rai stage III/IV: 24.3%; cytogenetic aberrations: 69.7% Stage/type of disease (control): Rai stage III/IV: 18.2%; cytogenetic aberrations: 60.6% Median age (experimental): median 57.8 (31.2 to 69.2) Median age (control): median 57.3 (40.5 to 70.5) Sex of male percentage (experimental): 66.70% Sex of male percentage (control): 60.60%
Interventions	Numbers of experimental group (rituximab): 33 Numbers of control group (observation): 33 Experimental intervention: rituximab 375 mg/m^2, administered intravenously for 8 cycles every 12 weeks Control intervention: observation
Outcomes	Primary outcome: progression-free survival

Robak 2018 *(Continued)*

	Secondary outcomes: • Complete response rate • Partial response rate • Overall response rate • Grade 3 and 4 adverse events • All adverse events • Minimal residual disease rate Reported and relevant outcomes for this review: PFS, CRR, ORR, grade 3/4 AE, TD Final numbers of primary outcome analysis (experimental): 33 Final numbers of primary outcome analysis (control): 33 Outcome assessment design: intention-to-treat
Notes	Conflict of interest: TR and JZB received grants and personal fees from Roche, GSK, and Novartis. PS received grants and personal fees from Roche, TW; Roche advisory board, honoraria, grants; Novartis: advisory board, honoraria. MF is an employee of Roche. ABS, MP, JR, JK, LB, BKB, UW and AU do not declare conflicts of interest. Funding: supported by grant funding from Roche Poland ML21283 trial, the Medical University of Lodz, Poland: Grants: No. 503/1-152-01/503-11-002, 503/1-093-01/503-11-003, and 502-03/8-093-01/502-64-032. Summary conclusion: 66 participants were randomised to the rituximab maintenance arm (n = 33) or the observational arm (n = 33). CR rates were 57.1% in the maintenance group vs 50% in the observation group. PFS was significantly longer in the rituximab maintenance vs observation arm (P = 0.028). Maintenance therapy with rituximab (P < 0.001) significantly decreased the hazard ratio of disease progression. The study confirmed the high efficacy and acceptable safety profile of induction therapy with RCC and maintenance therapy with rituximab in previously untreated patients with CLL.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote" Patients were randomly assigned to treatment groups through a central randomization process". Comment: unkown randomization sequence generation.
Allocation concealment (selection bias)	Unclear risk	Quote: The investigator sent a fax containing patient data to the Randomization Centre at least five days prior to randomization. All information included in the randomization form was verified by the data manager before confirmation of the randomization was sent to the site" Comment:unkown allocation concealment process.
Blinding of participants and personnel (performance bias): Objective outcomes	Low risk	Participants and personnel were not blinded to treatment, no effect on subjective outcomes expected
Blinding of participants and personnel (performance bias): Subjective outcomes	High risk	This was an open label trial. The study is at risk of performance bias.

Robak 2018 *(Continued)*

Blinding of outcome assessment (detection bias): Objective outcomes	Low risk	Investigator assessment of total population, subjective outcomes assumed reliable
Blinding of outcome assessment (detection bias): Subjective outcomes	High risk	This was an open-label trial. No blinding of outcome assessors was reported. Potential risk of detection bias in all objective outcomes is due to knowledge of allocated interventions by outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	66 participants were randomly assigned to both study arms. In study analysis only included 39 participants
Selective reporting (reporting bias)	Low risk	All outcomes mentioned in the protocol are reported
Other bias	High risk	The randomization was prolonged two times due to the very slow enrolment, which could suggest that the results are not representative. The low number of patients, less than assumed in the initial protocol (for maintenance study, estimated enorllment number:148 patients, but only recruited 66 patients) Comments: high risk of bias

Schweighofer 2009

Study characteristics	
Methods	**Long-term follow-up of a randomised phase 3 trial of the German CLL Study Group (GCLLSG)** Study design: international, 13 centres in 2 countries; open-label, randomised phase 3 trial. Language of publication: English Major induction regimen: 6 cycles of fludarabine (25 mg/m^2 days 1 to 5 IV every 28 days) or fludarabine plus cyclophosphamide (F 30 mg/m^2 days 1 to 3 IV, C 250 mg/m^2 days 1 to 3 IV every 28 days) Randomisation: 2 arms: alemtuzumab versus observation Recruitment period: not available Median follow-up time: 48.0 months Sample size calculation: treatment efficacy was assumed upon achievement of a 25% improvement of PFS at 2 years. Based on a sample size of 45 participants for each arm, the log-rank test was calculated to have an 80% power to detect differences between the survival curves based on 23 months accrual time and 47 months overall length of the trial.
Participants	Inclusion criteria: participants were stratified according to induction treatment (fludarabine or fludarabine/cyclophosphamide) and response to induction treatment and randomised to treatment with alemtuzumab or observation. Number of included participants (overall): 21 Numbers of SLL/CLL: 21 Study sites: multiple centres in Germany and Austria Stage/type of disease (experimental): Binet B: 72.7%; Binet C: 0% Stage/type of disease (control): Binet B: 60%; Binet C: 30%

Schweighofer 2009 *(Continued)*

	Median age (experimental): median 60 years (38 to 63) Median age (control): median 54.1 years (37 to 66) Sex of male percentage (experimental): 72.70% Sex of male percentage (control): 70%
Interventions	Numbers of experimental group (alemtuzumab): 11 Numbers of control group (observation): 10 Experimental intervention: alemtuzumab 30 mg, 3 times per week, intravenously, for up to 12 weeks. The starting dose of alemtuzumab was 3 mg on day 1; if well tolerated, dose was increased to 10 mg on day 2 and to the target dose of 30 mg on day 3. The 30 mg dose was subsequently given 3 times per week for a maximum of 12 weeks. Control intervention: observation
Outcomes	Primary outcome: progression-free survival Secondary outcomes: • Response improvement • Grade 3 and 4 adverse events • All adverse events • Haematologic grade 3 and 4 adverse events • Infectious complication • MRD negative Reported and relevant outcomes for this review: CRR, ORR Final numbers of primary outcome analysis (experimental): 11 Final numbers of primary outcome analysis (control): 10 Outcome assessment design: intention-to-treat
Notes	**Randomised phase 3 trial of the German CLL Study Group (GCLLSG)** **Original report: Wendtner 2004, DOI: 10.1038/sj.leu.2403354** **Long-term follow-up report: Schweighofer 2019, DOI: 10.1111/j.1365-2141.2008.07394.x** Conflict of interest: not available Funding: grant from Schering AG and ILEX Oncology Summary conclusion: alemtuzumab has shown considerable activity in untreated and relapsed CLL. The authors report their long-term experience in 21 participants within a randomised phase 3 trial investigating the role of alemtuzumab for consolidation therapy after first-line fludarabine ± cyclophosphamide, which was stopped prematurely due to severe infections. However, after a median follow-up of 48 months, progression-free survival was significantly prolonged for participants receiving alemtuzumab consolidation compared to those with no further treatment (P = 0.004). MRD levels were persistently reduced after consolidation. Therefore, despite toxicity, MRD reduction by alemtuzumab consolidation translates into a significantly improved long-term clinical outcome.

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**

Schweighofer 2009 *(Continued)*

Random sequence generation (selection bias)	Unclear risk	Unkown allocation sequence generation and randomization.
Allocation concealment (selection bias)	Unclear risk	Unkown allocation sequence concealment.
Blinding of participants and personnel (performance bias): Objective outcomes	Low risk	Participants and personnel were not blinded to treatment, no effect on subjective outcomes expected
Blinding of participants and personnel (performance bias): Subjective outcomes	High risk	This was an open-label trial. The study is at risk of performance bias.
Blinding of outcome assessment (detection bias): Objective outcomes	Low risk	Investigator assessment of total population, subjective outcomes assumed reliable
Blinding of outcome assessment (detection bias): Subjective outcomes	High risk	This was an open-label trial. No blinding of outcome assessors was reported. Potential risk of detection bias in all objective outcomes is due to knowledge of allocated interventions by outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote:"All 21 randomized patients represented the intent-to-treat population" Comment: measure by ITT population, no lost of follow up.
Selective reporting (reporting bias)	High risk	Lack of pre-specified results:"quality of life, resource use assessment" Comment: at risk of reporting bias
Other bias	Unclear risk	Stop prematurely due to severe infection

Van Oers 2019

Study characteristics	
Methods	**Final analysis of the PROLONG study** Study design: international, multicentre, 130 centres in 24 countries, open-label, randomised phase 3 trial Language of publication: English Major induction regimen: chemo-immunotherapy in 386 (80%) participants, primarily fludarabine, cyclophosphamide, and rituximab (53%) and bendamustine and rituximab (24%), while only 23 (5%) participants had received alkylating monotherapy Randomisation: 2 arms: ofatumumab versus observation Recruitment period: May 2010 and October 2014 Median follow-up time: median 40.9 months Sample size calculation: 280 events of disease progression or death occurred, which was needed to detect the targeted 40% improvement in PFS (hazard ratio, 0.71) difference with 80% power and 5% two-sided α level.

Van Oers 2019 *(Continued)*

Participants	Inclusion criteria: CLL in complete remission (CR) or partial remission (PR) (according to the International Workshop on CLL updated National Cancer Institute-Working Group guidelines) after second- or third-line treatment, achieved with a response of PR or CR after induction treatment. Number of included participants (overall): 480 Numbers of SLL/CLL: 480 Study sites: international, 24 countries Stage/type of disease (experimental): Binet B: 22.6%; Binet C: 13.8%; cytogenetic aberrations: 73.4% Stage/type of disease (control): Binet B: 16.1%; Binet C: 17.7%; cytogenetic aberrations: 54.4% Median age (experimental): median 64.0 (33 to 86) Median age (control): median 64.5 (39 to 87) Sex of male percentage (experimental): 67% Sex of male percentage (control): 67%
Interventions	Numbers of experimental group (ofatumumab): 240 Numbers of control group (observation): 240 Experimental intervention: ofatumumab started 300 mg IV, followed 1 week later by 1000 mg IV, every 8 weeks for up to 2 years Control intervention: observation
Outcomes	Primary outcome: progression-free survival Secondary outcomes: • Overall survival • Time to next treatment • PFS after next-line therapy (defined as the time from randomisation until progression or death following next-line therapy) • Grade 3 and 4 adverse events • All adverse events • Treatment discontinuation • Health-related quality of life Reported and relevant outcomes for this review: PFS, OS, grade 3/4 AEs, TD, HRQoL Final numbers of primary outcome analysis (experimental): 240 Final numbers of primary outcome analysis (control): 240 Outcome assessment design: intention-to-treat
Notes	**Van Oers reported interim analysis of PROLONG trial (DOI: 10.1016/ S1470-2045(15)00143-6)** Conflict of interest: MVO. serves as a consulting or advisory role at Roche, ISA Therapeutics, and Immunicum and provides expert testimony for Roche. LS and M-DL have received honoraria, consulting fees and travel, accommodation and expenses from Roche, Janssen, AbbVie, and Gilead. Both also served on speakers' bureau for Gilead. JD and PH are both employed by and hold stock or other ownership in Novartis. TS and HB are employees of Novartis. MP, FO, SG, and CG have no relationships to disclose. Funding: funded by GlaxoSmithKline (GSK) and later by Novartis. GSK provided the study drug and worked with the HOVON and Nordic CLL group in the development of the study design. GSK and Novartis worked with the HOVON and Nordic CLL group for the collection and interpretation of data.

Van Oers 2019 *(Continued)*

	Summary conclusion: a total of 480 participants with CLL in complete or partial remission after second- or third-line treatment were randomised 1:1 to ofatumumab (300 mg first week, followed by 1000 mg every 8 weeks for up to 2 years) or observation. Median follow-up duration was 40.9 months. Median progression-free survival was 34.2 and 16.9 months for ofatumumab and observation arms, respectively (hazard ratio 0.55, 95% confidence interval 0.43 to 0.70; P < 0.001). Median time to next treatment for ofatumumab and observation arms, respectively, was 37.4 and 27.6 months (0.72, 0.57 to 0.91; P = 0.0044). Overall survival was similar in both arms; median was not reached (0.99, 0.72 to 1.37). Grade ≥ 3 adverse events occurred in 62% and 51% of participants in the ofatumumab and observation arms, respectively, the most common being neutropenia (23% and 10%), pneumonia (13% and 12%), and febrile neutropenia (6% and 4%). Up to 60 days after the last treatment, 4 deaths were reported in the ofatumumab arm versus 6 in the observation arm, none of which were considered to be related to ofatumumab. Ofatumumab maintenance significantly prolonged progression-free survival in people with relapsed CLL and was well tolerated.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote"Randomisation was done with a randomisation list generated by a central computerised system operated from GlaxoSmithKline (Research Triangle Park, NC, USA)." Comment: low risk of bias of random sequence generation
Allocation concealment (selection bias)	Low risk	Quote"No investigator was involved in the generation of the randomisation lists. Investigators enrolled patients and then received centrally allocated randomisation codes through an interactive voice recognition system." Comment: low risk of bias of allocation concealment
Blinding of participants and personnel (performance bias): Objective outcomes	Low risk	Participants and personnel were not blinded to treatment, no effect on subjective outcomes expected
Blinding of participants and personnel (performance bias): Subjective outcomes	High risk	This was an open-label trial. The study is at risk of performance bias
Blinding of outcome assessment (detection bias): Objective outcomes	Low risk	Investigator assessment of total population, subjective outcomes assumed reliable
Blinding of outcome assessment (detection bias): Subjective outcomes	High risk	This was an open-label trial. No blinding of outcome assessors was reported. Potential risk of detection bias in all objective outcomes is due to knowledge of allocated interventions by outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote:"All 480 randomized patients represented the intent-to-treat population" Comment:clear CONSORT flow diagram, low risk of bias
Selective reporting (reporting bias)	Low risk	All outcomes mentioned in the protocol are reported
Other bias	Low risk	No other sources of bias were detected

Williams 2016

Study characteristics	
Methods	Study design: multiple centres, randomised phase 3 trial Language of publication: English Major induction regimen: rituximab 375 mg/m^2, intravenously, weekly in weeks 1 to 4 Randomisation: 2 arms: maintenance rituximab versus observation and re-treatment rituximab upon progression Recruitment period: November 2003 and September 2008 Median follow-up time: median 4.3 years Sample size calculation: not available
Participants	Inclusion criteria: patients with partial response, complete response (CR), or CR unconfirmed were randomised to maintenance rituximab or re-treatment rituximab Number of included participants (overall): 52 Numbers of SLL/CLL: 13 with 7 patients in maintenance rituximab and 6 patients in re-treatment rituximab Study sites: USA, multiple centres Stage/type of disease (experimental): ISS stage III/IV: 96.5% Stage/type of disease (control): ISS stage III/IV: 99.9% Median age (experimental): median 65.5 (38.5 to 85.4) Median age (control): median 61.3 (47.3 to 86.3) Sex of male percentage (experimental): 51.70% Sex of male percentage (control): 30.40%
Interventions	Numbers of experimental group (maintenance rituximab): 30 Numbers of control group (observation and re-treatment rituximab): 28 Experimental intervention: rituximab 375 mg/m^2 every 3 months until treatment failure Control intervention: observed and re-treated with rituximab 375 mg weekly upon disease progression. Re-treatment was repeated until treatment failure.
Outcomes	Primary outcome: time to treatment failure (TTTF) Secondary outcomes: • Overall survival • Grade 3 and 4 adverse events • Time to first cytotoxic therapy • Overall health-related quality of life Reported and relevant outcomes for this review: nil Final numbers of primary outcome analysis (experimental): 29 Final numbers of primary outcome analysis (control): 23

Williams 2016 *(Continued)*

	Outcome assessment design: intention-to-treat
Notes	Conflict of interest: Michael E Williams: research funding from Genentech/Roche. Fangxin Hong, Lynne I Wagner, John C Krauss, Thomas M Habermann, Lode J Swinnnen, Stephen J Schuster, Christopher G Peterson, Mark D Sborov, S Eric Martin, Matthias Weiss, and W Christopher Ehmann: none. Randy D Gascoyne and Brad S Kahl: consulting fees and research funding from Genentech/Roche. Sandra J Horning: employment Genentech/Roche. Funding: supported in part by Public Health Service Grants CA21115, CA23318, CA66636, CA49957, CA21076, CA17145, and CA13650 from the National Cancer Institute, National Institutes of Health, and the Department of Health and Human Services. Summary conclusion: patients responding to rituximab weekly were randomised to maintenance rituximab (MR) (single dose rituximab every 3 months until treatment failure) or re-treatment rituximab (RR) (rituximab weekly x 4) at the time of each progression until treatment failure. Patients with SLL (n = 57) received induction rituximab. The overall response rate was 40% (95% confidence interval 31% to 49%); all 52 responders were randomised. At a median of 43 years from randomisation, treatment failure occurred in 18/23 RR and 15/29 MR. The median TTTF was 14 years for RR and 48 years for MR (P = 0.012). In low-tumour-burden SLL and marginal zone lymphoma patients responding to rituximab induction, MR significantly improved TTTF when compared with RR.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unkown allocation sequence generation and randomization.
Allocation concealment (selection bias)	Unclear risk	Unkown allocation sequence concealment.
Blinding of participants and personnel (performance bias): Objective outcomes	Low risk	Participants and personnel were not blinded to treatment, no effect on subjective outcomes expected
Blinding of participants and personnel (performance bias): Subjective outcomes	High risk	This was an open-label trial. The study is at risk of performance bias
Blinding of outcome assessment (detection bias): Objective outcomes	Low risk	Investigator assessment of total population, subjective outcomes assumed reliable
Blinding of outcome assessment (detection bias): Subjective outcomes	High risk	This was an open-label trial. No blinding of outcome assessors was reported. Potential risk of detection bias in all objective outcomes is due to knowledge of allocated interventions by outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	58 participants were randomly assigned to both study arms. In study analysis only 52 participants (exclude 6 stable disease patient without clear reason).
Selective reporting (reporting bias)	Low risk	All outcomes mentioned in the protocol are reported
Other bias	Low risk	No other sources of bias were detected

AE: adverse event, CLB-R: chlorambucil plus rituximab, CLL: chronic lymphocytic leukaemia, CR: complete response, CRR: complete response rate, FCR: fludarabine plus cyclophosphamide plus rituximab, FCR+L: fludarabine plus cyclophosphamide plus rituximab plus lenalidomide, FR: fludarabine, FR+L: fludarabine plus rituximab plus lenalidomide, HRQoL: health-related quality of life, IGHV: immunoglobulin heavy chain gene, IQR: interquartile range, ISS: international staging system, IV: intravenous, IVPB: intravenous piggyback, MRD: minimal residual disease, nPR: non-partial response, PFS: progression-free survival, PR: partial response, OR: odds ratio, ORR: overall response rate, OS: overall survival, RCC: rituximab plus cladribine plus cyclophosphamide. SAE: serious adverse event, SD: standard deviation, SLL: small lymphocytic leukaemia, TD: treatment discontinuation, TRM: treatment-related mortality

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
Abrisqueta 2013	Single-arm trial
Chang 2016	Single-arm trial
Egle 2018	Single-arm trial
EUCTR2005-001569-33-ES	Single-arm trial
EUCTR2008-001823-71-IT	Single-arm trial
EUCTR2010-018519-14-IT	Single-arm trial
EUCTR2014-000569-35-DE	Single-arm trial
EUCTR2014-000580-40-DE	Single-arm trial
EUCTR2014-000582-47-DE	Single-arm trial
EUCTR2014-000590-39-DE	Single-arm trial
EUCTR2015-004985-27-NL	Single-arm trial
Flinn 2016	Single-arm trial
Hainsworth 2003	Single-arm trial
Huang 2014	Non-randomised historic comparison group
ISRCTN63375144	Withdrawn prior to recruitment statement (alemtuzumab licence withdrawn by Genyme Sanofi on 8 August 2012)
Kater 2019	No eligible intervention strategy (sequential treatment, induction chemotherapy with chlorambucil + rituximab + lenalidomide and subsequent re-induction chemotherapy with lenalidomide)
Kaufman 2011	Single-arm trial
Keller 1986	Single-arm trial
Kempin 2019	Single-arm trial
Khouri 2017	No eligible intervention regimen (maintenance therapy without previous induction therapy) No eligible population (CLL patients post-allogenic transplant without evidence of residual disease)

Study	Reason for exclusion
Lamanna 2009	Single-arm trial
Lampson 2019	Single-arm trial
Lin 2010	Single-arm trial
Mato 2015	Single-arm trial
Mauro 2003	No eligible intervention regimen (a-interferon is not an intervention of interest)
Moccia 2008	Other study type (review article)
Montillo 2002	Single-arm trial
Montillo 2006	Single-arm trial
Moreno 2019	No eligible intervention strategy (ibrutinib treat until progressive disease, no maintenance therapy; induction therapy with ibrutinib + obinutuzumab and ibrutinib treat until progressive disease compared with chlorambucil + obinutuzumab)
NCT00336206	Single-arm trial
NCT00587847	Single-arm trial, prematurely discontinued (insufficient recruitment)
NCT00771602	No comparator group (only included one participant, n of 1 trial)
NCT00860457	Single-arm trial
NCT00974233	Single-arm trial
NCT01258933	Single-arm trial
NCT01392079	Single-arm trial
NCT01403246	Single-arm trial (phase 1 plus 2); prematurely discontinued (insufficient recruitment)
NCT01465230	Single-arm trial (phase 2); prematurely discontinued (insufficient recruitment)
NCT01496976	Single-arm study (phase 2)
NCT01521689	Single-arm trial
NCT01600053	Single-arm trial (phase 2); prematurely discontinued (interim analysis showed that the trial would not meet the interim endpoint)
NCT01754857	Single-arm trial (phase 2)
NCT01754870	Single-arm trial (phase 2); prematurely discontinued (insufficient recruitment)
NCT01809847	Single-arm trial (phase 2)
NCT02013817	Single-arm trial (phase 2)
NCT03847727	Other study type (prospective case-control study)

Study	Reason for exclusion
Noy 2001	Single-arm trial
O'Brien 1995	No eligible intervention regimen (a-interferon is not an intervention of interest)
O'Brien 2003	Single-arm trial
O'Brien 2011	Other study type (review article)
Osterborg 2015	Single-arm trial
Pan 2003	No eligible intervention regimen (cyclophosphamide is not an intervention of interest)
Pleyer 2020	Single-arm trial
Rai 2002	Single-arm trial
Rosenthal 2017	Single-arm trial
Ruppert 2019	Other study type (review article)
Scaramucci 2004	Other study type (case series)
Schey 1999	Single-arm trial (phase 2)
Sehn 2012	Single-arm trial (phase 1)
Shadman 2016	Single-arm trial. No eligible intervention regimen (131I-tositumomab is not an intervention of interest)
Shanafelt 2013	Non-randomised historic comparison group
Shanafelt 2019	No eligible intervention strategy (ibrutinib treat until progressive disease, no maintenance therapy; induction therapy with ibrutinib + rituximab and ibrutinib treat until progression compared with fludarabine + cyclophosphamide + rituximab)
Sharman 2019	No eligible intervention strategy (idelalisib treat until progressive disease, no maintenance therapy; induction therapy with idelalisib + rituximab and idelalisib treat until progression compared with placebo + rituximab)
Sharman 2020	No eligible intervention strategy (acalabrutinib treat until progressive disease, no maintenance therapy; induction therapy with acalabrutinib + obinutuzumab and acalabrutinib treat until progression versus acalabrutinib alone treat until progression versus induction therapy with obinutuzumab + chlorambucil)
Srock 2007	Single-arm trial
Strati 2016	Non-randomised historic comparison group
Strati 2017	Single-arm trial (phase 2)
Thieblemont 2004	Non-randomised historic comparison group
von Tresckow 2019	Single-arm trial
Walewski 2020	No eligible intervention regimen (R-CVP vs R-CHOP, both followed by rituximab maintenance).

Study	Reason for exclusion
	Low ratio of CLL, and no subgroup data provided for these patients (SLL: 21)
Weiss 2000	Single-arm trial. No eligible intervention regimen (cyclophosphamide is not an intervention of interest)
Zinzani 1994	No eligible intervention regimen (a-interferon is not an intervention of interest)
Zinzani 1997	No eligible intervention regimen (a-interferon is not an intervention of interest)

CLL: chronic lymphocytic leukaemia
R-CHOP: cyclophosphamide, doxorubicin, vincristine, prednisone + rituximab
R-CVP: cyclophosphamide, vincristine, prednisone
SLL: small lymphocytic lymphoma

Characteristics of studies awaiting classification *[ordered by study ID]*

Jaksic 1988

Methods	Randomised IGCI clinical trial
Participants	B-cell chronic lymphocytic leukaemia patients
Interventions	Regimen A: high-dose continuous chlorambucil Regimen B: intermittent chlorambucil plus prednisone
Outcomes	• Remission rate • Survival
Notes	Limited information from abstract. Subject to confirmation of randomisation and intervention (maintenance or not). Waiting for response from author (August 2022)

Characteristics of ongoing studies *[ordered by study ID]*

NCT03280160

Study name	Protocol GELLC-7: Ibrutinib followed by ibrutinib consolidation in combination with ofatumumab
Methods	Multicentre, non-randomised, open-label, double agent, phase 2 study of the Spanish Group of CLL (GELLC)
Participants	Planned enrolment: 84 participants
Interventions	Ibrutinib will be administered orally 420 mg (3 x 140 mg capsules) once daily on a continuous schedule on an outpatient basis until disease progression or unacceptable toxicity. After 12 cycles of ibrutinib, participants who do not achieve a complete response (CR) will be treated with the combination of ibrutinib and ofatumumab. Participants in CR after 12 cycles of ibrutinib will continue with ibrutinib alone. Ofatumumab will be administered by intravenous infusion, 300 mg on day 1 and 1000 mg on day 8 of cycle 13, followed by 5 monthly infusions of 1000 mg (day 1 of subsequent 28-day cycles for cycles C14, C15, C16, C17, and C18).

NCT03280160 *(Continued)*

Outcomes	Efficacy in terms of response rate [Time Frame: 18 months]: to determine the complete response rate obtained with the combination of ibrutinib and ofatumumab
Starting date	Actual study start date: 9 September 2017 Actual primary completion date: 13 July 2018 Estimated study completion date: September 2022
Contact information	PETHEMA Foundation
Notes	Recruitment status: active, not recruiting Date first received: 12 September 2017 Planned enrolment: 84 Estimated completion date: September 2022 Trial status: we have been unable to establish the current status of this trial and have not identified any associated publications

Oughton 2017

Study name	GA101 (obinutuzumab) monocLonal Antibody as Consolidation Therapy In CLL (GALACTIC) trial: study protocol for a phase II/III randomised controlled trial
Methods	Type of study: a seamless phase 2/3, multicentre, randomised, controlled, open, parallel-group trial Location: UK Intended duration of follow-up: 3 years postrandomisation
Participants	Inclusion criteria: Target gender: male and female • At least 18 years old • Previous confirmation of B-CLL with a characteristic immunophenotype (e.g. CD5+, CD19+, CD23+ lymphoproliferative disorder) on peripheral blood flow cytometry • Maximum of 3 prior therapies received for CLL treatment and between 3 and 24 months post-therapy at registration • Response to most recent chemotherapy treatment for CLL with PR, CRi, or CR • World Health Organization (WHO) performance status (PS) of 0 or 1 • Able to provide written informed consent • Peripheral B-cell count < 5 x 10^9 L • For randomisation, the first MRD-positive peripheral blood sample (disease level found in peripheral blood is greater than 0.01%) must be between 3 and 12 months since completing most recent therapy for CLL • Absence of clinically or radiologically evident lymphadenopathy (largest lymph node 1.5 cm or less in minimum diameter) • Creatinine and bilirubin < 2 times upper limit of normal unless secondary to direct infiltration of the liver by CLL or haemolysis Exclusion criteria: • Disease progression after response to latest therapy • Active infection • Past history of anaphylaxis following exposure to rat- or mouse-derived CDR-grafted humanised monoclonal antibodies

Oughton 2017 *(Continued)*

	• Previous treatment with obinutuzumab • CNS involvement with CLL • Mantle cell lymphoma • Moderate or severe cardiac disease that would preclude treatment with obinutuzumab • Other severe concurrent diseases or mental disorders that could interfere with ability to participate • Known HIV positivity • Active secondary malignancy excluding basal cell carcinoma • Active haemolysis • Patients previously treated with allogeneic stem cell transplant • Pregnancy, lactation, or women of childbearing potential unwilling to use medically approved contraception while receiving treatment and for 12 months after treatment has finished • Men whose partners are capable of having children but who are not willing to use appropriate medically approved contraception while receiving treatment and for 12 months after treatment has finished, unless they are surgically sterile • Persisting severe pancytopenia (neutrophils < 0.5 x 10^9/L or platelets < 50 x 10^9/L) or transfusion-dependent anaemia • Positive serology for hepatitis B (HB) defined as a positive test for HBsAg. In addition, if negative for HBsAg but HBcAb positive (regardless of HBsAb status), an HB DNA test will be performed, and if positive the individual will be excluded. • Positive serology for hepatitis C (HC) defined as a positive test for HC antibody, in which case reflexively perform an immunoblot assay (RIBA) on the same sample to confirm the result
Interventions	Intervention arm: obinutuzumab, intravenous infusion on days 1 and 2, then weekly (days 8, 15, and 22) and fortnightly (days 26, 50, 64, 78) up to 3 months Comparator arm: no consolidation therapy
Outcomes	Primary outcome: progression-free survival (defined as the time from randomisation to first documented evidence of progression or death) Secondary outcomes: • Undetectable MRD • CR, CRi, and overall response (at least a PR) • Overall survival • Treatment-free survival • Serious adverse events • Mean quality of life scores
Starting date	Opened to recruitment in February 2015
Contact information	Kaixun Hu (phone: +86 01066947129; email: hukaixun@sohu.co) Huisheng Ai (phone: +86 01066947126; email: aihuisheng@sohu.com)
Notes	Date first received: 8 January 2015 Planned enrolment: 188 (94 per arm) Estimated completion date: not reported Trial status: we have been unable to establish the current status of this trial and have not identified any associated publications

B-CLL: B-cell chronic lymphocytic leukaemia, CDR: complementarity determining regions, CNS: central nervous system, CR: complete response, CRi: complete response with incomplete count recovery, HBcAb: hepatitis B core antibody, HBsAb: hepatitis B surface antibody, HBsAg: hepatitis B surface antigen, MRD: minimal residual disease, PR: partial response, RIBA: recombinant immunoblot assay

DATA AND ANALYSES

Comparison 1. Anti-CD20 monoclonal antibody maintenance therapy versus observation

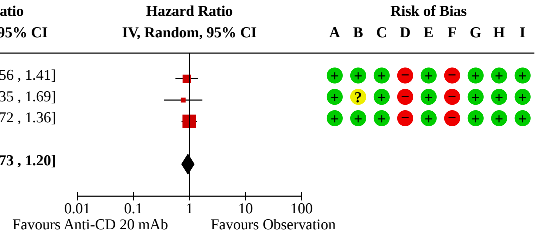
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 Overall survival (time to event)	3	1152	Hazard Ratio (IV, Random, 95% CI)	0.94 [0.73, 1.20]
1.2 Overall survival (time to event): type of drug	3		Hazard Ratio (IV, Random, 95% CI)	Subtotals only
1.2.1 Rituximab	2	672	Hazard Ratio (IV, Random, 95% CI)	0.86 [0.58, 1.28]
1.2.2 Ofatumumab	1	480	Hazard Ratio (IV, Random, 95% CI)	0.99 [0.72, 1.36]
1.3 Health-related quality of life	1	480	Mean Difference (IV, Random, 95% CI)	-1.70 [-8.59, 5.19]
1.4 Grade 3 and 4 adverse events	5	1284	Rate Ratio (IV, Random, 95% CI)	1.34 [1.06, 1.71]
1.5 Grade 3 and 4 adverse events: type of drug	5		Rate Ratio (IV, Random, 95% CI)	Subtotals only
1.5.1 Rituximab	4	804	Rate Ratio (IV, Random, 95% CI)	1.30 [0.93, 1.82]
1.5.2 Ofatumumab	1	480	Rate Ratio (IV, Random, 95% CI)	1.43 [1.06, 1.93]
1.6 Progression-free survival (time to event)	5	1255	Hazard Ratio (IV, Random, 95% CI)	0.61 [0.50, 0.73]
1.7 Progression-free survival (time to event): type of drug	5		Hazard Ratio (IV, Random, 95% CI)	Subtotals only
1.7.1 Rituximab	3	738	Hazard Ratio (IV, Random, 95% CI)	0.50 [0.37, 0.68]
1.7.2 Ofatumumab	2	517	Hazard Ratio (IV, Random, 95% CI)	0.66 [0.50, 0.86]
1.8 Treatment-related mortality	4	1189	Risk Ratio (M-H, Random, 95% CI)	0.82 [0.39, 1.71]
1.9 Treatment-related mortality: type of drug	4		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
1.9.1 Rituximab	2	672	Risk Ratio (M-H, Random, 95% CI)	1.12 [0.44, 2.88]
1.9.2 Ofatumumab	2	517	Risk Ratio (M-H, Random, 95% CI)	0.52 [0.25, 1.08]
1.10 Treatment discontinuation	6	1321	Risk Ratio (M-H, Random, 95% CI)	0.93 [0.72, 1.20]
1.11 Treatment discontinuation: type of drug	6		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
1.11.1 Rituximab	4	804	Risk Ratio (M-H, Random, 95% CI)	0.87 [0.58, 1.31]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.11.2 Ofatumumab	2	517	Risk Ratio (M-H, Random, 95% CI)	0.93 [0.77, 1.12]
1.12 All adverse events	6	1321	Rate Ratio (IV, Random, 95% CI)	1.23 [1.03, 1.47]
1.13 All adverse events: type of drug	6		Rate Ratio (IV, Random, 95% CI)	Subtotals only
1.13.1 Rituximab	4	804	Rate Ratio (IV, Random, 95% CI)	1.26 [0.97, 1.64]
1.13.2 Ofatumumab	2	517	Rate Ratio (IV, Random, 95% CI)	1.16 [0.96, 1.40]
1.14 Complete response rate	2	132	Risk Ratio (M-H, Random, 95% CI)	1.42 [0.81, 2.48]
1.15 Overall response rate	2	132	Risk Ratio (M-H, Random, 95% CI)	1.53 [1.05, 2.24]

Analysis 1.1. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 1: Overall survival (time to event)

Study or Subgroup	log[Hazard Ratio]	SE	Anti-CD 20 mAb Total	Observation Total	Weight	Hazard Ratio IV, Random, 95% CI
Dartigeas 2018	-0.117	0.236	202	207	29.0%	0.89 [0.56 , 1.41]
Greil 2016	-0.261	0.402	134	129	10.0%	0.77 [0.35 , 1.69]
Van Oers 2019	-0.0101	0.1625	240	240	61.1%	0.99 [0.72 , 1.36]
Total (95% CI)			**576**	**576**	**100.0%**	**0.94 [0.73 , 1.20]**

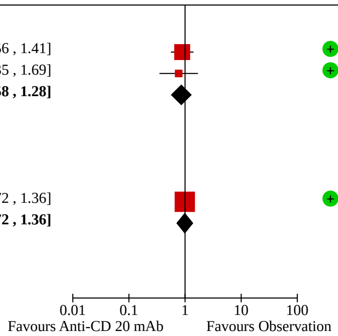
Heterogeneity: Tau² = 0.00; Chi² = 0.40, df = 2 (P = 0.82); I² = 0%
Test for overall effect: Z = 0.52 (P = 0.60)
Test for subgroup differences: Not applicable



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.2. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 2: Overall survival (time to event): type of drug

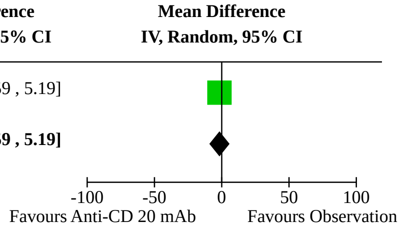
Study or Subgroup	log[Hazard Ratio]	SE	Anti-CD 20 mAb Total	Observation Total	Weight	Hazard Ratio IV, Random, 95% CI	A	B	C	D	E	F	G	H	I
1.2.1 Rituximab															
Dartigeas 2018	-0.117	0.236	202	207	74.4%	0.89 [0.56 , 1.41]	+	+	+	−	+	−	+	+	+
Greil 2016	-0.261	0.402	134	129	25.6%	0.77 [0.35 , 1.69]	+	?	+	−	+	−	+	+	+
Subtotal (95% CI)			**336**	**336**	**100.0%**	**0.86 [0.58 , 1.28]**									
Heterogeneity: Tau² = 0.00; Chi² = 0.10, df = 1 (P = 0.76); I² = 0%															
Test for overall effect: Z = 0.76 (P = 0.45)															
1.2.2 Ofatumumab															
Van Oers 2019	-0.0101	0.1625	240	240	100.0%	0.99 [0.72 , 1.36]	+	+	+	−	+	−	+	+	+
Subtotal (95% CI)			**240**	**240**	**100.0%**	**0.99 [0.72 , 1.36]**									
Heterogeneity: Not applicable															
Test for overall effect: Z = 0.06 (P = 0.95)															



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.3. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 3: Health-related quality of life

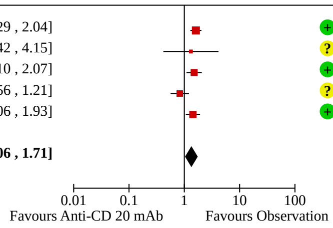
Study or Subgroup	Anti-CD 20 mAb Mean	SD	Total	Observation Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Van Oers 2019	0.2	38.5	239	1.9	38.5	241	100.0%	-1.70 [-8.59 , 5.19]
Total (95% CI)			**239**			**241**	**100.0%**	**-1.70 [-8.59 , 5.19]**
Heterogeneity: Not applicable								
Test for overall effect: Z = 0.48 (P = 0.63)								
Test for subgroup differences: Not applicable								



Analysis 1.4. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 4: Grade 3 and 4 adverse events

Study or Subgroup	log[Rate Ratio]	SE	Anti-CD 20 mAb Total	Observation Total	Weight	Rate Ratio IV, Random, 95% CI	Risk of Bias A	B	C	D	E	F	G	H	I
Dartigeas 2018	0.4817	0.117143119	202	207	28.9%	1.62 [1.29 , 2.04]	+	+	+	−	+	−	+	+	+
Foa 2014	0.2758	0.585540044	34	32	3.9%	1.32 [0.42 , 4.15]	?	?	+	−	+	−	+	+	−
Greil 2016	0.4095	0.162598618	134	129	23.2%	1.51 [1.10 , 2.07]	+	?	+	−	+	−	+	+	+
Robak 2018 (1)	-0.1892	0.195127427	33	33	19.6%	0.83 [0.56 , 1.21]	?	?	+	−	+	−	−	+	−
Van Oers 2019	0.3582	0.151780509	239	241	24.5%	1.43 [1.06 , 1.93]	+	+	+	−	+	−	+	+	+
Total (95% CI)			**642**	**642**	**100.0%**	**1.34 [1.06 , 1.71]**									

Heterogeneity: Tau² = 0.04; Chi² = 9.03, df = 4 (P = 0.06); I² = 56%
Test for overall effect: Z = 2.42 (P = 0.02)
Test for subgroup differences: Not applicable



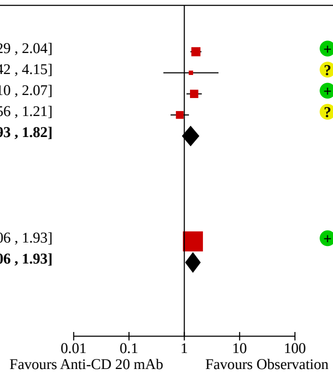
Footnotes
(1) total number of grade 3/4 adverse events is 48 at rituximab arm and 58 at observation arn for Robk 2018

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.5. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 5: Grade 3 and 4 adverse events: type of drug

Study or Subgroup	log[Rate Ratio]	SE	Anti-CD 20 mAb Total	Observation Total	Weight	Rate Ratio IV, Random, 95% CI	Risk of Bias A	B	C	D	E	F	G	H	I
1.5.1 Rituximab															
Dartigeas 2018	0.4817	0.117143119	202	207	35.2%	1.62 [1.29 , 2.04]	+	+	+	−	+	−	+	+	+
Foa 2014	0.2758	0.585540044	34	32	7.1%	1.32 [0.42 , 4.15]	?	?	+	−	+	−	+	+	−
Greil 2016	0.4095	0.162598618	134	129	30.5%	1.51 [1.10 , 2.07]	+	?	+	−	+	−	+	+	+
Robak 2018 (1)	-0.1892	0.195127427	33	33	27.2%	0.83 [0.56 , 1.21]	?	?	+	−	+	−	−	+	−
Subtotal (95% CI)			**403**	**401**	**100.0%**	**1.30 [0.93 , 1.82]**									
Heterogeneity: Tau² = 0.07; Chi² = 9.00, df = 3 (P = 0.03); I² = 67%															
Test for overall effect: Z = 1.54 (P = 0.12)															
1.5.2 Ofatumumab															
Van Oers 2019	0.3582	0.151780509	239	241	100.0%	1.43 [1.06 , 1.93]	+	+	+	−	+	−	+	+	+
Subtotal (95% CI)			**239**	**241**	**100.0%**	**1.43 [1.06 , 1.93]**									
Heterogeneity: Not applicable															
Test for overall effect: Z = 2.36 (P = 0.02)															

Test for subgroup differences: Chi² = 0.00, df = 1 (P < 0.00001), I² = 0%

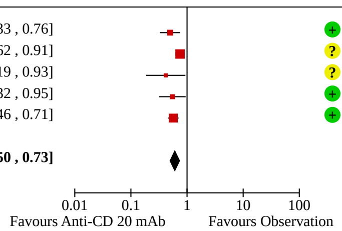


Footnotes
(1) total number of grade 3/4 adverse events is 48 at rituximab arm and 58 at observation arn for Robk 2018

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.6. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 6: Progression-free survival (time to event)

Study or Subgroup	log[Hazard Ratio]	SE	Anti-CD 20 mAb Total	Observation Total	Weight	Hazard Ratio IV, Random, 95% CI	Risk of Bias A	B	C	D	E	F	G	H	I
Greil 2016	-0.6931	0.212	134	129	15.4%	0.50 [0.33 , 0.76]	+	?	+	-	+	-	+	+	+
Osterborg 2016	-0.2877	0.0971	24	13	36.3%	0.75 [0.62 , 0.91]	?	?	+	-	+	-	-	-	?
Robak 2018	-0.8723	0.4104	33	33	5.1%	0.42 [0.19 , 0.93]	?	?	+	-	+	-	-	+	-
Dartigeas 2018	-0.598	0.2763	202	207	10.2%	0.55 [0.32 , 0.95]	+	+	+	-	+	-	+	+	+
Van Oers 2019	-0.5621	0.1094	240	240	33.1%	0.57 [0.46 , 0.71]	+	+	+	-	+	-	+	+	+
Total (95% CI)			**633**	**622**	**100.0%**	**0.61 [0.50 , 0.73]**									

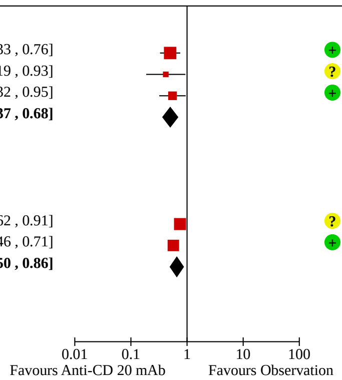


Heterogeneity: $Tau^2 = 0.02$; $Chi^2 = 6.49$, df = 4 (P = 0.17); $I^2 = 38\%$
Test for overall effect: Z = 5.16 (P < 0.00001)
Test for subgroup differences: Not applicable

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.7. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 7: Progression-free survival (time to event): type of drug

Study or Subgroup	log[Hazard Ratio]	SE	Anti-CD 20 mAb Total	Observation Total	Weight	Hazard Ratio IV, Random, 95% CI	Risk of Bias A	B	C	D	E	F	G	H	I
1.7.1 Rituximab															
Greil 2016	-0.6931	0.212	134	129	53.9%	0.50 [0.33 , 0.76]	+	?	+	-	+	-	+	+	+
Robak 2018	-0.8723	0.4104	33	33	14.4%	0.42 [0.19 , 0.93]	?	?	+	-	+	-	-	+	-
Dartigeas 2018	-0.598	0.2763	202	207	31.7%	0.55 [0.32 , 0.95]	+	+	+	-	+	-	+	+	+
Subtotal (95% CI)			**369**	**369**	**100.0%**	**0.50 [0.37 , 0.68]**									
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.31$, df = 2 (P = 0.86); $I^2 = 0\%$															
Test for overall effect: Z = 4.43 (P < 0.00001)															
1.7.2 Ofatumumab															
Osterborg 2016	-0.2877	0.0971	24	13	51.7%	0.75 [0.62 , 0.91]	?	?	+	-	+	-	-	-	?
Van Oers 2019	-0.5621	0.1094	240	240	48.3%	0.57 [0.46 , 0.71]	+	+	+	-	+	-	+	+	+
Subtotal (95% CI)			**264**	**253**	**100.0%**	**0.66 [0.50 , 0.86]**									
Heterogeneity: $Tau^2 = 0.03$; $Chi^2 = 3.52$, df = 1 (P = 0.06); $I^2 = 72\%$															
Test for overall effect: Z = 3.06 (P = 0.002)															

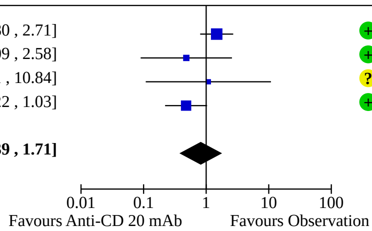


Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.8. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 8: Treatment-related mortality

Study or Subgroup	Anti-CD 20 mAbs Events	Anti-CD 20 mAbs Total	Observation Events	Observation Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G	H	I
Dartigeas 2018	23	202	16	207	41.5%	1.47 [0.80 , 2.71]	+	+	+	-	+	-	+	+	+
Greil 2016	2	134	4	129	14.5%	0.48 [0.09 , 2.58]	+	?	+	-	+	-	+	+	+
Osterborg 2016	2	24	1	13	8.7%	1.08 [0.11 , 10.84]	?	?	+	-	+	-	-	-	?
Van Oers 2019	9	239	19	241	35.4%	0.48 [0.22 , 1.03]	+	+	+	-	+	-	+	+	+
Total (95% CI)		**599**		**590**	**100.0%**	**0.82 [0.39 , 1.71]**									
Total events:	36		40												

Heterogeneity: Tau² = 0.25; Chi² = 5.67, df = 3 (P = 0.13); I² = 47%
Test for overall effect: Z = 0.53 (P = 0.60)
Test for subgroup differences: Not applicable

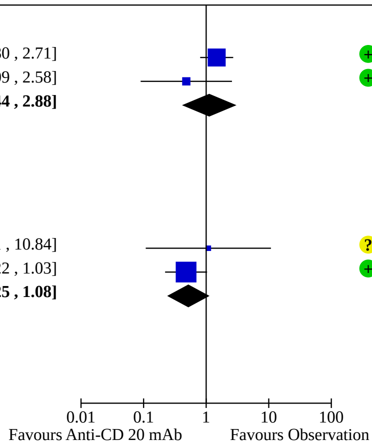


Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.9. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 9: Treatment-related mortality: type of drug

Study or Subgroup	Anti-CD 20 mAbs Events	Anti-CD 20 mAbs Total	Observation Events	Observation Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G	H	I
1.9.1 Rituximab															
Dartigeas 2018	23	202	16	207	75.4%	1.47 [0.80 , 2.71]	+	+	+	-	+	-	+	+	+
Greil 2016	2	134	4	129	24.6%	0.48 [0.09 , 2.58]	+	?	+	-	+	-	+	+	+
Subtotal (95% CI)		**336**		**336**	**100.0%**	**1.12 [0.44 , 2.88]**									
Total events:	25		20												
Heterogeneity: Tau² = 0.21; Chi² = 1.51, df = 1 (P = 0.22); I² = 34%															
Test for overall effect: Z = 0.23 (P = 0.82)															
1.9.2 Ofatumumab															
Osterborg 2016	2	24	1	13	10.1%	1.08 [0.11 , 10.84]	?	?	+	-	+	-	-	-	?
Van Oers 2019	9	239	19	241	89.9%	0.48 [0.22 , 1.03]	+	+	+	-	+	-	+	+	+
Subtotal (95% CI)		**263**		**254**	**100.0%**	**0.52 [0.25 , 1.08]**									
Total events:	11		20												
Heterogeneity: Tau² = 0.00; Chi² = 0.44, df = 1 (P = 0.51); I² = 0%															
Test for overall effect: Z = 1.76 (P = 0.08)															

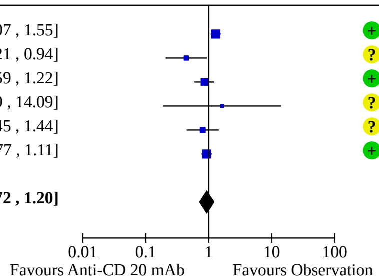
Test for subgroup differences: Chi² = 0.00, df = 1 (P < 0.00001), I² = 0%



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.10. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 10: Treatment discontinuation

Study or Subgroup	Anti-CD 20 mAb Events	Anti-CD 20 mAb Total	Observation Events	Observation Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G	H	I
Dartigeas 2018	122	202	97	207	28.8%	1.29 [1.07 , 1.55]	+	+	+	-	+	-	+	+	+
Foa 2014	7	34	15	32	8.7%	0.44 [0.21 , 0.94]	?	?	+	-	+	-	+	+	-
Greil 2016	38	134	43	129	20.2%	0.85 [0.59 , 1.22]	+	?	+	-	+	-	+	+	+
Osterborg 2016	3	24	1	13	1.4%	1.63 [0.19 , 14.09]	?	?	+	-	+	-	-	-	?
Robak 2018	12	33	15	33	12.3%	0.80 [0.45 , 1.44]	?	?	+	-	+	-	-	+	-
Van Oers 2019	110	239	120	241	28.6%	0.92 [0.77 , 1.11]	+	+	+	-	+	-	+	+	+
Total (95% CI)		**666**		**655**	**100.0%**	**0.93 [0.72 , 1.20]**									
Total events:	292		291												

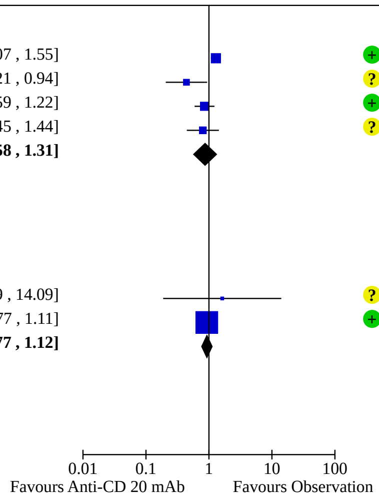


Heterogeneity: Tau² = 0.05; Chi² = 14.06, df = 5 (P = 0.02); I² = 64%
Test for overall effect: Z = 0.56 (P = 0.57)
Test for subgroup differences: Not applicable

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.11. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 11: Treatment discontinuation: type of drug

Study or Subgroup	Anti-CD 20 mAb Events	Anti-CD 20 mAb Total	Observation Events	Observation Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G	H	I
1.11.1 Rituximab															
Dartigeas 2018	122	202	97	207	34.1%	1.29 [1.07 , 1.55]	+	+	+	-	+	-	+	+	+
Foa 2014	7	34	15	32	16.4%	0.44 [0.21 , 0.94]	?	?	+	-	+	-	+	+	-
Greil 2016	38	134	43	129	28.5%	0.85 [0.59 , 1.22]	+	?	+	-	+	-	+	+	+
Robak 2018	12	33	15	33	21.0%	0.80 [0.45 , 1.44]	?	?	+	-	+	-	-	+	-
Subtotal (95% CI)		**403**		**401**	**100.0%**	**0.87 [0.58 , 1.31]**									
Total events:	179		170												
Heterogeneity: Tau² = 0.12; Chi² = 11.95, df = 3 (P = 0.008); I² = 75%															
Test for overall effect: Z = 0.67 (P = 0.50)															
1.11.2 Ofatumumab															
Osterborg 2016	3	24	1	13	0.7%	1.63 [0.19 , 14.09]	?	?	+	-	+	-	-	-	?
Van Oers 2019	110	239	120	241	99.3%	0.92 [0.77 , 1.11]	+	+	+	-	+	-	+	+	+
Subtotal (95% CI)		**263**		**254**	**100.0%**	**0.93 [0.77 , 1.12]**									
Total events:	113		121												
Heterogeneity: Tau² = 0.00; Chi² = 0.26, df = 1 (P = 0.61); I² = 0%															
Test for overall effect: Z = 0.78 (P = 0.43)															



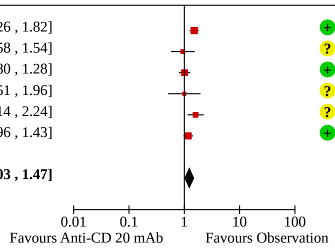
Test for subgroup differences: Chi² = 0.00, df = 1 (P < 0.00001), I² = 0%

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.12. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 12: All adverse events

Study or Subgroup	log[Rate Ratio]	SE	Anti-CD 20 mAb Total	Observation Total	Weight	Rate Ratio IV, Random, 95% CI	Risk of Bias A	B	C	D	E	F	G	H	I
Dartigeas 2018	0.4122	0.093992481	202	207	24.6%	1.51 [1.26 , 1.82]	+	+	+	−	+	−	+	+	+
Foa 2014	-0.0606	0.250001001	34	32	9.5%	0.94 [0.58 , 1.54]	?	?	+	−	+	−	+	+	−
Greil 2016	0.0111	0.118505526	134	129	21.3%	1.01 [0.80 , 1.28]	+	?	+	−	+	−	+	+	+
Osterborg 2016	0	0.344368616	24	13	5.8%	1.00 [0.51 , 1.96]	?	?	+	−	+	−	−	−	?
Robak 2018	0.47	0.171887913	33	33	15.3%	1.60 [1.14 , 2.24]	?	?	+	−	+	−	−	+	−
Van Oers 2019	0.1601	0.102489446	239	241	23.4%	1.17 [0.96 , 1.43]	+	+	+	−	+	−	+	+	+
Total (95% CI)			**666**	**655**	**100.0%**	**1.23 [1.03 , 1.47]**									

Heterogeneity: Tau² = 0.03; Chi² = 11.38, df = 5 (P = 0.04); I² = 56%
Test for overall effect: Z = 2.27 (P = 0.02)
Test for subgroup differences: Not applicable

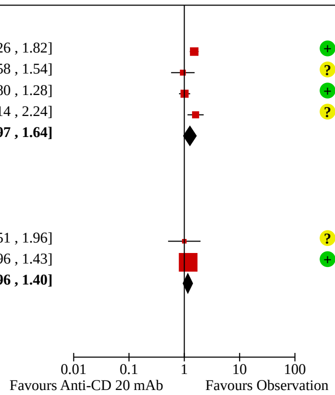


Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.13. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 13: All adverse events: type of drug

Study or Subgroup	log[Rate Ratio]	SE	Anti-CD 20 mAb Total	Observation Total	Weight	Rate Ratio IV, Random, 95% CI	Risk of Bias A	B	C	D	E	F	G	H	I
1.13.1 Rituximab															
Dartigeas 2018	0.4122	0.093992481	202	207	31.7%	1.51 [1.26 , 1.82]	+	+	+	−	+	−	+	+	+
Foa 2014	-0.0606	0.250001001	34	32	16.2%	0.94 [0.58 , 1.54]	?	?	+	−	+	−	+	+	−
Greil 2016	0.0111	0.118505526	134	129	29.0%	1.01 [0.80 , 1.28]	+	?	+	−	+	−	+	+	+
Robak 2018	0.47	0.171887913	33	33	23.1%	1.60 [1.14 , 2.24]	?	?	+	−	+	−	−	+	−
Subtotal (95% CI)			**403**	**401**	**100.0%**	**1.26 [0.97 , 1.64]**									
Heterogeneity: Tau² = 0.05; Chi² = 10.16, df = 3 (P = 0.02); I² = 70%															
Test for overall effect: Z = 1.75 (P = 0.08)															
1.13.2 Ofatumumab															
Osterborg 2016	0	0.344368616	24	13	8.1%	1.00 [0.51 , 1.96]	?	?	+	−	+	−	−	−	?
Van Oers 2019	0.1601	0.102489446	239	241	91.9%	1.17 [0.96 , 1.43]	+	+	+	−	+	−	+	+	+
Subtotal (95% CI)			**263**	**254**	**100.0%**	**1.16 [0.96 , 1.40]**									
Heterogeneity: Tau² = 0.00; Chi² = 0.20, df = 1 (P = 0.66); I² = 0%															
Test for overall effect: Z = 1.50 (P = 0.13)															

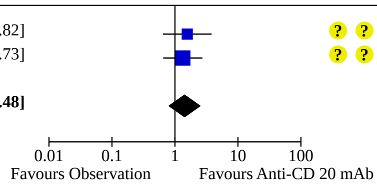
Test for subgroup differences: Chi² = 0.00, df = 1 (P < 0.00001), I² = 0%



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.14. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 14: Complete response rate

Study or Subgroup	Anti-CD 20 mAb Events	Anti-CD 20 mAb Total	Observation Events	Observation Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G	H	I
Foa 2014	10	34	6	32	39.4%	1.57 [0.64 , 3.82]	?	?	+	−	+	−	+	+	−
Robak 2018	12	33	9	33	60.6%	1.33 [0.65 , 2.73]	?	?	+	−	+	−	−	+	−
Total (95% CI)		**67**		**65**	**100.0%**	**1.42 [0.81 , 2.48]**									
Total events:	22		15												

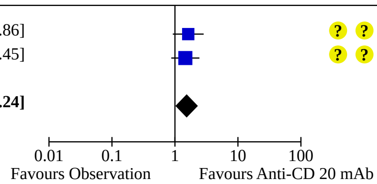


Heterogeneity: Tau² = 0.00; Chi² = 0.08, df = 1 (P = 0.78); I² = 0%
Test for overall effect: Z = 1.23 (P = 0.22)
Test for subgroup differences: Not applicable

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 1.15. Comparison 1: Anti-CD20 monoclonal antibody maintenance therapy versus observation, Outcome 15: Overall response rate

Study or Subgroup	Anti-CD 20 mAb Events	Anti-CD 20 mAb Total	Observation Events	Observation Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G	H	I
Foa 2014	19	34	11	32	45.4%	1.63 [0.92 , 2.86]	?	?	+	−	+	−	+	+	−
Robak 2018	19	33	13	33	54.6%	1.46 [0.87 , 2.45]	?	?	+	−	+	−	−	+	−
Total (95% CI)		**67**		**65**	**100.0%**	**1.53 [1.05 , 2.24]**									
Total events:	38		24												



Heterogeneity: Tau² = 0.00; Chi² = 0.07, df = 1 (P = 0.78); I² = 0%
Test for overall effect: Z = 2.21 (P = 0.03)
Test for subgroup differences: Not applicable

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

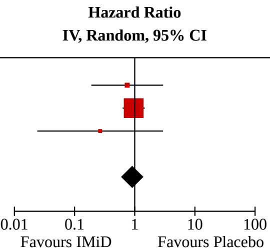
Comparison 2. Immunomodulatory drug maintenance therapy versus placebo

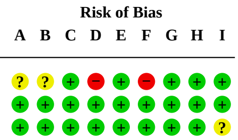
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 Overall survival (time to event)	3	461	Hazard Ratio (IV, Random, 95% CI)	0.91 [0.61, 1.35]
2.2 Grade 3 and 4 adverse events	2	400	Rate Ratio (IV, Random, 95% CI)	1.82 [1.38, 2.38]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.3 Progression-free survival (time to event)	3	461	Hazard Ratio (IV, Random, 95% CI)	0.37 [0.19, 0.73]
2.4 Treatment-related mortality	3	458	Risk Ratio (M-H, Random, 95% CI)	1.22 [0.35, 4.29]
2.5 Treatment discontinuation	2	400	Risk Ratio (M-H, Random, 95% CI)	0.71 [0.47, 1.05]
2.6 All adverse events	3	458	Rate Ratio (IV, Random, 95% CI)	1.41 [1.28, 1.54]

Analysis 2.1. Comparison 2: Immunomodulatory drug maintenance therapy versus placebo, Outcome 1: Overall survival (time to event)

Study or Subgroup	log[Hazard Ratio]	SE	IMiD Total	Placebo Total	Weight	Hazard Ratio IV, Random, 95% CI
Byrd 2018	-0.2877	0.7005	31	27	8.4%	0.75 [0.19 , 2.96]
Chanan-Khan 2017	-0.041	0.215	160	154	88.9%	0.96 [0.63 , 1.46]
Fink 2017	-1.324	1.227	60	29	2.7%	0.27 [0.02 , 2.95]
Total (95% CI)			**251**	**210**	**100.0%**	**0.91 [0.61 , 1.35]**





Heterogeneity: Tau² = 0.00; Chi² = 1.14, df = 2 (P = 0.56); I² = 0%
Test for overall effect: Z = 0.48 (P = 0.63)
Test for subgroup differences: Not applicable

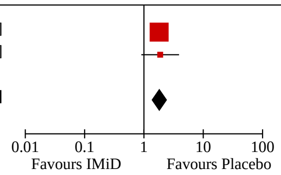
Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 2.2. Comparison 2: Immunomodulatory drug maintenance therapy versus placebo, Outcome 2: Grade 3 and 4 adverse events

Study or Subgroup	log[Rate Ratio]	SE	IMiD Total	Placebo Total	Weight	Rate Ratio IV, Random, 95% CI	A	B	C	D	E	F	G	H	I
Chanan-Khan 2017	0.5908	0.149555239	157	154	86.2%	1.81 [1.35 , 2.42]	+	+	+	+	+	+	+	+	+
Fink 2017	0.6311	0.373741274	60	29	13.8%	1.88 [0.90 , 3.91]	+	+	+	+	+	+	+	+	?
Total (95% CI)			**217**	**183**	**100.0%**	**1.82 [1.38 , 2.38]**									

Heterogeneity: Tau² = 0.00; Chi² = 0.01, df = 1 (P = 0.92); I² = 0%
Test for overall effect: Z = 4.29 (P < 0.0001)
Test for subgroup differences: Not applicable

Rate Ratio IV, Random, 95% CI



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
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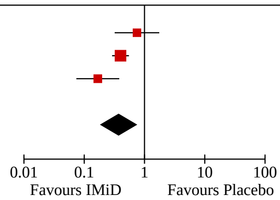
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 2.3. Comparison 2: Immunomodulatory drug maintenance therapy versus placebo, Outcome 3: Progression-free survival (time to event)

Study or Subgroup	log[Hazard Ratio]	SE	IMiD Total	Placebo Total	Weight	Hazard Ratio IV, Random, 95% CI	A	B	C	D	E	F	G	H	I
Byrd 2018	-0.2877	0.4346	31	27	27.4%	0.75 [0.32 , 1.76]	?	?	+	−	+	−	+	+	+
Chanan-Khan 2017	-0.9163	0.1641	160	154	44.2%	0.40 [0.29 , 0.55]	+	+	+	+	+	+	+	+	+
Fink 2017	-1.7838	0.4183	60	29	28.4%	0.17 [0.07 , 0.38]	+	+	+	+	+	+	+	+	?
Total (95% CI)			**251**	**210**	**100.0%**	**0.37 [0.19 , 0.73]**									

Heterogeneity: Tau² = 0.24; Chi² = 6.34, df = 2 (P = 0.04); I² = 68%
Test for overall effect: Z = 2.89 (P = 0.004)
Test for subgroup differences: Not applicable

Hazard Ratio IV, Random, 95% CI



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
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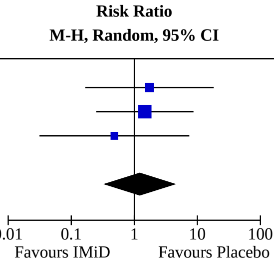
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 2.4. Comparison 2: Immunomodulatory drug maintenance therapy versus placebo, Outcome 4: Treatment-related mortality

Study or Subgroup	IMiD Events	IMiD Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G	H	I
Byrd 2018	2	31	1	27	28.8%	1.74 [0.17 , 18.16]	?	?	+	−	+	−	+	+	+
Chanan-Khan 2017	3	157	2	154	50.1%	1.47 [0.25 , 8.68]	+	+	+	+	+	+	+	+	+
Fink 2017	1	60	1	29	21.1%	0.48 [0.03 , 7.46]	+	+	+	+	+	+	+	+	?
Total (95% CI)		**248**		**210**	**100.0%**	**1.22 [0.35 , 4.29]**									
Total events:	6		4												

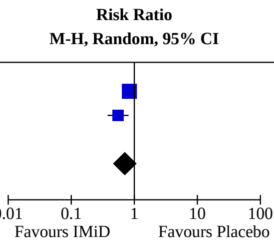


Heterogeneity: Tau² = 0.00; Chi² = 0.57, df = 2 (P = 0.75); I² = 0%
Test for overall effect: Z = 0.31 (P = 0.76)
Test for subgroup differences: Not applicable

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 2.5. Comparison 2: Immunomodulatory drug maintenance therapy versus placebo, Outcome 5: Treatment discontinuation

Study or Subgroup	IMiD Events	IMiD Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G	H	I
Chanan-Khan 2017	103	157	121	154	59.5%	0.83 [0.73 , 0.96]	+	+	+	+	+	+	+	+	+
Fink 2017	24	60	21	29	40.5%	0.55 [0.38 , 0.81]	+	+	+	+	+	+	+	+	?
Total (95% CI)		**217**		**183**	**100.0%**	**0.71 [0.47 , 1.05]**									
Total events:	127		142												



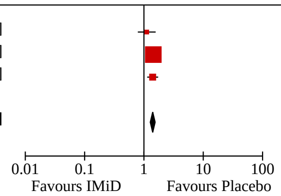
Heterogeneity: Tau² = 0.07; Chi² = 4.01, df = 1 (P = 0.05); I² = 75%
Test for overall effect: Z = 1.70 (P = 0.09)
Test for subgroup differences: Not applicable

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 2.6. Comparison 2: Immunomodulatory drug maintenance therapy versus placebo, Outcome 6: All adverse events

Study or Subgroup	log[Rate Ratio]	SE	IMiD Total	Placebo Total	Weight	Rate Ratio IV, Random, 95% CI	Risk of Bias A	B	C	D	E	F	G	H	I
Byrd 2018	0.1092	0.176755452	31	27	7.4%	1.12 [0.79 , 1.58]	?	?	+	−	+	−	+	+	+
Chanan-Khan 2017	0.364	0.056168897	157	154	73.1%	1.44 [1.29 , 1.61]	+	+	+	+	+	+	+	+	+
Fink 2017	0.3389	0.108595939	60	29	19.5%	1.40 [1.13 , 1.74]	+	+	+	+	+	+	+	+	?
Total (95% CI)			**248**	**210**	**100.0%**	**1.41 [1.28 , 1.54]**									

Heterogeneity: Tau² = 0.00; Chi² = 1.89, df = 2 (P = 0.39); I² = 0%
Test for overall effect: Z = 7.09 (P < 0.00001)
Test for subgroup differences: Not applicable



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
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(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

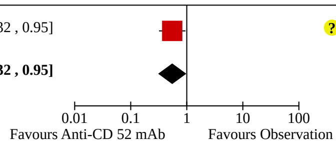
Comparison 3. Anti-CD52 monoclonal antibody maintenance therapy versus observation

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.1 Progression-free survival (time to event)	1	21	Hazard Ratio (IV, Random, 95% CI)	0.55 [0.32, 0.95]
3.2 Complete response rate	1	21	Risk Ratio (M-H, Random, 95% CI)	1.36 [0.28, 6.56]
3.3 Overall response rate	1	21	Risk Ratio (M-H, Random, 95% CI)	1.41 [0.92, 2.14]

Analysis 3.1. Comparison 3: Anti-CD52 monoclonal antibody maintenance therapy versus observation, Outcome 1: Progression-free survival (time to event)

Study or Subgroup	log[Hazard Ratio]	SE	Anti-CD 52 mAb Total	Observation Total	Weight	Hazard Ratio IV, Random, 95% CI	Risk of Bias A	B	C	D	E	F	G	H	I
Schweighofer 2009	-0.5978	0.2763	11	10	100.0%	0.55 [0.32 , 0.95]	?	?	+	−	+	−	+	−	?
Total (95% CI)			**11**	**10**	**100.0%**	**0.55 [0.32 , 0.95]**									

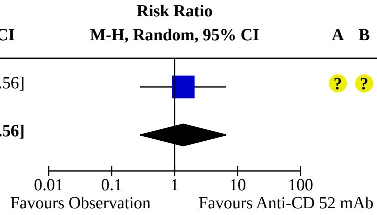
Heterogeneity: Not applicable
Test for overall effect: Z = 2.16 (P = 0.03)
Test for subgroup differences: Not applicable



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 3.2. Comparison 3: Anti-CD52 monoclonal antibody maintenance therapy versus observation, Outcome 2: Complete response rate

Study or Subgroup	Anti-CD 52 mAb Events	Anti-CD 52 mAb Total	Observation Events	Observation Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G	H	I
Schweighofer 2009	3	11	2	10	100.0%	1.36 [0.28 , 6.56]	?	?	+	−	+	−	+	−	?
Total (95% CI)		**11**		**10**	**100.0%**	**1.36 [0.28 , 6.56]**									
Total events:	3		2												

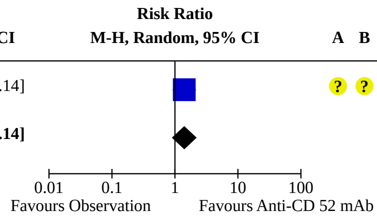


Heterogeneity: Not applicable
Test for overall effect: Z = 0.39 (P = 0.70)
Test for subgroup differences: Not applicable

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
(E) Blinding of outcome assessment (detection bias): Objective outcomes
(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

Analysis 3.3. Comparison 3: Anti-CD52 monoclonal antibody maintenance therapy versus observation, Outcome 3: Overall response rate

Study or Subgroup	Anti-CD 52 mAb Events	Anti-CD 52 mAb Total	Observation Events	Observation Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G	H	I
Schweighofer 2009	11	11	7	10	100.0%	1.41 [0.92 , 2.14]	?	?	+	−	+	−	+	−	?
Total (95% CI)		**11**		**10**	**100.0%**	**1.41 [0.92 , 2.14]**									
Total events:	11		7												



Heterogeneity: Not applicable
Test for overall effect: Z = 1.59 (P = 0.11)
Test for subgroup differences: Not applicable

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias): Objective outcomes
(D) Blinding of participants and personnel (performance bias): Subjective outcomes
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(F) Blinding of outcome assessment (detection bias): Subjective outcomes
(G) Incomplete outcome data (attrition bias)
(H) Selective reporting (reporting bias)
(I) Other bias

ADDITIONAL TABLES

Table 1. Sensitivity analyses

Comparison	Outcome	Random-effects model				Fixed-effect model				Random sequence	Allocation concealment	Blinding
		HR/risk ratio/rate ratio	95% CI low	95% CI high	P value	HR/risk ratio/rate ratio	95% CI low	95% CI high	P value	Test for subgroup differences	Test for subgroup differences	Test for subgroup differences
Anti-CD20 mAbs versus observation	OS	0.94	0.73	1.2	P = 0.60	0.94	0.73	1.2	P = 0.60	No high risk of bias trial	No high risk of bias trial	No high risk of bias trial
	HRQoL	Only one trial										
	Grade 3/4 AEs	1.34	1.06	1.71	P = 0.02	1.4	1.21	1.62	P < 0.001	No high risk of bias trial	No high risk of bias trial	No low risk of bias trial
	PFS	0.61	0.5	0.73	P < 0.001	0.63	0.56	0.72	P < 0.001	No high risk of bias trial	No high risk of bias trial	No low risk of bias trial
	TRM	0.82	0.39	1.71	P = 0.60	0.89	0.58	1.37	P = 0.60	No high risk of bias trial	No high risk of bias trial	No high risk of bias trial
	TD	0.93	0.72	1.2	P = 0.57	1	0.89	1.13	P = 0.94	No high risk of bias trial	No high risk of bias trial	No high risk of bias trial
	All AEs	1.23	1.03	1.47	P = 0.02	1.26	1.13	1.4	P < 0.001	No high risk of bias trial	No high risk of bias trial	No low risk of bias trial
	ORR	1.42	0.81	2.48	P = 0.22	1.43	0.82	2.5	P = 0.21	No high risk of bias trial	No high risk of bias trial	No low risk of bias trial
	CRR	1.53	1.05	2.24	P = 0.03	1.54	1.05	2.25	P = 0.03	No high risk of bias trial	No high risk of bias trial	No low risk of bias trial
	MRD	No data available										
IMiD versus observation	OS	0.91	0.61	1.35	P = 0.63	0.91	0.61	1.35	P = 0.63	No high risk of bias trial	No high risk of bias trial	P = 0.82, I^2 = 0
	HRQoL	No data available										

Table 1. Sensitivity analyses *(Continued)*

	Grade 3/4 AEs	1.82	1.38	2.38	P < 0.001	1.82	1.38	2.38	P < 0.001	No high risk of bias trial	No high risk of bias trial	No high risk of bias trial
	PFS	0.37	0.19	0.73	P = 0.004	0.39	0.29	0.51	P < 0.001	No high risk of bias trial	No high risk of bias trial	P = 0.11, I^2 = 61.3
	TRM	1.22	0.35	4.29	P = 0.76	1.24	0.36	4.2	P = 0.73	No high risk of bias trial	No high risk of bias trial	P = 0.72, I^2 = 0
	TD	0.71	0.47	1.05	P = 0.09	0.78	0.68	0.89	P < 0.001	No high risk of bias trial	No high risk of bias trial	No high risk of bias trial
	All AEs	1.41	1.28	1.54	P < 0.001	1.41	1.28	1.54	P < 0.001	No high risk of bias trial	No high risk of bias trial	P = 0.17, I^2 = 45.8
	ORR	No data available										
	CRR	No data available										
	MRD	No data available										
Anti-CD52 mAbs versus observation	**OS**	No data available										
	HRQoL	No data available										
	Grade 3/4 AEs	No data available										
	PFS	Only one trial										
	TRM	No data available										
	TD	No data available										
	All AEs	No data available										
	ORR	Only one trial										
	CRR	Only one trial										
	MRD	No data available										

AEs: adverse events; **CI**: confidence interval; **CRR**: complete response rate; **HR**: hazard ratio; **HRQoL**: health-related quality of life; **IMiD:** immunomodulatory drug; **mAb:** monoclonal antibody; **MRD**: minimal residual disease; **ORR**: overall response rate; **OS**: overall survival; **PFS**: progression-free survival; **TD**: treatment discontinuation; **TRM**: treatment-related mortality

APPENDICES

Appendix 1. CENTRAL search strategy

1. MeSH descriptor Leukemia, B-Cell explode all trees
2. MeSH descriptor Leukemia, Lymphocytic, Chronic, B-Cell explode all trees
3. (lymph*) OR (leu*em*):TI,AB,KY
4. (lymphoblast*) OR (lymphocyt*) OR (linfoid*) OR (b-cell*):TI,AB,KY
5. (chronic*) OR (cronic*) OR (chroniq*) OR (well-differential*):TI,AB,KY
6. CLL
7. (#1 OR #2 OR #3 OR #4 OR #5 OR #7)
8. Mesh descriptor Maintenance explode all trees
9. maintenanc* OR (post-remission*):TI,AB,KY
10. (#8 OR #9)
11. (#7 AND #10)
12. MeSH descriptor Antibodies, Monoclonal explode all trees
13. MeSH descriptor Antigens, CD20 explode all trees
14. (CD20) OR (CD-20) OR (CD 20) OR (ANTI-CD20) OR (ANTI CD20) OR (ANTICD20 OR (ANTI-CD-20) OR (ANTICD-20):TI,AB,KY
15. (antibod*) OR (monoclonal*):TI,AB,KY
16. (mabt*) OR (ritux*):TI,AB,KY
17. (ofatumumab*) OR (arzerr*):TI,AB,KY
18. (obinutuzumab*) OR (gazyva*):TI,AB,KY
19. MeSH descriptor Antigens, CD52 explode all trees
20. (CD52) OR (CD-52) OR (CD 52) OR (ANTI-CD52 or ANTI CD52) OR (ANTICD52 or ANTI-CD-52 or ANTICD-52):TI,AB,KY
21. (alemtuzumab*) OR (campath*):TI,AB,KY
22. (lenalidomide*) OR (revlimid*):TI,AB,KY
23. (#12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22)
24. (#11 AND #23)

Appendix 2. MEDLINE Ovid search strategy

1. exp LEUKEMIA, B-CELL/
2. exp LEUKEMIA, LYMPHOCYTIC, CHRONIC, B-CELL/
3. ((leuk?em$ or leu?em$ or lymph$) adj (lymphocyt$ or lymphoblast$ or linfoid$ or b-cell$)).tw,kf,ot.
4. (chronic$ or cronic$ or chroniq$ or well-differential$).tw,kf,ot.
5. 3 and 4
6. cll.tw.
7. 1 or 2 or 5 or 6
8. exp MAINTENANCE/
9. maintenanc$.tw,kf,ot.
10. ((post-remission$ or postremission$) adj2 therap$).tw,kf,ot.
11. or/8-10
12. 7 and 11
13. exp ANTIBODIES, MONOCLONAL/
14. (antibod$ adj2 monoclonal$).tw,kf,ot.
15. exp LENALIDOMIDE
16. 13 or 14 or 15
17. exp ANTIGENS, CD20/
18. ((CD20 or CD-20 or CD 20) adj3 antibod$).tw,kf,nm,ot.
19. (ANTI-CD20 or ANTI CD20).tw,kf,ot,nm.
20. (ANTICD20 or ANTI-CD-20 or ANTICD-20).tw,kf,nm,ot.
21. mabt$.tw,kf,ot,nm.
22. ritux$.tw,kf,ot,nm.

23.ofatumumab$.tw,kf,ot,nm.
24.arzerr$.tw,kf,ot.
25.obinutuzumab$.tw,kf,ot.
26.gazyva$.tw,kf,ot.
27.exp ANTIGENS, CD52/
28.((CD52 or CD-52 or CD 52) adj3 antibod$).tw,kf,ot,nm.
29.(ANTI-CD52 or ANTI CD52).tw,kf,ot,nm.
30.(ANTICD52 or ANTI-CD-52 or ANTICD-52).tw,kf,ot,nm.
31.alemtuzumab$.tw,kf,nm,ot.
32.campath$.tw,kf,nm,ot.
33.lenalidomide$.tw,kf,nm.
34.revlimid$.tw,kf,ot,nm.
35.or/17-34
36.12 and (16 or 35)
37.randomized controlled trial.pt.
38.controlled clinical trial.pt.
39.randomized.ab.
40.placebo.ab.
41.drug therapy.fs.
42.randomly.ab.
43.trial.ab.
44.groups.ab.
45.or/37-44
46.36 and 45
47.humans.sh.
48.46 and 47

Appendix 3. Embase search strategy

1. exp B CELL LEUKEMIA/
2. exp CHRONIC LYMPHATIC LEUKEMIA/
3. ((lymphocyt* or lymphoblast* or b-cell*) adj (leukemia*)).tw.
4. (chronic* or well-differential*).tw.
5. 3 and 4
6. cll.tw.
7. 1 or 2 or 5 or 6
8. exp MAINTENANCE/
9. maintenanc*.tw.
10. ((post-remission*) adj2 therap*).tw.
11. or/8-10
12. 7 and 11
13. 'crossover procedure':de OR 'double-blind procedure':de OR 'randomized controlled trial':de OR 'single-blind procedure':de OR (random* OR factorial* OR crossover* OR cross NEXT/1 over* OR placebo* OR doubl* NEAR/1 blind* OR singl* NEAR/1 blind* OR assign* OR allocat* OR volunteer*):de,ab,ti
14. 12 and 13
15. exp ANTIBODIES, MONOCLONAL/
16. (antibod* adj2 monoclonal*).tw.
17. exp LENALIDOMIDE
18. 15 or 16 or 17
19. 14 and 18
20. exp ANTIGENS, CD20/
21. ((CD20 or CD-20 or CD 20) adj3 antibod*).tw.
22. (ANTI-CD20 or ANTI CD20).tw.
23. (ANTICD20 or ANTI-CD-20 or ANTICD-20).tw.

24.mabt*.tw.
25.ritux*.tw.
26.ofatumumab*.tw.
27.arzerr*.tw.
28.obinutuzumab*.tw.
29.gazyva*.tw.
30.exp ANTIGENS, CD52/
31.((CD52 or CD-52 or CD 52) adj3 antibod*).tw.
32.(ANTI-CD52 or ANTI CD52).tw.
33.(ANTICD52 or ANTI-CD-52 or ANTICD-52).tw.
34.alemtuzumab*.tw.
35.campath*.tw.
36.lenalidomide*.tw.
37.revlimid*.tw
38.or/20-37
39.19 and 38

Appendix 4. ClinicalTrials.gov search strategy

Conditions: leukemia, lymphocytic, chronic, b-cell OR leukemia, b-cell

Interventions: alemtuzumab OR lenalidomide OR antigen cd20 OR rituximab OR protein-tyrosine kinases OR BCR inhibitor OR ibrutinib OR idelalisib OR BCL 2 inhibitor OR venetoclax OR maintenanceRecruitment: All studies

Study type: Interventional studies

Appendix 5. WHO ICTRP search strategy

Condition: chronic lymphocytic leukemia OR B-cell leukemia

Intervention: alemtuzumab OR lenalidomide OR antigen cd20 OR rituximab OR protein-tyrosine kinases OR BCR inhibitor OR ibrutinib OR idelalisib OR BCL 2 inhibitor OR venetoclax OR maintenance

Recruitment status: ALL

Appendix 6. EU Clinical Trials Register search strategy

("chronic lymphocytic leukemia" OR "B-cell leukemia") AND (alemtuzumab OR lenalidomide OR antigen cd20 OR rituximab OR protein-tyrosine kinases OR BCR inhibitor OR ibrutinib OR idelalisib OR BCL 2 inhibitor OR venetoclax) AND maintenance

HISTORY

Protocol first published: Issue 11, 2019

CONTRIBUTIONS OF AUTHORS

- Wrote the protocol and full review (Cho-Hao Lee, Po-Huang Chen)
- Designed the data input methods (Chin Lin)
- Contributed clinical expertise (Tzu-Chuan Huang, Po-Huang Chen, Yi-Ying Wu)
- Developed the search strategy (Yi-Fen Zou, Hong-Jie Jhou)
- Designed the content input methods (Ju-Chun Cheng)
- Wrote the protocol and supervised the final review (Ching-Liang Ho)

DECLARATIONS OF INTEREST

Cho-Hao Lee: none known

Yi-Ying Wu: none known

Tzu-Chuan Huang: none known

Chin Lin: none known

Yi-Fen Zou: none known

Ju-Chun Cheng: none known

Po-Huang Chen: none known

Hong-Jie Jhou: none known

Ching-Liang Ho: none known

SOURCES OF SUPPORT

Internal sources

- Research Grant | Tri-Service General Hospital/National Defense Medical Center (No. TSGH-C110-128, 2021), Taiwan

  Hospital grant
- Research Grant | Tri-Service General Hospital/National Defense Medical Center (No. TSGH-D-111158, 2022), Taiwan

  Hospital grant

External sources

- No external sources of support, Taiwan

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

- In Types of studies, we included justifications for excluding variants of randomised trials and non-randomised studies. We also excluded cluster-RCTs due to the potential for bias based on how individual participants were identified and recruited within clusters.
- In Types of participants, we added a description clarifying that patients who received curative intent therapy (stem cell transplant) for CLL and did not experience relapse or refractory were excluded from the review.
- In Types of interventions, we altered the order of comparisons specified in the protocol based on their importance in clinical practice. Furthermore, we clarified that we excluded drugs that are no longer in use and are only of historical interest for the treatment of CLL.
- In Primary outcomes, we changed PFS to a secondary outcome and added grade 3 and 4 adverse events and health-related quality of life as our primary outcomes based on suggestions from review authors.
- In Secondary outcomes, we excluded event-free survival (EFS) as an outcome measure because CLL progresses slowly and treatment may only delay, rather than cure, the disease. EFS measures the time from treatment initiation to specific events or death and is commonly used to evaluate the risk of disease worsening, but may have biased definitions. Our goal was to assess the impact of maintenance therapy on CLL progression, but EFS may not accurately evaluate the effectiveness of maintenance therapy, thus we opted to exclude it.
- In Electronic searches, we added a search strategy for the WHO ICTRP and EU Clinical Trials Register. We also added a search for treatment guidelines and removed the need to contact drug manufacturers. We removed the Database of Abstracts of Reviews of Effects (DARE) from our search due to it no longer being part of the Cochrane Library.
- In Assessment of risk of bias in included studies, we added definitions of low, high, and unclear risk of bias and specified the risk of bias tool that we used. We separated subjective and objective outcomes when reporting findings for the performance and detection bias domains of the risk of bias tables.
- In Measures of treatment effect, we added a description of how to estimate hazard ratios if they were not reported in the studies. We also added a description of how to deal with rate ratios.
- In Measures of treatment effect, we added the following sentence: "If data were only available that could be extracted from figures, we used GetData Graph Digitizer software (GetData Graph Digitizer)".
- In Data synthesis, we added more information on how to incorporate multi-arm studies into the synthesis.
- In Subgroup analysis and investigation of heterogeneity, we amended the prespecified variables based on review authors' comments by adding the potential effect modifier 'type of drug' and removing prognostic factors such as Rai and Binet disease stage, response to induction, and the influence of prognostic factors.
- In Sensitivity analysis, we revised our methodology to include a comparison between fixed-effect model and random-effects model. Additionally, we conducted separate comparisons between high and low risk of bias in the domains of random sequence generation, allocation concealment, and blinding.
- In Summary of findings and assessment of the certainty of the evidence under 'Tabulated summary of findings', we added the outcomes HRQoL and TRM, and removed the outcomes ORR and MRD, based on their importance.
- In Results, under 'All adverse events', we reported rate ratio as treatment measurements.